UNIT 4

Computer Systems and Networks

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings & Distractor Breakdowns)
- Questions 91–100: Create Performance Task Style Written Responses (9 Points Each)
- Complete Answer Key and Detailed Worked Solutions at the End

Examination Structure

This paper contains 100 continuously numbered questions designed for complete mastery of AP Computer Science Principles Unit 4: Computer Systems and Networks (Big Idea 4: CSN, weighted 11%–15% of the AP Exam). All answers, step-by-step worked solutions, and 9-point scoring rubrics appear after Question 100.

★ PRIORITY RATING SYSTEM

★★★★★ **Very Important**	High-frequency exam concepts: Open protocols (IETF), Packet switching & dynamic routing, IPv4 vs IPv6 128-bit scaling, TCP reliability vs UDP speed, DNS resolution hierarchy & spoofing, Fault tolerance & network graph SPOF cuts, Parallel speedup formula & Amdahl's Law, and Asymmetric public-key encryption (RSA).
★★★★ **High Priority**	Core networking standards: End-to-end principle, Physical transmission media, Time-To-Live (TTL) loop prevention, HTTP vs HTTPS with TLS/SSL, Bandwidth vs Latency calculations, Bottlenecks in series, Volunteer distributed computing, Rogue Wi-Fi access points, and Multi-Factor Authentication (MFA).
★★★ **Mandatory**	Essential systems mechanics: Device vs system vs network hierarchy, Private IP and NAT translation, Jitter in real-time streaming, Diagnostic tools (Ping, Traceroute), and Horizontal vs Vertical cloud scaling.

Topic Coverage (Big Idea 4: CSN-1 through CSN-2)

4.1 The Internet Computing Systems, Networks, Open Protocols & Layering

4.1b IP Addressing IPv4 (32-bit), IPv6 (128-bit), Routers & Hierarchical Addressing

4.1c Packets & Transport Packet Headers, Payloads, TCP Reliability & UDP Low Latency

4.1d Web Systems Domain Name System (DNS), HTTP/HTTPS, URLs & Certificates

4.2 Fault Tolerance Redundancy, Network Graph Topologies, SPOF & Mesh Resilience

4.3 Parallel Computing Concurrency, Speedup ($S = T_{seq}/T_{par}$), Amdahl's Law & Overhead

4.3b Distributed Systems Volunteer Science, Cloud Elasticity, Data Locality & DDoS Botnets

4.4 Cybersecurity Symmetric vs Asymmetric Crypto, Digital Signatures, VPNs & MFA

AP Exam Pro Tip: Remember: The Internet is an open, distributed packet-switched network with *no central authority*. Dynamic routing allows packets to take independent paths. **Bandwidth** is bitrate (capacity); **Latency** is propagation delay (time). When adding processors, **Amdahl's Law** dictates that speedup is strictly limited by the sequential fraction. And in public-key encryption: *encrypt with recipient's public key, decrypt with recipient's private key*!

- **Questions 1–90:** Multiple choice, four options (A–D), exactly one correct answer.
- **Questions 91–100:** Create Performance Task Written Response scenarios, 9 points each, with full rubrics.
- **Difficulty Mix:** Authentic Medium AP Exam Level ❖ Rigorously 3-Pass Verified.
- **Answer Section:** Begins after Question 100 — master answer key, detailed MCQ solutions with distractor failure analysis, and model FRQ solutions with official point allocations.

Question 01

★★★★★ VERY IMPORTANT

FUNDAMENTAL ARCHITECTURE OF THE INTERNET

Which of the following statements best describes the fundamental architecture of the Internet according to computer science principles?

(A) A globally distributed network of interconnected computing devices that communicate using open, standardized protocols.

(B) A single software application that runs in web browsers to display hyperlinked multimedia documents.

(C) A centralized supercomputer managed by an international governing board that routes all global web requests.

(D) A proprietary network owned and operated by telecommunications corporations using confidential communication rules.

Question 02

★★★★★ VERY IMPORTANT

OPEN STANDARDS & PROTOCOL GOVERNANCE (IETF)

The Internet Engineering Task Force (IETF) is a loosely organized collection of engineers and computer scientists that develops internet standards. Why is it essential that the core communication protocols of the Internet (such as IP, TCP, and HTTP) are **OPEN STANDARDS**?

(A) Open standards restrict internet access exclusively to certified academic and governmental research institutions.

(B) Open standards ensure that all data transmitted across the Internet is automatically encrypted and confidential.

(C) Open standards allow devices manufactured by different vendors and running different operating systems to communicate seamlessly with one another without proprietary licensing restrictions.

(D) Open standards prevent hackers from accessing source code, thereby eliminating all cybersecurity vulnerabilities.

Question 03

★★★★ HIGH PRIORITY

CENTRALIZED VS. DECENTRALIZED VS. DISTRIBUTED NETWORKS

Consider three network topologies connecting 8 user nodes:

- Network X (Centralized): All 8 nodes connect directly to a single central server hub.
- Network Y (Decentralized): Nodes connect to regional hubs, which connect to each other.
- Network Z (Distributed / Mesh): Every node connects to multiple neighboring nodes with redundant paths.

Which of the following describes a key vulnerability of Network X that Network Z successfully avoids?

(A) In Network X, data packets cannot be routed sequentially, whereas Network Z eliminates all routing delays.

(B) In Network X, bandwidth increases exponentially as nodes fail, whereas Network Z experiences server collapse.

(C) In Network X, computing devices cannot use open protocols, whereas Network Z requires proprietary software.

(D) In Network X, if the central hub fails, all communication across the entire network ceases (Single Point of Failure), whereas Network Z can route around failed nodes.

Question 04 ★★★★★ VERY IMPORTANT

PACKET SWITCHING VS. CIRCUIT SWITCHING

Traditional landline telephone networks utilized **circuit switching**, whereas the Internet utilizes **packet switching**.

Which of the following is a major advantage of packet switching over circuit switching for global computer communications?

(A) Packet switching prevents network congestion by disconnecting users when network traffic exceeds 50% capacity.

(B) Packet switching guarantees that all data arrives in perfectly sequential order without requiring sequence numbering.

(C) Packet switching establishes an unbroken physical wire connection between sender and receiver for the entire duration of the session.

(D) Packet switching allows multiple devices to dynamically share the same physical network cables simultaneously without reserving dedicated, idle communication channels.

Question 05 ★★★★★ VERY IMPORTANT

FUNCTION & ROLE OF NETWORK ROUTERS

What is the primary role of a **router** in the operation of the Internet?

(A) To inspect the destination IP address of each incoming packet and forward it along an optimal path toward its final destination.

(B) To assign legal copyright ownership to files uploaded by network clients.

(C) To convert digital HTML code into human-readable text and images on a user's screen.

(D) To provide physical battery backup power to client workstations during power outages.

Question 06 ★★★★ HIGH PRIORITY

THE END-TO-END ARCHITECTURAL PRINCIPLE

The architectural design of the Internet is governed by the **End-to-End Principle**. Which of the following best summarizes this design philosophy?

(A) Physical cables must connect every single computer directly to every other computer on Earth without intermediate hops.

(B) Internet service providers (ISPs) must inspect and approve all application software before it can transmit data.

(C) Complex application logic and error-checking are handled by the sender and receiver devices (the end hosts), keeping the core network routers simple, fast, and content-agnostic.

(D) All data packets must be encrypted twice: once at the beginning of the transmission and once at the end.

Question 07 ★★★★ HIGH PRIORITY

COMPUTING DEVICE VS. SYSTEM VS. NETWORK

The AP CSP curriculum makes formal distinctions between a computing device, a computer system, and a computer network.

Which of the following correctly pairs an example with its proper classification?

(A) A laptop is a **computer network**; the internet is a **computing device**; a smart sensor is a **computer system**.

(B) A supercomputer is a **computer network**; a local Wi-Fi router is a **computer system**; a server rack is a **computing device**.

(C) A smart thermostat is a **computing device**; the thermostat connected to a furnace controller is a **computer system**; five smart homes connected to a utility grid server form a **computer network**.

(D) A smartphone is a **computer system**; a single CPU is a **computer network**; an ethernet cable is a **computing device**.

Question 08 ★★★★★ VERY IMPORTANT

PROTOCOL HIERARCHY & LAYERED ABSTRACTION

Internet protocols are organized in a **layered hierarchy** (e.g., Application layer: HTTP; Transport layer: TCP; Internet layer: IP; Link layer: Ethernet/Wi-Fi).

What is the primary engineering advantage of organizing internet protocols into this hierarchical abstraction?

(A) It eliminates the need for IP addresses when sending web requests.

(B) Higher-level protocols can be designed and updated without needing to know or manage the physical details of how bits are transmitted across copper wires, fiber optics, or radio waves.

(C) It ensures that all files transmitted are compressed by exactly 50%.

(D) It prevents governments from monitoring any packet headers.

Question 09 ★★★★★ VERY IMPORTANT

SCALABILITY OF DISTRIBUTED COMPUTER NETWORKS

The Internet has expanded from connecting just 4 universities in 1969 to connecting over 5 billion devices today. What term describes a system's ability to maintain performance and handle increasing demands by adding resources without redesigning the core architecture?

(A) Abstraction

(B) Fault tolerance

(C) Quantization

(D) Scalability

Question 10

★★★★ HIGH PRIORITY

PHYSICAL TRANSMISSION MEDIA TRADE-OFFS

Computers transmit bits over physical media using electrical signals, light pulses, or radio waves. A network engineer is designing a communications link connecting two data centers separated by 500 miles.

Which of the following physical media is best suited for this long-distance, high-bandwidth connection, and what is its primary transmission mechanism?

(A) Radio frequency Wi-Fi; transmits physical electrons through the upper atmosphere without physical infrastructure.

(B) Unshielded twisted-pair Ethernet cable; limited to line-of-sight laser beams through the air.

(C) Copper coaxial cable; transmits bits as sound waves through solid metal wires.

(D) Fiber-optic cable; transmits bits as light pulses through glass fibers, providing extremely high bandwidth and low signal degradation over long distances.

Question 11

★★★★★ VERY IMPORTANT

STRUCTURE & CAPACITY OF IPV4 ADDRESSES

An IPv4 address (such as `172.217.14.206`) is used to uniquely identify devices on the Internet.

Which of the following correctly states the bit length of an IPv4 address and the maximum number of unique devices it can theoretically address?

(A) 128 bits; providing 2^{128} (undecillions of) unique addresses.

(B) 64 bits; providing 2^{64} unique addresses.

(C) 32 bits; providing 2^{32} (approximately 4.3 billion) unique addresses.

(D) 16 bits; providing 2^{16} (65,536) unique addresses.

Question 12

★★★★★ VERY IMPORTANT

IPV4 ADDRESS EXHAUSTION & THE NEED FOR IPV6

In the 1990s, computer scientists recognized that the Internet was rapidly running out of available IPv4 addresses (a crisis known as IPv4 address exhaustion).

What was the primary technological driver that caused IPv4 addresses to become exhausted?

(A) Computer manufacturers stopped producing 32-bit computer processors.

(B) Web browsers discontinued support for decimal numbers.

(C) The exponential explosion of internet-connected mobile phones, laptops, smart home IoT devices, and global web servers far exceeded the 4.3 billion addresses available in IPv4.

(D) Fiber-optic cables cannot physically transmit 32-bit packets.

Question 13 ★★★★★ VERY IMPORTANT

STRUCTURE & CAPACITY OF IPV6 ADDRESSES

Consider the following sample network address:

`2001:0db8:85a3:0000:0000:8a2e:0370:7334`

Which of the following correctly identifies the protocol version of this address and its total bit length?

(A) IPv6; it consists of 128 bits represented as 8 hexadecimal groups separated by colons.

(B) IPv4; it consists of 32 bits represented as 4 decimal groups separated by periods.

(C) Domain Name; it consists of alphanumeric characters resolved by HTTP.

(D) MAC address; it consists of 48 bits represented as 6 octets.

Question 14 ★★★★ HIGH PRIORITY

MATHEMATICAL COMPARISON OF IPV4 VS. IPV6 CAPACITY

How many times larger is the address space of IPv6 (128 bits) compared to the address space of IPv4 (32 bits)?

(A) 2^{96} times larger (an astronomical factor of approximately 7.9×10^{28} times larger).

(B) 2^{4} times larger (16 times larger).

(C) 96 times larger (since 128 - 32 = 96).

(D) 4 times larger (since 128 / 32 = 4).

Question 15 ★★★★★ VERY IMPORTANT

DYNAMIC PACKET ROUTING MECHANICS

When a user streams a 5-minute video, the video file is segmented into thousands of individual data packets. Why do different packets from the same video file frequently travel along completely different physical routes across the Internet?

(A) Video streaming files are programmed to intentionally scramble their paths to confuse web browsers.

(B) Network security rules mandate that every packet must visit a different country before reaching the destination.

(C) Internet service providers assign each user a different communication satellite for every second of video playback.

(D) Routers use dynamic routing protocols to send packets along alternate paths based on real-time network traffic congestion, link availability, and transmission costs.

Question 16

★★★★ HIGH PRIORITY

TIME-TO-LIVE (TTL) AND ROUTING LOOP PREVENTION

Every IP packet header includes an 8-bit field called **Time-To-Live (TTL)**, which is initialized to a number (such as 64) by the sender.

What happens to the TTL value as the packet traverses the Internet, and what occurs if it reaches 0?

(A) TTL increases by 1 at each router; if it reaches 100, the packet is granted administrative priority.

(B) TTL represents the exact time in seconds the user has left on their monthly mobile cellular data plan.

(C) TTL measures the physical temperature of the router's central processing unit.

(D) Each router decrements TTL by 1; if TTL reaches 0 before the packet reaches its destination, the packet is discarded to prevent it from circulating endlessly in an infinite routing loop.

Question 17

★★★★★ VERY IMPORTANT

HIERARCHICAL STRUCTURE OF IP ADDRESSES



How is this hierarchy reflected in the structure of an IP address?

(A) The first bit always indicates whether the user is on a mobile phone or a desktop computer.

(B) The middle two bytes always contain the user's geographic GPS coordinates.

(C) The leading bits (network prefix) identify the specific network or Internet Service Provider (ISP), while the trailing bits identify the specific host device on that network.

(D) All IP addresses are assigned entirely at random without any grouping or regional structure.

Question 18

★★★★ HIGH PRIORITY

ROUTER FORWARDING TABLE DECISIONS

Consider a network router that receives an IP packet. How does the router determine which physical port (network interface) to forward the packet through?

(A) It compares the packet's destination IP address against its internal routing table to find the longest matching network prefix.

(B) It opens the data payload and reads the user's plain-text message to understand where it wants to go.

(C) It broadcasts the packet out of every single port simultaneously to ensure delivery.

(D) It asks the user's web browser to pick the outgoing cable.

Question 19

★★★★ HIGH PRIORITY

PUBLIC VS. PRIVATE IP ADDRESSES & NAT

Many home and school Wi-Fi networks assign devices IP addresses that begin with `192.168.x.x` or `10.x.x.x`.

Why can multiple different homes across the world all use the exact same IP address (such as `192.168.1.10`) simultaneously without causing addressing collisions on the global Internet?

(A) These are designated private IP addresses used only inside local area networks; the home router translates them to a single unique public IP address using Network Address Translation (NAT) when accessing the global Internet.

(B) Only one home in the world can be connected to the Internet at any given time.

(C) Private IP addresses bypass the Internet entirely and communicate using satellite telepathy.

(D) The Internet assigns each home a unique encryption key that changes the numbers automatically every second.

Question 20

★★★★★ VERY IMPORTANT

IP SPOOFING & NETWORK SECURITY VULNERABILITIES

What is **IP address spoofing**, and how does an attacker exploit it in a computer network?

(A) An attacker registers a copyright claim over an existing website's domain name.

(B) An attacker crafts data packets with a falsified source IP address in the packet header to conceal their identity or impersonate a trusted computing device.

(C) An attacker increases the bandwidth of a client computer without paying the ISP.

(D) An attacker physically cuts undersea fiber-optic cables to reroute all internet traffic.

Question 21

★★★★★ VERY IMPORTANT

STRUCTURE OF A NETWORK PACKET: HEADER VS. PAYLOAD

Every data packet transmitted across the Internet consists of two main sections: the **Packet Header** and the **Packet Payload**.

Which of the following correctly describes the contents and role of these two components?

(A) The header is discarded by the sender before transmission, and only the payload travels across routers.

(B) The header contains the user's credit card information, while the payload contains the operating system code.

(C) The header contains routing metadata (such as source IP, destination IP, packet size, and sequence number), while the payload contains the actual user data being transmitted.

(D) The header contains the entire user file, while the payload contains legal copyright agreements.

Question 22

★★★★★ VERY IMPORTANT

OUT-OF-ORDER PACKET ARRIVAL & TCP REASSEMBLY

A high-resolution photograph is broken into 500 packets and transmitted across the Internet. Because the packets travel along different physical paths, they arrive at the destination computer out of order (for example: packet #14 arrives before packet #2).

How does the receiving computer reconstruct the original photograph correctly?

(A) The receiving computer discards all packets that arrive out of order and asks the user to restart the computer.

(B) The receiving computer displays the image pixels in the scrambled order they arrived, creating visual noise.

(C) The router automatically delays all packets until they line up in perfect sequential order before entering the home.

(D) The Transmission Control Protocol (TCP) on the receiving computer reads the sequence number in each packet's header and reorders them into their original sequence before displaying the image.

Question 23

★★★★★ VERY IMPORTANT

CHARACTERISTICS OF THE TRANSMISSION CONTROL PROTOCOL (TCP)

Which of the following best describes the core operational characteristics of the **Transmission Control Protocol (TCP)**?

(A) Connection-oriented and reliable; it establishes a connection, verifies that all packets arrive intact, requests retransmission of lost packets, and reassembles them in order.

(B) Connectionless and fast; it sends packets without checking for errors, acknowledgments, or packet loss.

(C) Physical and optical; it modulates light frequencies along undersea fiber cables.

(D) Proprietary and encrypted; it is owned exclusively by Microsoft and encrypts all web traffic.

Question 24

★★★★★ VERY IMPORTANT

CHARACTERISTICS OF THE USER DATAGRAM PROTOCOL (UDP)

Which of the following best describes the core operational characteristics of the **User Datagram Protocol (UDP)**?

(A) Connectionless and lightweight; it transmits packets rapidly with minimal overhead, without establishing a formal connection, tracking packet order, or retransmitting lost packets.

(B) A database query protocol used to search domain names on root servers.

(C) Connection-oriented and slow; it guarantees that zero packets are ever lost under any circumstances.

(D) A hardware specification governing the electrical voltage of Ethernet pins.

Question 25

★★★★★ VERY IMPORTANT

APPLICATION SELECTION: TCP VS. UDP

A software development firm is engineering two distinct networked applications:

• Application 1: An online banking platform for processing wire transfers between banks.
• Application 2: A live multiplayer first-person video game where players compete in real time.

Which transport protocols should the firm select for Application 1 and Application 2, respectively?

(A) Both applications must use UDP; TCP cannot transmit numbers.

(B) Application 1 should use TCP (guaranteeing 100% accuracy without data corruption); Application 2 should use UDP (minimizing latency and input lag where dropping an occasional packet is acceptable).

(C) Application 1 should use UDP; Application 2 should use TCP.

(D) Both applications must use TCP; UDP is obsolete and forbidden on the modern Internet.

Question 26

★★★★ HIGH PRIORITY

NETWORK CONGESTION & PACKET DROPPING MECHANICS

During a major sporting event, millions of viewers simultaneously stream live video, causing severe network congestion at an intermediate router.

Why does a router drop packets when congested, and how does the network handle this loss?

(A) Dropped packets cause the entire Internet backbone to reboot.

(B) Routers never drop packets; instead, they permanently store infinite packets on internal hard drives.

(C) When incoming packets arrive faster than the router's outgoing link can transmit, the router's memory buffer overflows and drops incoming packets; TCP detects missing acknowledgments and retransmits them at a reduced sending rate.

(D) Routers drop packets to penalize users who have used too much internet bandwidth.

Question 27

★★★★ HIGH PRIORITY

THE TCP THREE-WAY HANDSHAKE

Before a client computer and a web server can exchange data using TCP, they must synchronize and establish a connection using the **Three-Way Handshake**.

Which of the following sequences represents the standard three-way handshake?

(A) 1. Client sends DATA
2. Server sends DATA
3. Client sends FINISH

(B) 1. Client sends PING
2. Server sends PONG
3. Client sends RESET

(C) 1. Client sends SYN (Synchronize)
2. Server responds with SYN-ACK (Synchronize-Acknowledge)
3. Client responds with ACK (Acknowledge)

(D) 1. Client sends GET
2. Server sends POST
3. Client sends DELETE

Question 28

★★★★★ VERY IMPORTANT

PACKET FRAGMENTATION & RETRANSMISSION EFFICIENCY

A 100-megabyte software installer is transmitted across the Internet. Rather than sending the entire 100 MB file as a single massive continuous transmission, the operating system fragments the file into approximately 70,000 small packets of roughly 1,500 bytes each.

Why is breaking data into small packets significantly more efficient than transmitting a single massive file?

(A) Small packets prevent internet service providers from charging data fees.

(B) Small packets bypass all routing tables and travel faster than the speed of light.

(C) Computer memory cannot store files larger than 2,000 bytes.

(D) If a single packet is corrupted or dropped due to network noise, only that individual 1,500-byte packet needs to be retransmitted, rather than re-sending the entire 100 MB file.

Question 29

★★★★ HIGH PRIORITY

PACKET CHECKSUMS & BIT ERROR DETECTION

Every TCP and IP packet header includes a **checksum** field. What is the purpose of this checksum?

(A) To allow the receiving device to perform a mathematical verification to detect whether any bits in the packet were corrupted or altered during transmission.

(B) To count the number of advertisements contained in the payload.

(C) To display the author's copyright license to the recipient.

(D) To measure the physical distance between the sender and receiver in miles.

Question 30

★★★★★ VERY IMPORTANT

PROTOCOL DATA ENCAPSULATION HIERARCHY

When a user types a web address into a browser, an HTTP GET request is generated. As this message travels down the protocol stack toward physical transmission, it undergoes **encapsulation**.

Which of the following correctly describes the hierarchical order of encapsulation from application down to physical link?

(A) The Ethernet frame is placed inside a TCP segment, which is placed inside an IP packet.

(B) The HTTP message is placed inside a TCP segment, which is placed inside an IP packet, which is placed inside an Ethernet frame.

(C) The IP packet is placed inside an HTTP message, which is placed inside a TCP segment.

(D) The TCP segment is placed inside an HTTP message, which is transmitted as raw Wi-Fi.

Question 31

★★★★★ VERY IMPORTANT

DOMAIN NAME SYSTEM (DNS) CORE ROLE

When a student types `www.collegeboard.org` into a web browser, the browser cannot immediately transmit data until it determines the destination computer's numerical IP address.

Which of the following describes the primary function of the Domain Name System (DNS) in this process?

(A) It encrypts the student's personal password before transmitting it across public Wi-Fi networks.

(B) It physically routes electrical signals through underground fiber-optic cables between cities.

(C) It translates human-readable domain names (such as `www.collegeboard.org`) into machine-readable numerical IP addresses (such as `198.51.100.45`).

(D) It compresses high-definition video files into smaller zip archives to preserve network bandwidth.

Question 32

★★★★★ VERY IMPORTANT

HIERARCHICAL ARCHITECTURE OF THE DNS SYSTEM

The Domain Name System is organized in a hierarchical, distributed tree rather than storing every global website on a single centralized supercomputer.

What is the correct sequence of DNS servers queried during a full recursive lookup for the domain `apstudents.collegeboard.org` if the record is not locally cached?

(A) Internet Service Provider (ISP) central router → Local Wi-Fi router → End-user laptop.

(B) Top-Level Domain (TLD) server (`.org`) → Root DNS server → Browser history log.

(C) Root DNS server → Top-Level Domain (TLD) server (`.org`) → Authoritative DNS server (`collegeboard.org`).

(D) Authoritative DNS server (`collegeboard.org`) → Root DNS server → Top-Level Domain (TLD) server (`.org`).

Question 33

★★★★ HIGH PRIORITY

DNS CACHING & TIME-TO-LIVE (TTL)

To reduce redundant queries across global root servers, operating systems and local DNS resolvers store recently resolved domain-to-IP pairings in a local **DNS Cache** for a duration specified by a **Time-To-Live (TTL)** value.

What is the primary benefit of DNS caching?

(A) It decreases latency for recurring page visits and significantly reduces network query traffic on upstream DNS root and TLD servers.

(B) It allows end users to browse websites without being connected to any physical local network.

(C) It guarantees that an attacker can never intercept unencrypted web traffic.

(D) It completely eliminates the need for websites to ever update their web hosting servers or IP addresses.

Question 34

★★★★★ VERY IMPORTANT

DNS SPOOFING & CACHE POISONING CYBERATTACK

A cyber attacker manages to corrupt the cache of a local recursive DNS server by inserting a fraudulent IP address for a popular banking website `securebank.com`. When legitimate users in the district type `securebank.com` into their browsers, they are silently redirected to a clone website operated by the attacker.

Which cybersecurity threat is demonstrated in this scenario?

(A) Physical fiber-optic cable severance.

(B) Keylogger hardware implant.

(C) Distributed Denial of Service (DDoS) bandwidth flood.

(D) DNS Spoofing (also known as DNS Cache Poisoning).

Question 35

★★★★★ VERY IMPORTANT

HYPERTEXT TRANSFER PROTOCOL: HTTP VS. HTTPS

A student is shopping online while connected to an open, unencrypted Wi-Fi hotspot at a coffee shop.

Why is it essential that the student's browser connects to the shopping website using **HTTPS** rather than standard **HTTP**?

(A) HTTPS prevents physical damage to the coffee shop's wireless router in the event of an electrical power surge.

(B) HTTPS encrypts the data exchanged between the browser and web server using TLS/SSL, preventing unauthorized eavesdroppers on the open Wi-Fi network from reading credit card numbers and passwords.

(C) HTTPS ensures that the student's laptop battery will not drain while downloading large images.

(D) HTTPS allows the user to browse the web at double the normal bandwidth speed regardless of ISP limits.

Question 36

★★★★ HIGH PRIORITY

ANATOMY OF A UNIFORM RESOURCE LOCATOR (URL)

Consider the following Uniform Resource Locator (URL):

`https://apstudents.collegeboard.org:443/courses/csp/exam-prep?year=2026`

Which of the following correctly maps the components of this URL to their respective networking roles?

(A) Protocol: `apstudents` | Domain: `https` | Port: `2026` | Path: `exam-prep` | Query Parameter: `443`

(B) Protocol: `collegeboard` | Domain: `org` | Port: `none` | Path: `courses` | Query Parameter: `csp`

(C) Protocol: `https` | Domain: `apstudents.collegeboard.org` | Port: `443` | Path: `/courses/csp/exam-prep` | Query Parameter: `year=2026`

(D) Protocol: `courses/csp` | Domain: `443` | Port: `https` | Path: `apstudents` | Query Parameter: `collegeboard.org`

Question 37

★★★★ HIGH PRIORITY

STATELESSNESS OF HTTP AND SESSION MANAGEMENT VIA COOKIES

By design, HTTP is a **stateless protocol**, meaning that the web server treats every individual request as completely independent and retains no memory of previous requests from the same user.

How do modern e-commerce websites maintain items in a customer's shopping cart as the customer navigates between different pages?

(A) The server sends a unique session identifier stored in a client-side **HTTP Cookie** or local storage, which the browser automatically attaches to subsequent requests.

(B) The user's Internet Service Provider permanently updates global DNS records with the shopping cart contents.

(C) Routers along the Internet backbone save the user's shopping cart in their internal routing tables.

(D) The physical copper wire connecting the home router retains an analog magnetic memory of every clicked link.

Question 38

★★★★★ VERY IMPORTANT

DIGITAL CERTIFICATES AND CERTIFICATE AUTHORITIES (CAS)

When a browser establishes an HTTPS connection with a website, the website presents a **Digital Certificate** signed by an accredited **Certificate Authority (CA)**.

What is the primary role of the Certificate Authority in securing Internet communications?

(A) It verifies the authentic identity of the website owner and issues a cryptographically signed certificate linking the website's public key to its domain name.

(B) It automatically pays the website's electricity bills and internet subscription fees.

(C) It physically hosts all web files on government-owned data storage arrays.

(D) It monitors the user's browsing activity to report illegal downloads to national law enforcement.

Question 39

★★★★ HIGH PRIORITY

DNS SCALABILITY, ANYCAST ROUTING, AND REDUNDANCY

There are only 13 logical root server IP addresses in the global DNS system (named letters `a.root-servers.net` through `m.root-servers.net`). Yet, billions of Internet users make trillions of DNS queries every day without crashing the root system.

Which of the following architectural strategies allows the DNS root system to handle this massive load and resist hardware failures?

(A) Every single home laptop is forced by law to act as an unmonitored global root DNS server.

(B) Each logical root address is implemented using hundreds of physical servers distributed globally that share the same IP address through Anycast routing and aggressive hierarchical caching.

(C) DNS queries are scheduled to run only during overnight hours when website traffic is lowest.

(D) All Internet users are restricted to making a maximum of five DNS queries per day.

Question 40

★★★★★ VERY IMPORTANT

MATCHING INTERNET PROTOCOLS WITH FUNCTIONAL ROLES

A computer science teacher presents a table comparing four core Internet protocols:

Protocol	Primary Responsibility
Protocol 1	Formats web page requests and responses between browsers and web servers
Protocol 2	Translates human-friendly domain names into numerical IP addresses
Protocol 3	Provides reliable, ordered, and error-checked delivery of a stream of packets
Protocol 4	Assigns network addresses and routes individual packets across interconnected networks

Which of the following correctly identifies Protocols 1, 2, 3, and 4?

(A) Protocol 1: TCP | Protocol 2: IP | Protocol 3: HTTP | Protocol 4: DNS

(B) Protocol 1: IP | Protocol 2: TCP | Protocol 3: DNS | Protocol 4: HTTP

(C) Protocol 1: DNS | Protocol 2: HTTP | Protocol 3: IP | Protocol 4: TCP

(D) Protocol 1: HTTP | Protocol 2: DNS | Protocol 3: TCP | Protocol 4: IP

Question 41

★★★★★ VERY IMPORTANT

DEFINITION AND MEASUREMENT OF BANDWIDTH

In computer networking, **Bandwidth** is one of the most fundamental metrics used to characterize transmission capacity.

Which of the following best defines network bandwidth?

(A) The time of day when a user's computer is turned on and receiving electricity.

(B) The physical length of the copper wire connecting a computer to the wall outlet, measured in meters.

(C) The maximum rate at which data can be transferred across a network connection in a given amount of time, typically measured in bits per second (bps, Mbps, or Gbps).

(D) The total amount of hard drive storage space remaining on a local laptop computer.

Question 42

★★★★★ VERY IMPORTANT

DEFINITION AND MEASUREMENT OF NETWORK LATENCY

While playing an online multiplayer video game, a student notices a significant delay between pressing a controller button and seeing the character respond on screen, even though their home has a high-speed fiber-optic connection.

Which network metric directly explains this delay, and how is it measured?

(A) **Latency**, which measures the time elapsed for a data packet to travel from the sender to the receiver, typically expressed in milliseconds (ms).

(B) **Pixel density**, which measures the sharpness of the laptop monitor in dots per inch (dpi).

(C) **Clock speed**, which measures the number of instructions executed per second by the CPU in gigahertz (GHz).

(D) **Battery voltage**, which measures the electrical potential of the power supply in volts (V).

Question 43

★★★★ HIGH PRIORITY

DISTINGUISHING BETWEEN BANDWIDTH AND LATENCY

Consider a rural research station in Antarctica that connects to the Internet via a geostationary satellite link. The connection can download large data files at 100 Mbps, but every web request takes approximately 650 milliseconds before any response begins returning.

Which statement correctly characterizes this network connection?

(A) The connection exhibits **high bandwidth** (high volume of data transmitted per second) but **high latency** (long propagation delay due to orbital distance).

(B) The connection exhibits **low bandwidth** and **low latency**.

(C) The connection has low latency because satellite radio waves travel faster than the speed of light.

(D) The connection exhibits **zero bandwidth** because satellite signals cannot pass through clouds.

Question 44

★★★★★ VERY IMPORTANT

MATHEMATICAL CALCULATION OF NETWORK TRANSFER TIME

A software developer needs to transfer a raw video file with a size of **150 Megabytes (MB)** over a dedicated network connection with a bandwidth of **60 Megabits per second (Mbps)**.

Assuming optimal conditions with zero packet loss and neglecting protocol overhead, approximately how long will the transfer take?
(Note: 1 Byte = 8 bits)

(A) **150 seconds**

(B) **20 seconds**

(C) **9,000 seconds**

(D) **2.5 seconds**

Question 45

★★★★★ VERY IMPORTANT

IDENTIFYING NETWORK BOTTLENECKS IN SEQUENTIAL PATHS

Data travels from Server A to Client D through three sequential network links connected in series:



What is the maximum theoretical data transfer rate between Server A and Client D?

(A) **1,000 Mbps**, because the initial link sets the permanent speed for all downstream links.

(B) **15 Mbps**, because the overall end-to-end bandwidth is constrained by the slowest link (the bottleneck) along the transmission path.

(C) **371.6 Mbps**, because the bandwidth is calculated as the average of the three links.

(D) **1,115 Mbps**, because the bandwidth of all three links is added together.

Question 46

★★★★ HIGH PRIORITY

DISTINGUISHING BETWEEN BANDWIDTH AND ACTUAL THROUGHPUT

A family subscribes to an Internet plan advertised by their ISP as having a maximum bandwidth of **500 Mbps**. However, when running an actual file download test, their measured transfer rate is **380 Mbps**.

Which of the following factors best explains why actual **Throughput** is frequently lower than theoretical **Bandwidth**?

(A) The ISP automatically cuts Internet speed by half whenever it detects video files.

(B) Throughput is measured in Bytes, which are physically smaller than bits.

(C) The computer's monitor refresh rate limits the speed at which electrical signals pass through cables.

(D) Throughput accounts for real-world factors such as protocol header overhead, network congestion from competing users, and retransmissions caused by packet loss.

Question 47

★★★★ HIGH PRIORITY

NETWORK JITTER AND PACKET ARRIVAL CONSISTENCY

During a live video conference call, a teacher notices that voices occasionally stutter, speed up, or sound robotic, even though the average bandwidth is sufficient.

Which networking phenomenon describes this inconsistency in packet arrival times?

(A) **Jitter**, which represents the variance or fluctuation in packet arrival latency over time.

(B) **Attenuation**, which is the total permanent loss of computer electricity.

(C) **Modulation**, which is the process of converting digital data into analog radio waves.

(D) **Duplexing**, which allows two computers to communicate simultaneously.

Question 48

★★★★ HIGH PRIORITY

NETWORK DIAGNOSTIC UTILITIES: PING AND TRACEROUTE

A network administrator suspects that a slow connection between New York and London is caused by a faulty intermediate router in Ireland. The administrator runs the `traceroute` command-line utility from the New York terminal.

What information does `traceroute` provide to help diagnose the issue?

(A) It displays the complete personal passwords and email addresses of all network users in Ireland.

(B) It displays the list of every intermediate router (hop) along the packet's path, along with the round-trip latency measured to each router.

(C) It physically re-wires the transatlantic fiber-optic cable without human intervention.

(D) It permanently deletes all spam emails stored on the destination web server.

Question 49

★★★★ HIGH PRIORITY

NETWORK CONGESTION AND BUFFER QUEUING DELAYS

During a major live sports broadcast, millions of viewers simultaneously stream video from a single content distribution network. Intermediate routers experience traffic volumes that far exceed their outgoing transmission link capacities.

What happens inside the routers when this **network congestion** occurs?

(A) The router converts all packets into optical light and emits them into the room air.

(B) Packets accumulate in router memory queues, increasing queuing latency; once memory buffers are completely filled, incoming packets are dropped and must be retransmitted.

(C) The router instantly prints out paper copies of all packets to save memory space.

(D) The router shuts down permanently and must be replaced by a physical technician.

Question 50

★★★★★ VERY IMPORTANT

COMPARING FILE TRANSFER SCENARIOS ACROSS DIVERSE NETWORKS

A school district compares four different network configurations for backing up a **400 Megabyte (MB)** educational database to a regional cloud server:

Configuration	Bandwidth	Average Latency
Option 1	16 Mbps	15 ms
Option 2	32 Mbps	80 ms
Option 3	64 Mbps	40 ms
Option 4	128 Mbps	250 ms

Assuming bulk continuous transfer where propagation latency is negligible compared to total data transmission time, which option will complete the backup in the **shortest total time**, and approximately how long will it take?

(A) **Option 1**, completing the backup in approximately 200 seconds because it has the lowest latency.

(B) **Option 2**, completing the backup in approximately 100 seconds.

(C) **Option 4**, completing the backup in approximately **25 seconds**.

(D) **Option 3**, completing the backup in approximately 50 seconds.

Question 51

★★★★★ VERY IMPORTANT

DEFINITION OF FAULT TOLERANCE IN COMPUTER NETWORKS

The Internet was engineered from its inception to maintain communication even during severe physical disruptions such as severed cables or downed regional routers.

Which of the following best defines a **Fault-Tolerant** system in computer science?

(A) A software program that automatically deletes itself whenever a programming error or bug is detected.

(B) A computer network that operates without requiring any electrical power or battery backup.

(C) A system that can continue to function properly and support communication even in the event of the failure of one or more of its components.

(D) A system that is guaranteed to never experience any hardware wear, wire degradation, or component failures.

Question 52

★★★★★ VERY IMPORTANT

ROLE OF REDUNDANCY IN ACHIEVING FAULT TOLERANCE

An engineer adds multiple extra fiber-optic communication cables connecting critical data centers across a state, even though a single cable is physically capable of handling normal daily traffic.

Why is adding this **Redundancy** essential for building a robust network?

(A) It converts standard digital text files into physical handwritten letters.

(B) It halves the electrical cost of running the data center by sharing electricity across both cables.

(C) It prevents users from having to create passwords for their personal email accounts.

(D) It provides alternative transmission routes so that if a primary cable is severed, data traffic can be dynamically rerouted without interrupting network services.

Question 53

★★★★★ VERY IMPORTANT

SINGLE POINT OF FAILURE (SPOF) IDENTIFICATION

Consider the following diagram of a corporate computer network where letters represent routers and lines represent physical connections:



If Router R experiences an internal motherboard failure, which of the following best describes the resulting network impact?

(A) The network is fully fault-tolerant, so data will continue flowing between Branch Offices and Server Farms without delay.

(B) The failure of Router R will cause all connected computers to permanently erase their operating systems.

(C) Only Branch Office A will lose connection, while Branch Office B will automatically connect directly to Server Farm 1.

(D) Router R is a **Single Point of Failure (SPOF)**; its failure completely disconnects both Branch Offices from both Server Farms.

Question 54

★★★★★ VERY IMPORTANT

MINIMUM NUMBER OF LINK FAILURES TO DISCONNECT A DEVICE

A school district designs a network graph connecting six server nodes (A, B, C, D, E, F). Node A is directly connected by individual physical cables to Node B, Node C, and Node D.

What is the **minimum number of cable cuts** required to completely isolate Node A from communicating with all other nodes in the network?

(A) **6 cuts**, because there are six total server nodes in the school district.

(B) **3 cuts**, because Node A has exactly 3 direct physical connections (to B, C, and D), and all 3 must fail to isolate it.

(C) **1 cut**, because cutting any single cable in a network immediately halts all global traffic.

(D) **0 cuts**, because wireless signals will automatically travel through the severed copper wires.

Question 55

★★★★★ VERY IMPORTANT

COUNTING ALTERNATIVE PATHS BETWEEN SOURCE AND DESTINATION

Consider the following network topology connecting Source Node **S** to Destination Node **D**:



How many distinct, independent paths exist for packets to travel from Node S to Node D?

(A) **0 paths**, because data cannot travel through intermediate nodes without being erased.

(B) **1 path**, because the Internet only permits a single predetermined route between two computers.

(C) **3 distinct paths** (`S → 1 → D`, `S → 2 → D`, and `S → 3 → D`).

(D) **9 paths**, because each node multiplies the total number of paths by 3.

Question 56

★★★★ HIGH PRIORITY

ENGINEERING TRADE-OFFS: ECONOMIC COST VS. REDUNDANCY

A city government is designing an emergency management network. The engineering team debates between building a **fully connected mesh network** (where every municipal building connects directly to every other building) versus a **partially connected mesh network** with select redundant loops.

Which of the following describes the primary engineering trade-off of building a fully connected mesh network?

(A) It requires zero routers, but can only transmit data during daylight hours.

(B) It guarantees zero latency, but causes every building to catch fire due to excessive cable friction.

(C) It maximizes fault tolerance and eliminates all single points of failure, but incurs extremely high financial cost and physical complexity to lay and maintain cables between all pairs of buildings.

(D) It makes the network completely immune to all software malware, but prevents users from saving files.

Question 57

★★★★★ VERY IMPORTANT

DYNAMIC REROUTING AND AUTONOMOUS FAULT RECOVERY

While a hospital database is transferring a patient record across a regional health network, an underground construction excavator accidentally cuts through the primary fiber link between Router 2 and Router 3.

How do dynamic routing protocols handle this sudden failure without losing the ongoing session?

(A) Routers detect the broken link via missing keepalive signals, immediately update their routing tables, and automatically forward subsequent packets along an alternative redundant path.

(B) The severed cable emits radio waves into the dirt that wirelessly leap across the gap.

(C) The sender's computer prints an error message instructing the doctor to manually reconnect the physical wires.

(D) The entire regional network shuts down permanently until construction workers repair the glass fibers.

Question 58

★★★★ HIGH PRIORITY

COMPARATIVE FAULT TOLERANCE: STAR VS. MESH TOPOLOGIES

A school IT department compares two different Local Area Network (LAN) topologies for the high school computer lab:

- **Topology X (Star):** All 30 student computers connect exclusively to a single central network switch.
- **Topology Y (Ring with Cross-links):** Computers connect to neighboring computers in a ring with redundant cross-links between rows.

What is a major vulnerability of **Topology X** compared to **Topology Y**?

(A) Topology X cannot transmit text files, whereas Topology Y can only transmit images.

(B) Topology X causes computers to overheat rapidly due to excess radio interference.

(C) Topology X requires 100 times more cables than Topology Y.

(D) If the central switch in Topology X fails, all 30 computers lose network access simultaneously; whereas in Topology Y, failure of a single device can be bypassed through alternative links.

Question 59

★★★★★ VERY IMPORTANT

WHY THE INTERNET IS DESIGNED AS A DECENTRALIZED DISTRIBUTED MESH

In the 1960s, early ARPANET researchers chose to build a **decentralized, distributed mesh network** rather than a **centralized network** governed by a single master routing hub.

What was the primary military and engineering motivation behind choosing a decentralized architecture?

(A) A decentralized network prevents users from ever receiving spam or computer viruses.

(B) Decentralized networks run on free solar energy without requiring commercial electrical grids.

(C) Centralized networks are physically impossible to build using electrical copper cables.

(D) A centralized network could be completely incapacitated if enemy forces destroyed the single central hub, whereas a decentralized mesh can survive the destruction of multiple major nodes by routing around the damage.

Question 60

★★★★★ VERY IMPORTANT

IDENTIFYING THE CRITICAL BOTTLENECK NODE IN A NETWORK GRAPH

Consider the following network connection diagram between five routers:



- Router 1 is connected to Router 3.
- Router 2 is connected to Router 3.
- Router 3 is connected to Router 4.
- Router 3 is connected to Router 5.

Which of the following actions would successfully make this entire network fault-tolerant against the failure of **Router 3**?

(A) Increasing the bandwidth of Router 3 from 100 Mbps to 1,000 Mbps.

(B) Replacing all copper cables with faster fiber-optic cables without adding any new connections.

(C) Adding direct backup connections between Router 1 and Router 4, and between Router 2 and Router 5, creating alternative routes that bypass Router 3 entirely.

(D) Renaming Router 3 to 'Master Server' in the network configuration file.

Question 61

★★★★★ VERY IMPORTANT

SEQUENTIAL VS. PARALLEL COMPUTING DEFINITIONS

A software company is redesigning its image processing software to improve performance on modern computers.

Which of the following correctly distinguishes between **Sequential Computing** and **Parallel Computing**?

(A) In sequential computing, multiple processors are used; in parallel computing, only a single processor can ever be used.

(B) In sequential computing, code is run exclusively on weekends; in parallel computing, code is run during weekdays.

(C) In sequential computing, operations are executed one after another in order on a single processor; in parallel computing, a program is split into multiple parts that execute simultaneously on multiple processors or cores.

(D) In sequential computing, all data is encrypted; in parallel computing, all data is made public.

Question 62

★★★★★ VERY IMPORTANT

MULTI-CORE PROCESSORS AND HARDWARE CONCURRENCY

A student purchases a laptop advertised as having an '8-core processor'.

What physical capability does this hardware specification provide to the computer?

(A) The computer has eight separate keyboards attached to the motherboard.

(B) The Central Processing Unit (CPU) contains eight independent physical execution units (cores) that can each read and execute program instructions concurrently.

(C) The computer battery will last exactly eight hours regardless of how the laptop is used.

(D) The computer can only store eight individual files on its hard drive before becoming completely full.

Question 63

★★★★★ VERY IMPORTANT

MATHEMATICAL CALCULATION OF PARALLEL SPEEDUP

A computer program processing satellite weather photos takes **120 seconds** to run when executed sequentially on a single-core processor. An engineer parallelizes the algorithm, and the new version takes **30 seconds** to complete on a 4-core processor.

What is the **Speedup** achieved by this parallel implementation?
(Formula: $\text{Speedup} = \frac{T_{\text{sequential}}}{T_{\text{parallel}}}$)

(A) **90.0**, because the time was reduced by 90 seconds.

(B) **4.0**, meaning the parallel program runs 4 times faster than the sequential version.

(C) **3,600.0**, because 120 multiplied by 30 equals 3,600.

(D) **0.25**, meaning the parallel program runs 4 times slower.

Question 64

★★★★★ VERY IMPORTANT

AMDAHL'S LAW AND THE SEQUENTIAL BOTTLENECK

An engineer tests a program on a computer with 4 cores and achieves a 3.5x speedup. Excited by the result, the engineer rents a supercomputer with 1,000 cores, expecting the program to run 1,000 times faster. However, the program only achieves a 4.8x speedup.

Which principle of computer science best explains why adding more processors produces diminishing returns?

(A) **Boyle's Law**, which states that electrical current increases pressure inside silicon transistors.

(B) **Metcalfe's Law**, which states that computer programs cannot run on more than 8 cores simultaneously.

(C) **Moore's Law**, which states that processors lose computational speed as more cores are added to a chip.

(D) **Amdahl's Law**, which states that the overall speedup of a program is strictly limited by the portion of the program that must be executed sequentially.

Question 65

★★★★★ VERY IMPORTANT

CALCULATING PARALLEL RUNTIME WITH A SEQUENTIAL COMPONENT

A data analysis task requires a total sequential runtime of **100 seconds**:
- **20 seconds** consists of loading and preparing data, which **must be executed sequentially** on a single core.
- **80 seconds** consists of analyzing data rows, which can be **perfectly parallelized** across any number of cores.

If this task is executed on a computer with **4 parallel cores**, what will be the new total execution time, and what is the resulting speedup?

(A) Total Time = **20 seconds** | Speedup = **5.0**

(B) Total Time = **25 seconds** | Speedup = **4.0**

(C) Total Time = **80 seconds** | Speedup = **1.25**

(D) Total Time = **40 seconds** | Speedup = **2.5**

Question 66

★★★★ HIGH PRIORITY

PARALLEL OVERHEAD AND COMMUNICATION COSTS

A programmer attempts to speed up a small sorting task that takes 0.05 seconds sequentially by distributing it across 16 parallel threads. Surprisingly, the parallel version takes **0.18 seconds**—more than three times slower than the sequential version!

What is the most likely cause for this performance regression?

(A) The sorting algorithm erased all 16 cores permanently from the motherboard.

(B) The computer's central processor ran out of electrical current and temporarily paused.

(C) The **parallel overhead** (time spent creating threads, allocating memory, and synchronizing communication between cores) exceeded the tiny computational time of the original task.

(D) Parallel computing is only capable of performing addition, not comparisons.

Question 67

★★★★ HIGH PRIORITY

DATA DEPENDENCIES AND CONCURRENCY HAZARDS

Consider the following algorithmic calculation:

`Step 1: Set X = 5`
`Step 2: Set Y = X * 2`
`Step 3: Set Z = Y + 10`

Why can these three steps **NOT** be executed simultaneously in parallel across three separate processor cores?

(A) Processors can only perform arithmetic operations if all variable names begin with the letter A.

(B) There is a strict **data dependency**; Step 2 cannot calculate `Y` until Step 1 computes `X`, and Step 3 cannot calculate `Z` until Step 2 computes `Y`.

(C) Parallel cores are forbidden by Internet protocols from reading variable values.

(D) Modern operating systems do not allow variables to be multiplied by the number 2 in parallel.

Question 68

★★★★ HIGH PRIORITY

EMBARRASSINGLY PARALLEL PROBLEMS VS. INHERENTLY SEQUENTIAL PROBLEMS

In computational computer science, some problems can be split into thousands of completely independent pieces with almost zero communication required between workers (known as **embarrassingly parallel** problems).

Which of the following problems is the best candidate for massive parallelization?

(A) Calculating a running account balance where each bank transaction updates the balance of the next.

(B) Rendering 10,000 individual, independent pixel colors in a 3D animated movie frame.

(C) Computing the 100th Fibonacci number where each term $F_n = F_{n-1} + F_{n-2}$ requires the immediate two preceding numbers.

(D) Reading a physical paperback book by turning pages strictly from front to back.

Question 69

★★★★ HIGH PRIORITY

PARALLEL EFFICIENCY AND PROCESSOR UTILIZATION

A research team runs a parallel simulation on an 8-processor workstation. The sequential version took **80 seconds**, and the parallel version takes **20 seconds**.

What is the **Parallel Efficiency** of this system, and how is it interpreted?
(Formula: $\text{Efficiency} = \frac{\text{Speedup}}{N_{\text{processors}}}$, where $\text{Speedup} = \frac{T_{\text{seq}}}{T_{\text{par}}}$)

(A) Speedup = 4.0 | Efficiency = **0.50 (50%)**, meaning each processor is effectively utilized at 50% of its theoretical maximum potential.

(B) Speedup = 60.0 | Efficiency = 7.5 (750%), meaning 8 processors run 60 times faster.

(C) Speedup = 1.0 | Efficiency = 1.0 (100%), meaning zero speedup occurred.

(D) Speedup = 0.25 | Efficiency = 2.0 (200%), meaning processors operate beyond physical laws.

Question 70

★★★★★ VERY IMPORTANT

EVALUATING ALGORITHMIC SPEEDUP POTENTIAL ACROSS CANDIDATE TASKS

A software architect evaluates four candidate algorithms to run on an **8-core processor**. Each algorithm takes a total of 100 seconds sequentially, with differing proportions of sequential and parallelizable code:

Algorithm	Mandatory Sequential Time	Parallelizable Time
Alg W	50 seconds	50 seconds
Alg X	30 seconds	70 seconds
Alg Y	10 seconds	90 seconds
Alg Z	2 seconds	98 seconds

Which algorithm will achieve the **fastest runtime** and **highest speedup** on the 8-core machine?

(A) **Alg X**, because 70 seconds is an even multiple of 8 cores.

(B) **Alg W**, because having equal sequential and parallel times balances processor heat evenly.

(C) All four algorithms will finish in exactly the same runtime because all processors run at the same clock frequency.

(D) **Alg Z**, because it has the smallest sequential bottleneck (only 2 seconds), allowing 98% of its workload to be divided across the 8 cores.

Question 71

★★★★★ VERY IMPORTANT

DEFINITION OF DISTRIBUTED COMPUTING

Astronomers analyzing signals from deep space receive petabytes of telescope data every week—far too much data for a single university supercomputer to process alone.

Which of the following describes how a **Distributed Computing** system solves this massive computational challenge?

(A) A model where a single computer is connected to ten computer mice so ten students can type simultaneously.

(B) A program that forces a computer to run backwards in time to reverse calculations.

(C) A system that replaces all digital files with microfiche film reels stored in a vault.

(D) A model in which multiple autonomous computers located in different physical areas communicate across a network to share the workload and collaborate on solving a massive computational problem.

Question 72

★★★★★ VERY IMPORTANT

KEY ARCHITECTURAL DIFFERENCES: PARALLEL VS. DISTRIBUTED COMPUTING

While both parallel computing and distributed computing involve simultaneous calculations, they operate under fundamentally different hardware constraints.

Which of the following correctly contrasts **Parallel Computing on a single multi-core computer** with **Distributed Computing across a network**?

(A) Parallel computing can only add numbers, while distributed computing can only subtract numbers.

(B) Parallel computing typically uses multiple cores sharing common high-speed local memory (RAM) with microscopic nanosecond communication latency, whereas distributed computing connects distinct computers over a network where communication latency is substantially higher (milliseconds).

(C) Distributed computing requires all computers to be physically glued together inside the same metal case.

(D) Parallel computing requires an Internet connection, whereas distributed computing cannot use any network.

Question 73 ★★★★★ VERY IMPORTANT

VOLUNTEER COMPUTING AND CITIZEN SCIENCE NETWORKS

Projects like **Folding@home** and **SETI@home** allow everyday citizens to download a background software client that runs simulations of human protein structures or searches for extraterrestrial radio signals whenever their personal laptops are idle.

What is this collaborative model called, and what is its primary advantage?

(A) **Sequential Paging**; it converts volunteer computers into analog fax machines.

(B) **Volunteer Computing (or Citizen Science)**; it enables scientific researchers to harness the aggregate processing power of millions of consumer devices at a fraction of the cost of building a dedicated supercomputer.

(C) **Phishing**; it tricks users into giving their bank passwords to universities.

(D) **Bandwidth Throttling**; it slows down the Internet across an entire continent.

Question 74 ★★★★ HIGH PRIORITY

CLOUD COMPUTING AND ELASTIC INFRASTRUCTURE

An online retail startup experiences a 50x surge in website traffic during Black Friday sales. Instead of purchasing thousands of physical servers that would sit idle for the rest of the year, the company uses a **Cloud Computing** provider.

What feature of cloud computing makes this strategy effective and economically advantageous?

(A) **Elasticity (On-Demand Scalability)**, which allows computing resources (virtual servers, storage, bandwidth) to automatically scale up dynamically during peak traffic and scale down when traffic subsides, paying only for resources used.

(B) Cloud providers guarantee that no user will ever return an unwanted purchased product.

(C) Cloud servers do not consume any electrical energy, making them completely free to operate.

(D) Cloud computing eliminates the need for websites to use HTML or web browsers.

Question 75 ★★★★★ VERY IMPORTANT

MALICIOUS DISTRIBUTED COMPUTING: BOTNETS AND DDOS ATTACKS

A cybercriminal infects 250,000 smart home webcams and routers with malware, creating an army of remotely controlled 'zombie' devices called a **Botnet**. On command, all 250,000 devices simultaneously flood a target hospital server with billions of junk HTTP requests, overwhelming the server's bandwidth and crashing its website.

Which type of cyberattack is described here?

(A) A physical Ethernet wiretap.

(B) A symmetric key cryptographic handshake.

(C) A **Distributed Denial of Service (DDoS)** attack.

(D) A brute-force dictionary attack on an offline password hash.

Question 76

★★★★ HIGH PRIORITY

SCALABILITY STRATEGIES: HORIZONTAL VS. VERTICAL SCALING

A social media platform's database server is slowing down as millions of new members register. The engineering team debates two scaling architectures:

- **Strategy 1:** Replace the single central server with a larger, more expensive enterprise machine featuring 128 CPU cores and 2 Terabytes of RAM.
- **Strategy 2:** Distribute the database across 50 standard, inexpensive server computers connected in a distributed cluster.

Which of the following correctly categorizes these two strategies?

(A) Strategy 1 is **Horizontal Scaling**, while Strategy 2 is **Vertical Scaling**.

(B) Both strategies are examples of sequential computing on a single core.

(C) Neither strategy can handle more users because network cables have a fixed maximum number of users.

(D) Strategy 1 is **Vertical Scaling (Scaling Up)**, while Strategy 2 is **Horizontal Scaling (Scaling Out)**.

Question 77

★★★★ HIGH PRIORITY

FAULT RECOVERY AND WORKER REDUNDANCY IN DISTRIBUTED SYSTEMS

In a distributed computing framework (such as MapReduce or Apache Spark), a master coordinator node divides a massive data sorting job into 1,000 sub-tasks and assigns them to 1,000 worker computers. Ten minutes into the job, worker node #482 experiences a power failure and stops responding.

How does a robust distributed computing system maintain fault tolerance?

(A) The coordinator permanently deletes the entire database and halts all 999 other workers.

(B) The system accepts corrupted partial results and outputs incorrect numbers.

(C) The coordinator sends high-voltage electricity through the network cable to revive worker #482's battery.

(D) The coordinator detects that worker #482 stopped sending heartbeat signals, marks its sub-task as incomplete, and reassigns that exact sub-task to an available healthy worker node without restarting the entire 1,000-task job.

Question 78

★★★★ HIGH PRIORITY

DATA LOCALITY AND NETWORK BANDWIDTH CONSTRAINTS

A distributed system analyzes a 100-Terabyte dataset stored across 100 cluster hard drives. An engineer notes that moving 100 Terabytes of raw data across local network switches to a central compute node would take several hours and saturate the network.

Which software architectural principle solves this network congestion problem?

(A) **Data Locality (Moving Computation to Data)**: Sending the lightweight analysis code to execute directly on the nodes where the data is already stored, transmitting only the tiny summary results back across the network.

(B) Converting the data into analog radio waves and transmitting them over FM commercial radio stations.

(C) Printing all 100 Terabytes on paper and having office staff manually type the numbers into a calculator.

(D) Compressing the 100 Terabytes into a single byte using lossy audio compression.

Question 79

★★★★ HIGH PRIORITY

SOCIETAL AND SCIENTIFIC APPLICATIONS OF DISTRIBUTED COMPUTING

Biomedical researchers seeking to discover new medications need to test how billions of potential chemical drug molecules bind to viral proteins. Testing each molecule in a physical wet chemistry laboratory would take centuries.

How does distributed computing accelerate this scientific discovery process?

(A) It allows molecular docking simulations for millions of candidate chemicals to be calculated simultaneously across thousands of networked computers, shrinking decades of physical lab screening into weeks.

(B) It eliminates the need for doctors or medicine in modern society.

(C) It chemically manufactures physical pills inside computer monitors.

(D) It automatically cures patients through Wi-Fi radio transmissions.

Question 80

★★★★★ VERY IMPORTANT

MATCHING COMPUTATIONAL TASKS WITH THE OPTIMAL COMPUTING PARADIGM

A technology institute evaluates four computational tasks and needs to select the most suitable computing model for each:

Task	Description
Task 1	Calculating the running total balance of a bank ledger where each line depends directly on the previous line
Task 2	Rendering a complex 3D animated film where each pixel is calculated independently on a high-end graphics workstation
Task 3	Analyzing 50 Petabytes of global climate telemetry using a cluster of 5,000 networked servers
Task 4	An automated botnet where thousands of compromised IoT devices flood a target server

Which of the following correctly pairs these tasks with their computing model?

(A) Task 1: Quantum Computing | Task 2: Analog Computing | Task 3: Optical Computing | Task 4: Mechanical Computing

(B) Task 1: Distributed Computing | Task 2: Sequential Computing | Task 3: Parallel Computing | Task 4: Single-core Computing

(C) Task 1: Sequential Computing | Task 2: Parallel Computing | Task 3: Distributed Computing | Task 4: Malicious Distributed Computing (DDoS)

(D) Task 1: Parallel Computing | Task 2: Distributed Computing | Task 3: Sequential Computing | Task 4: Volunteer Computing

Question 81

★★★★★ VERY IMPORTANT

SYMMETRIC ENCRYPTION AND THE KEY DISTRIBUTION PROBLEM

In classical cryptography, two communicating parties use **Symmetric Key Encryption** to protect their data.

Which of the following describes how symmetric key encryption operates, and what is its primary operational challenge across the Internet?

(A) Symmetric encryption requires computers to be connected by a physical gold wire.

(B) The sender and receiver use the exact same secret key to both encrypt and decrypt data; the primary challenge is securely transmitting that secret key to the receiver without an eavesdropper intercepting it.

(C) Symmetric encryption can only encrypt text if all letters are uppercase.

(D) The sender uses a public key that everyone can see, and the receiver uses a completely unrelated random number that changes every second.

Question 82

★★★★★ VERY IMPORTANT

ARCHITECTURE OF ASYMMETRIC (PUBLIC KEY) CRYPTOGRAPHY

To solve the key exchange dilemma on an open network, computer scientists developed **Asymmetric Key Encryption** (such as RSA and Elliptic Curve Cryptography).

Which of the following correctly describes the key architecture used in asymmetric cryptography?

(A) Asymmetric encryption requires the receiver to physically mail their hard drive to the sender.

(B) Each user generates a mathematically linked key pair: a **Public Key** that can be freely shared with the world, and a **Private Key** that is kept strictly secret and never shared.

(C) Every packet sent across the Internet must use the same password: 'admin'.

(D) Each user is assigned a single universal key published in a global newspaper.

Question 83

★★★★★ VERY IMPORTANT

ASYMMETRIC ENCRYPTION WORKFLOW: ALICE SENDING TO BOB

Alice wishes to transmit a confidential financial document to Bob across the public Internet using asymmetric cryptography.

Which key must Alice use to **encrypt** the document, and which key must Bob use to **decrypt** it?

(A) Alice encrypts the document using **her own Public Key**; Bob decrypts it using **his own Public Key**.

(B) Alice encrypts the document using **Bob's Private Key**; Bob decrypts it using **Alice's Private Key**.

(C) Alice encrypts the document using **Alice's Private Key**; Bob decrypts it using **Alice's Public Key**.

(D) Alice encrypts the document using **Bob's Public Key**; Bob decrypts the document using **Bob's Private Key**.

Question 84

★★★★★ VERY IMPORTANT

DIGITAL SIGNATURES AND MESSAGE INTEGRITY

A software vendor releases a critical operating system security update to millions of users. To ensure that attackers cannot tamper with the update or distribute malware disguised as the vendor, the vendor attaches a **Digital Signature**.

How does the vendor generate the digital signature, and how do client computers verify it?

(A) The vendor encrypts a hash of the software using the **vendor's Private Key**; client computers decrypt and verify the signature using the **vendor's Public Key**.

(B) The vendor physically signs each computer monitor with an ink marker at the factory.

(C) The vendor sends their private key in an email to all users so they can install the software.

(D) The vendor converts the software into an uncompressed MP3 audio file.

Question 85

★★★★★ VERY IMPORTANT

MAN-IN-THE-MIDDLE (MITM) ATTACKS AND PKI TRUST VERIFICATION

Suppose asymmetric encryption exists, but there are no Certificate Authorities or Public Key Infrastructure (PKI). An attacker secretly intercepts all traffic between a student and their bank.

When the student asks for the bank's public key, the attacker intercepts the request and sends the attacker's own public key instead. The student encrypts their password with the attacker's key, allowing the attacker to decrypt it, steal the funds, and re-encrypt it to the real bank.

Which type of attack occurred, and how do modern web browsers prevent it?

(A) A denial-of-service attack; prevented by unplugging the router from the wall.

(B) A physical hardware theft; prevented by locking the laptop in a metal safe.

(C) A **Man-in-the-Middle (MitM)** attack; prevented because browsers verify the bank's digital certificate signed by a trusted **Certificate Authority (CA)** to ensure the public key belongs to the real bank.

(D) A bit-flip error; prevented by increasing the monitor's brightness level.

Question 86

★★★★ HIGH PRIORITY

ROGUE WI-FI ACCESS POINTS AND EVIL TWIN ATTACKS

While waiting at an airport, a passenger opens their laptop's Wi-Fi menu and sees two networks:
- `Airport_Official_WiFi` (Requires login)
- `Free_Airport_HighSpeed_WiFi` (Open, no password required)

The passenger connects to the second network. Unknown to the passenger, this network is a **Rogue Access Point (Evil Twin)** set up by an attacker sitting nearby with a portable hotspot.

What immediate risk does the passenger face while connected to this rogue network?

(A) The laptop's battery will immediately explode due to excess radio frequency absorption.

(B) The attacker can monitor and capture all unencrypted network traffic (HTTP requests, DNS lookups, and cleartext credentials) transmitted by the laptop.

(C) The laptop will permanently lose the ability to connect to any other Wi-Fi network in the future.

(D) All files stored on the passenger's local SSD will be physically printed out on airport printers.

Question 87

★★★★★ VERY IMPORTANT

ROLE AND OPERATION OF VIRTUAL PRIVATE NETWORKS (VPNS)

To protect their privacy when forced to work from public coffee shops and hotel networks, an employee uses a **Virtual Private Network (VPN)**.

How does a VPN safeguard the employee's data when connected to an untrusted public network?

(A) It makes the laptop completely invisible to human eyes in the coffee shop.

(B) It establishes an encrypted tunnel between the employee's device and the VPN server, preventing anyone on the local Wi-Fi network or the local ISP from inspecting the contents or destinations of their web traffic.

(C) It physically replaces the coffee shop's router with an armored military switch.

(D) It permanently disables the employee's computer keyboard to prevent typing mistakes.

Question 88

★★★★ HIGH PRIORITY

MULTI-FACTOR AUTHENTICATION (MFA) PRINCIPLES

A university requires students to log into their academic portals using **Multi-Factor Authentication (MFA)**. When logging in, students must enter their password and then approve a notification prompt on their registered smartphone.

Which of the following correctly identifies the two authentication factors being utilized in this system?

(A) **Something you know** (the password) and **Something you have** (the registered smartphone).

(B) **Something you know** (the password) and **Something you know** (a security question answer).

(C) **Something you eat** (lunch) and **Something you wear** (a coat).

(D) **Something you are** (biometrics) and **Somewhere you live** (home address).

Question 89

★★★★ HIGH PRIORITY

HYBRID ENCRYPTION ARCHITECTURE IN TLS/HTTPS

While asymmetric encryption solves the key exchange problem, it is mathematically complex and approximately 1,000 times slower to execute than symmetric encryption.

How does modern HTTPS/TLS combine both forms of encryption to achieve both security and high performance?

(A) It requires the user to solve a complex math puzzle by hand before loading each web page.

(B) It encrypts the text of a website with symmetric encryption and discards all images unencrypted.

(C) It uses asymmetric encryption on odd-numbered days of the month and symmetric encryption on even-numbered days.

(D) It uses **asymmetric encryption** during the initial handshake to securely exchange a temporary secret session key, and then switches to fast **symmetric encryption** using that session key to encrypt all subsequent bulk web traffic.

Question 90

★★★★★ VERY IMPORTANT

COMPREHENSIVE CYBERSECURITY TECHNIQUES AND GUARANTEES SYNTHESIS

An AP Computer Science Principles class reviews four core cybersecurity technologies:

Technology	Primary Security Guarantee
Tech 1: Asymmetric Encryption	Securely exchanges confidential messages across untrusted networks without prior shared secrets
Tech 2: Digital Certificate	Cryptographically verifies that a public key belongs to an authentic, validated organization
Tech 3: Virtual Private Network (VPN)	Encrypts all outgoing device traffic to protect communications on untrusted local Wi-Fi networks
Tech 4: Multi-Factor Authentication (MFA)	Prevents unauthorized account logins even if an attacker discovers the user's password

Which of the following statements regarding these four technologies is accurate?

(A) Only Tech 1 is legal to use in the United States; the other three are banned by Internet engineering standards.

(B) If a user enables Tech 3 (VPN), they can safely share their bank passwords with strangers without risk.

(C) All four technologies work synergistically across different layers to provide comprehensive confidentiality, integrity, authentication, and access control across modern computer systems.

(D) Tech 2 (Digital Certificate) guarantees that a computer hard drive will never experience physical mechanical wear.

Question 91 · Written Response Scenario

★★★★★ 9 POINTS

NETWORK PACKET INSPECTION & FIREWALL FILTER

A student develops a network security simulation program called `NetGuard` that acts as an intelligent packet filtering firewall for a school district. The program intercepts incoming raw data packets, inspects packet header metadata (source IP, destination port, packet size, and protocol type), and automatically blocks malicious traffic originating from known blacklisted IP subnets or unauthorized ports while permitting legitimate educational traffic to pass through.

Written Response Prompts (Create Performance Task Style):

• **Part A (Purpose & Function):** Describe the overall purpose of the program and explain how the user or network administrator interacts with the program to monitor network traffic.

• **Part B (Data Abstraction):** Identify the primary list or collection used in your program. Explain what data is stored in this collection and explain how using this list manages program complexity (including what would happen if the list were not used).

• **Part C (Managing Complexity & Algorithm):** Provide and explain a student-developed procedure named 'filterPacketStream(packetList, blacklistedPrefixes)'. The procedure must include parameters, sequencing, selection (conditional logic), and iteration (looping), returning or storing a filtered list of permitted packets.

• **Part D (Testing & Edge Cases):** Describe two distinct test calls to your procedure with specific arguments that execute different code paths. Explain what condition is being tested in each call and the resulting outcome.

Question 92 · Written Response Scenario

★★★★★ 9 POINTS

DYNAMIC ROUTING & FAULT-TOLERANT PATH OPTIMIZER

A student develops a routing protocol simulator called `RouteMesh` designed to model autonomous packet forwarding across a dynamic mesh network. The program monitors network link latencies, updates routing tables when connections fail, and guarantees that packets find alternative redundant paths between source and destination nodes.

Written Response Prompts (Create Performance Task Style):
• **Part A (Purpose & Function):** Describe the overall purpose of the program and explain how the system functions when a physical link failure occurs.
• **Part B (Data Abstraction):** Identify the collection used in your program. Explain the data it holds and how it manages complexity compared to separate variables.
• **Part C (Managing Complexity & Algorithm):** Present a procedure named 'findOptimalPath(networkGraph, startNode, endNode, failedLinks)'. Explain how it uses sequencing, selection, and iteration to discover the lowest-latency active path.
• **Part D (Testing & Edge Cases):** Detail two test calls demonstrating successful rerouting around a severed connection versus handling a completely partitioned network.

Question 93 · Written Response Scenario

★★★★★ 9 POINTS

DNS CACHE RESOLVER & TTL INVALIDATION MANAGER

A student creates a local DNS resolver simulator called `SmartResolver` that caches domain name queries to minimize upstream root and TLD query traffic. The program checks local cache records, validates Time-To-Live (TTL) expiration timestamps, evicts stale records, and recursively queries upstream authoritative nameservers on cache misses.

Written Response Prompts (Create Performance Task Style):
- **Part A (Purpose & Function):** State the program purpose and how it accelerates web page loading for client devices.
- **Part B (Data Abstraction):** Identify the list 'dnsCacheTable' and explain how it stores and organizes records.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'resolveDomain(queryDomain, currentTime, dnsCacheTable)' explaining its selection, iteration, and cache eviction mechanics.
- **Part D (Testing & Edge Cases):** Test a valid cached hit versus an expired TTL triggering an authoritative recursive query.

Question 94 · Written Response Scenario

★★★★★ 9 POINTS

PARALLEL BATCH PROCESSING & SPEEDUP SIMULATOR

A student builds a performance benchmarking simulation called `ParallelSim` that models the execution of scientific image processing tasks across varying processor core counts (1, 2, 4, 8 cores). The program quantifies Amdahl's Law by tracking mandatory sequential setup time versus parallelized task queues and calculating total speedup and parallel efficiency.

Written Response Prompts (Create Performance Task Style):

- **Part A (Purpose & Function):** Explain the educational and engineering purpose of the simulation.
- **Part B (Data Abstraction):** Identify the list 'taskDurationList' and explain how it stores task runtimes.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'calculateParallelMetrics(taskDurationList, seqFraction, coreCount)' calculating runtime, speedup, and efficiency.
- **Part D (Testing & Edge Cases):** Test execution on 1 core (baseline) versus 8 cores, detailing the calculated speedup.

Question 95 · Written Response Scenario

★★★★★ 9 POINTS

DISTRIBUTED SENSOR DATA AGGREGATION & VOLUNTEER NETWORK

A student designs a citizen-science distributed application called `EcoGrid` that aggregates atmospheric air quality data from thousands of volunteer environmental sensors across a state. The program manages worker node status, dispatches calibration tasks, detects offline nodes, and aggregates regional pollution indices.

Written Response Prompts (Create Performance Task Style):

- **Part A (Purpose & Function):** State the environmental and computing purpose of 'EcoGrid'.
- **Part B (Data Abstraction):** Identify the collection 'sensorNodesList' and explain the telemetry data it stores.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'aggregateAirQuality(sensorNodesList, maxAgeSeconds, currentTimestamp)' handling dead node filtering and average calculation.
- **Part D (Testing & Edge Cases):** Test normal active sensor aggregation versus filtering out stale or disconnected sensors.

Question 96 · Written Response Scenario

★★★★★ 9 POINTS

SYMMETRIC VS ASYMMETRIC HYBRID KEY EXCHANGE MODULE

A student develops a secure communication module called `CryptoVault` implementing a hybrid cryptographic handshake. The program uses asymmetric RSA public-key encryption to securely authenticate and exchange a random symmetric AES session key, then transitions to fast symmetric encryption for transmitting message payloads.

Written Response Prompts (Create Performance Task Style):
- **Part A (Purpose & Function):** State the cybersecurity purpose of the hybrid key exchange.
- **Part B (Data Abstraction):** Identify the list 'trustedCertificatesList' and explain how it stores validated public keys.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'negotiateSecureSession(clientHello, serverCert, trustedCertificatesList)' verifying certificates and generating encrypted session keys.
- **Part D (Testing & Edge Cases):** Test a genuine certificate handshake versus a spoofed/untrusted certificate rejection.

Question 97 · Written Response Scenario

★★★★★ 9 POINTS

NETWORK LATENCY, BANDWIDTH & BOTTLENECK ANALYZER

A student writes a diagnostic network simulation called `NetPathSim` that computes end-to-end data transfer times across multi-hop communication routes. The program accounts for propagation delays (latency) and link bandwidths in series, identifies the narrowest bottleneck link, and calculates total transfer time for varying file sizes.

Written Response Prompts (Create Performance Task Style):

- **Part A (Purpose & Function):** State the networking performance purpose of 'NetPathSim'.
- **Part B (Data Abstraction):** Identify the list 'hopList' and explain how it stores link characteristics.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'calculateTransferTime(hopList, fileSizeBytes)' calculating bottleneck bandwidth, cumulative latency, and total transfer seconds.
- **Part D (Testing & Edge Cases):** Test a small packet where latency dominates versus a massive file where bottleneck bandwidth dominates.

Question 98 · Written Response Scenario

★★★★★ 9 POINTS

DDOS TRAFFIC DETECTION & RATE LIMITER

A student implements an anti-DDoS traffic throttling engine called `DefendDDoS` designed to protect a cloud server from botnet attacks. The program tracks the frequency of incoming HTTP requests from client IP addresses within a rolling time window, logs abnormal request spikes, and throttles or blocks offending IPs while ensuring legitimate users maintain uninterrupted access.

Written Response Prompts (Create Performance Task Style):

- **Part A (Purpose & Function):** State the cybersecurity purpose of 'DefendDDoS'.
- **Part B (Data Abstraction):** Identify the collection 'ipRequestLog' and explain the data it tracks.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'evaluateRateLimit(incomingIP, ipRequestLog, maxRequestsPerMinute, currentTime)' handling frequency checking and firewall bans.
- **Part D (Testing & Edge Cases):** Test a normal low-frequency user versus a high-frequency botnet flood trigger.

Question 99 · Written Response Scenario

★★★★★ 9 POINTS

TCP PACKET SEQUENCING & OUT-OF-ORDER REASSEMBLY BUFFER

A student writes a core transport-layer simulation called `TCPReassemble` that models how the Transmission Control Protocol (TCP) reconstructs fragmented, out-of-order data streams over the Internet. The program stores arriving packets in a reassembly buffer, sorts them by sequence number, detects missing packets to request retransmissions, and outputs the clean data stream once complete.

Written Response Prompts (Create Performance Task Style):
• **Part A (Purpose & Function):** State the transport-layer purpose of 'TCPReassemble'.
• **Part B (Data Abstraction):** Identify the collection 'packetBuffer' and explain the packet metadata it stores.
• **Part C (Managing Complexity & Algorithm):** Present procedure 'assembleDataStream(packetBuffer, expectedTotalPackets)' detecting missing sequence gaps and reconstructing message text.
• **Part D (Testing & Edge Cases):** Test an out-of-order packet array arriving successfully versus identifying a missing packet requiring retransmission.

Question 100 · Written Response Scenario

★★★★★ 9 POINTS

MULTI-FACTOR AUTHENTICATION (MFA) & SECURE TOKEN VALIDATOR

A student develops an identity security framework called `AuthShield` that enforces Multi-Factor Authentication (MFA) for an online banking portal. The program requires users to supply both knowledge-based credentials (hashed password) and possession-based credentials (a time-sensitive 6-digit cryptographic TOTP token). The program validates credentials against security policies, checks token freshness, and locks accounts upon repeated brute-force failures.

Written Response Prompts (Create Performance Task Style):

- **Part A (Purpose & Function):** State the cybersecurity purpose of the multi-factor authentication program.
- **Part B (Data Abstraction):** Identify the collection 'userAuthDatabase' and explain how it stores user authentication records.
- **Part C (Managing Complexity & Algorithm):** Present procedure 'validateMFA(username, enteredPasswordHash, enteredToken, currentTime, userAuthDatabase)' evaluating two distinct authentication factors and account lockouts.
- **Part D (Testing & Edge Cases):** Test a successful two-factor login versus an invalid time-expired token rejection.

SECTION II ◆ EXAMINATION KEY & SOLUTIONS

Answers, Worked Solutions & Scoring Rubrics

The following pages contain the complete 90-question Multiple Choice Answer Key Matrix, comprehensive step-by-step worked solutions with distractor failure analysis, and official 9-point College Board scoring rubrics for all 10 Create Performance Task Written Response questions.

Section I Answer Key Matrix

Quick-reference answer matrix for Multiple Choice Questions 1 through 90. All questions are weighted equally.

Q1 – Q23		Q24 – Q46		Q47 – Q68		Q69 – Q90	
Q01	A	Q24	A	Q47	A	Q69	A
Q02	C	Q25	B	Q48	B	Q70	D
Q03	D	Q26	C	Q49	B	Q71	D
Q04	D	Q27	C	Q50	C	Q72	B
Q05	A	Q28	D	Q51	C	Q73	B
Q06	C	Q29	A	Q52	D	Q74	A
Q07	C	Q30	B	Q53	D	Q75	C
Q08	B	Q31	C	Q54	B	Q76	D
Q09	D	Q32	C	Q55	C	Q77	D
Q10	D	Q33	A	Q56	C	Q78	A
Q11	C	Q34	D	Q57	A	Q79	A
Q12	C	Q35	B	Q58	D	Q80	C
Q13	A	Q36	C	Q59	D	Q81	B
Q14	A	Q37	A	Q60	C	Q82	B
Q15	D	Q38	A	Q61	C	Q83	D
Q16	D	Q39	B	Q62	B	Q84	A
Q17	C	Q40	D	Q63	B	Q85	C
Q18	A	Q41	C	Q64	D	Q86	B
Q19	A	Q42	A	Q65	D	Q87	B
Q20	B	Q43	A	Q66	C	Q88	A
Q21	C	Q44	B	Q67	B	Q89	D
Q22	D	Q45	B	Q68	B	Q90	C
Q23	A	Q46	D				

Question 01 · Fundamental Architecture of the Internet

Priority ★★★★★

CORRECT ANSWER: (A) A globally distributed network of interconnected computing devices that communicate using open, standardized protocols.

- Analyze the formal definition of the Internet in the AP CSP curriculum:
- • The Internet is an interconnected network of computing networks (an 'inter-network') consisting of billions of independent devices.
- • It operates using open, non-proprietary protocols (such as IP, TCP, and UDP) that are published publicly and free to use.
- • It is fundamentally decentralized and distributed; no single central computer, company, or government controls or routes all internet traffic.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Confuses the World Wide Web (a software application system of linked documents) with the underlying Internet (the physical and logical hardware network).
✕ **(C):** The Internet has no centralized hub or central supercomputer; it is a distributed network of interconnected routers.
✕ **(D):** Internet protocols are open and standardized (managed by the IETF), not proprietary or confidential.

Question 02 · Open Standards & Protocol Governance (IETF)

Priority ★★★★★

CORRECT ANSWER: (C) Open standards allow devices manufactured by different vendors and running different operating systems to communicate seamlessly with one another without proprietary licensing restrictions.

- Examine the role of open protocols in system interoperability:
- • An open protocol is a communication standard that is publicly documented, freely accessible, and not owned by any single commercial entity.
- • Because protocols like IP and TCP are open, an Apple smartphone, a Windows laptop, an Android tablet, and a Linux server can all exchange packets flawlessly.
- • Any developer or manufacturer can build compatible hardware or software without paying licensing royalties to a third party.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Open protocols democratize access for the general public, not restrict access.
✕ **(B):** Core protocols like early IP and HTTP transmit plaintext; encryption requires distinct security layers like TLS/HTTPS.
✕ **(D):** Open standards mean the specifications are publicly visible to everyone; security relies on cryptographic math, not secrecy of the protocol.

Question 03 · Centralized vs. Decentralized vs. Distributed Networks

Priority ★★★★

CORRECT ANSWER: (D) In Network X, if the central hub fails, all communication across the entire network ceases (Single Point of Failure), whereas Network Z can route around failed nodes.

- Compare architectural resilience across network models:
- • In a **Centralized network** (Network X), the central hub is a **Single Point of Failure (SPOF)**. If that one server or router goes offline, no devices can communicate.
- • In a **Distributed network** (Network Z), there is no single central controller. Nodes are interconnected with multiple redundant physical links.
- • If one node or cable in Network Z is destroyed, dynamic routing protocols immediately re-route packets through alternate available neighbors.
- • This resilience was the primary design motivation behind ARPANET (the precursor to the modern Internet).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Network Z still experiences transmission latency and routing delays as packets traverse hops.
✕ **(B):** Bandwidth does not increase when nodes fail; centralized networks suffer complete outage upon hub failure.
✕ **(C):** Network topology does not dictate whether protocols are open or proprietary.

Question 04 · Packet Switching vs. Circuit Switching

Priority ★★★★★

CORRECT ANSWER: (D) Packet switching allows multiple devices to dynamically share the same physical network cables simultaneously without reserving dedicated, idle communication channels.

- Analyze the fundamental mechanics of packet switching:
- • In **circuit switching**, a dedicated physical circuit is reserved exclusively for a call. Even when both parties are silent, the wire cannot be used by anyone else.
- • In **packet switching**, data is broken into small chunks called **packets**. Each packet travels independently across shared routers and wires.
- • If one channel is quiet, other devices' packets use that bandwidth, dramatically increasing the efficiency, capacity, and scalability of shared physical infrastructure.
- • Packets from hundreds of different users interleave across the same transatlantic fiber cable without interference.

WHY OTHER CHOICES ARE INCORRECT:

- ✕ **(A):** Packet switching manages congestion dynamically through buffering and packet dropping, not by forcibly disconnecting users.
- ✕ **(B):** Packets can take different paths and arrive out of order; TCP uses sequence numbers in the header to reassemble them.
- ✕ **(C):** Reserving an unbroken physical wire for the entire duration is the definition of circuit switching, not packet switching.

Question 05 · Function & Role of Network Routers

Priority ★★★★★

CORRECT ANSWER: (A) To inspect the destination IP address of each incoming packet and forward it along an optimal path toward its final destination.

- Analyze router functionality in computer networks:
- • A router is a specialized computing device that connects two or more computer networks.
- • When a packet arrives, the router reads the metadata in the packet header—specifically the **destination IP address**.
- • The router consults its internal routing table to determine the best next network 'hop' and sends the packet on its way.
- • Routers act like postal sorting centers, forwarding packages toward their destination without needing to understand or alter the contents of the payload.

WHY OTHER CHOICES ARE INCORRECT:

- ✕ **(B):** Copyright law and digital rights management are legal and software issues unrelated to router hardware.
- ✕ **(C):** Rendering HTML into visual elements is the job of a client web browser, not a network router.
- ✕ **(D):** Power delivery is handled by uninterruptible power supplies (UPS), not routers.

Question 06 · The End-to-End Architectural Principle

Priority ★★★★

CORRECT ANSWER: (C) Complex application logic and error-checking are handled by the sender and receiver devices (the end hosts), keeping the core network routers simple, fast, and content-agnostic.

- Examine the End-to-End design principle:
- • In the Internet architecture, the 'core' of the network (routers and switches) is designed to do only one thing: move packets from point A to point B as quickly as possible.
- • Complex logic—such as verifying data integrity, handling lost packets, and interpreting applications (e.g., video streaming, email, web browsing)—is implemented entirely at the 'edges' (the end hosts: client and server).
- • This principle ensures that anyone can invent a brand new internet application (like Zoom or Spotify) without having to modify or update the millions of routers across the globe.

WHY OTHER CHOICES ARE INCORRECT:

- ✕ **(A):** Direct physical connections between all devices would be an impossibly massive complete graph ($O(n^2)$ cables); the internet uses intermediate routing nodes.
- ✕ **(B):** The internet is open and permissionless; ISPs do not approve application software.
- ✕ **(D):** Encryption is handled by transport/application layers (TLS), not a mandate of the core network architecture.

Question 07 · Computing Device vs. System vs. Network

Priority ★★★★

CORRECT ANSWER: (C) A smart thermostat is a **computing device; the thermostat connected to a furnace controller is a **computer system**; five smart homes connected to a utility grid server form a **computer network**.**

- Review the hierarchical definitions in AP CSP:
- • **Computing Device:** A physical artifact that can run a program (e.g., computer, smartphone, smart sensor, thermostat).
- • **Computer System:** A group of computing hardware devices and programs working together for a common purpose (e.g., a thermostat interacting with sensor inputs and furnace actuators).
- • **Computer Network:** A group of interconnected computing devices capable of sending or receiving data (e.g., multiple homes communicating with a central server).
- • Option A accurately demonstrates this hierarchy from device to system to network.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** A laptop is a single computing device, while the Internet is a global computer network.
✕ **(B):** A supercomputer is a computing device (or cluster system), not a network.
✕ **(D):** A CPU is a hardware component within a device, and an ethernet cable is a transmission medium, not a device.

Question 08 · Protocol Hierarchy & Layered Abstraction

Priority ★★★★★

CORRECT ANSWER: (B) Higher-level protocols can be designed and updated without needing to know or manage the physical details of how bits are transmitted across copper wires, fiber optics, or radio waves.

- Analyze the role of abstraction in protocol stacks:
- • Layering is an application of computational abstraction.
- • When a software engineer writes a web app using HTTP, they only need to worry about web requests (GET/POST) and URLs.
- • The application layer delegates data segmentation and reliability to TCP, which delegates addressing and routing to IP, which delegates electrical/optical signaling to the physical layer.
- • If engineers upgrade from 4G to 5G cellular, or from copper to fiber-optic cables, the higher-level protocols (HTTP, TCP, IP) continue functioning without modifying a single line of code.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** IP addresses are fundamental to packet routing and cannot be eliminated by layering.
✕ **(C):** Compression is an independent algorithmic process, not a property of protocol layering.
✕ **(D):** Protocol layering does not hide packet headers; IP headers must remain readable by intermediate routers.

Question 09 · Scalability of Distributed Computer Networks

Priority ★★★★★

CORRECT ANSWER: (D) Scalability

- Define network scalability:
- • **Scalability** is the capacity of a system, network, or algorithm to handle growing amounts of work or to be readily enlarged to accommodate that growth.
- • Because the Internet uses decentralized routing, open protocols, and hierarchical addressing (like IPv6 and DNS), new routers, servers, and billions of new smartphones can join the network continuously without bringing down existing traffic or requiring a complete rebuild.
- • Scalable architecture ensures long-term viability under massive user expansion.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Abstraction is hiding detail to manage complexity.
✕ **(B):** Fault tolerance is the ability to continue operating despite the failure of individual components (redundancy).
✕ **(C):** Quantization is the process of mapping continuous analog signals into discrete digital values in data representation.

Question 10 · Physical Transmission Media Trade-offs

Priority ★★★★

CORRECT ANSWER: (D) Fiber-optic cable; transmits bits as light pulses through glass fibers, providing extremely high bandwidth and low signal degradation over long distances.

- Compare physical networking transmission media:
- • **Fiber-optic cables** transmit data as pulses of light (photons) bounced through pure optical glass fibers using total internal reflection.
- - Advantages: Immense bandwidth, immune to electromagnetic interference, and minimal signal loss (attenuation) over hundreds of miles.
- - It is the standard medium for undersea cables and long-distance terrestrial backbones.
- • **Copper wire (Ethernet / Coaxial):** Transmits electrical voltage signals; suffers significant attenuation and interference over distances greater than ~100 meters.
- • **Radio waves (Wi-Fi / Cellular / Satellite):** Wireless transmission using electromagnetic waves; useful for mobile mobility but subject to physical obstruction and weather interference.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Wi-Fi transmits electromagnetic radio waves, not physical electrons, and has a range of only ~100 feet.
✕ **(B):** Twisted-pair Ethernet uses copper electrical wiring, not laser beams, and is restricted to ~100 meters.
✕ **(C):** Copper cables transmit electrical voltage, not sound waves.

Question 11 · Structure & Capacity of IPv4 Addresses

Priority ★★★★★

CORRECT ANSWER: (C) 32 bits; providing 2^{32} (approximately 4.3 billion) unique addresses.

- Analyze IPv4 addressing specifications in the AP CSP curriculum:
- • An IPv4 address consists of exactly **32 bits** (4 bytes / octets).
- • It is typically written in dotted-decimal notation: four decimal numbers (each ranging from 0 to 255) separated by periods (e.g., `192.168.1.1`).
- • With 32 binary bits, the total number of unique permutations is $2^{32} = 4,294,967,296$ (approximately 4.3 billion possible addresses).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** 128 bits is the length of an IPv6 address, not IPv4.
✕ **(B):** IPv4 has never used 64-bit addresses.
✕ **(D):** 16 bits yields only 65,536 addresses, which was far too small even for early ARPANET networks.

Question 12 · IPv4 Address Exhaustion & The Need for IPv6

Priority ★★★★★

CORRECT ANSWER: (C) The exponential explosion of internet-connected mobile phones, laptops, smart home IoT devices, and global web servers far exceeded the 4.3 billion addresses available in IPv4.

- Examine the causes of IPv4 exhaustion:
- • When IPv4 was standardized in 1981, 4.3 billion addresses seemed virtually inexhaustible for a handful of university and military research labs.
- • However, the rise of the commercial World Wide Web, global consumer adoption, and the proliferation of mobile devices, smartphones, and IoT smart devices (watches, cameras, cars, appliances) meant there were more connected devices than humans on Earth.
- • Because $2^{32} \approx 4.3$ billion was insufficient for the global population, the IETF developed **IPv6** to provide a virtually limitless address pool.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Processor architecture (32-bit vs 64-bit CPUs) is unrelated to network IP protocol versions.
✕ **(B):** Dotted-decimal notation is simply human-friendly formatting; computers always process raw binary bits.
✕ **(D):** Fiber-optic cables transmit raw light pulses representing arbitrary binary bit streams regardless of protocol version.

Question 13 · Structure & Capacity of IPv6 Addresses

Priority ★★★★★

CORRECT ANSWER: (A) IPv6; it consists of 128 bits represented as 8 hexadecimal groups separated by colons.

- Identify IPv6 format and bit specifications:
- • **IPv6** addresses are **128 bits** long.
- • Because 128 binary bits would be too long to write in decimal, they are formatted as 8 groups of 4 **hexadecimal digits** separated by colons (`:`).
- • Each hexadecimal digit represents 4 bits ($8 \times 4 \times 4 = 128$ bits).
- • Total capacity is $2^{128} \approx 3.4 \times 10^{38}$ unique addresses, ensuring humanity will never run out of network addresses.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** IPv4 is 32 bits and uses decimal numbers separated by periods.
✕ **(C):** A Domain Name is a human-readable text address like `www.example.com`.
✕ **(D):** A MAC address is a 48-bit hardware identifier on a local network card, formatted in 6 pairs (e.g. `00:1A:2B:3C:4D:5E`).

Question 14 · Mathematical Comparison of IPv4 vs. IPv6 Capacity

Priority ★★★★

CORRECT ANSWER: (A) 2^{96} times larger (an astronomical factor of approximately 7.9×10^{28} times larger).

- Calculate the exponential growth of binary address spaces:
- • The number of possible IPv4 addresses is 2^{32}.
- • The number of possible IPv6 addresses is 2^{128}.
- • To find how many times larger IPv6 is compared to IPv4, divide the two quantities:
- $$\frac{2^{128}}{2^{32}} = 2^{128 - 32} = 2^{96}$$
- • Each additional bit doubles the address space. Adding 96 bits doubles the address space 96 times!
- • $2^{96} \approx 7.92 \times 10^{28}$, which is nearly 80 octillion times larger than the entire IPv4 address space.
- • (Common student trap: Dividing 128 by 32 to get 4 completely overlooks that address space grows exponentially in base 2, not linearly).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Calculates $2^4 = 16$, which is an incorrect exponent calculation.
✕ **(C):** Subtracts exponents without applying base 2 (2^{96}, not 96).
✕ **(D):** Divides 128 by 32, confusing the bit length ratio with the exponential capacity ratio ($2^{128} \ne 4 \times 2^{32}$).

Question 15 · Dynamic Packet Routing Mechanics

Priority ★★★★★

CORRECT ANSWER: (D) Routers use dynamic routing protocols to send packets along alternate paths based on real-time network traffic congestion, link availability, and transmission costs.

- Analyze dynamic routing in packet-switched networks:
- • The Internet does not maintain fixed, static pathways for data streams.
- • Routers continuously communicate with neighboring routers to monitor network health, queue delays, and cable cuts.
- • If Router A detects heavy congestion or a broken fiber link on Route 1, it will immediately forward subsequent packets along Route 2.
- • This **dynamic routing** maximizes overall network throughput and ensures continuous delivery even during component failures.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Path diversity is an emergent property of independent router decisions, not an intentional browser-scrambling mechanism.
✕ **(B):** Packets seek the most efficient path; there is no rule requiring visits to different countries.
✕ **(C):** Satellites are only one specific transmission medium and are not swapped per second.

Question 16 · Time-To-Live (TTL) and Routing Loop Prevention

Priority ★★★★

CORRECT ANSWER: (D) Each router decrements TTL by 1; if TTL reaches 0 before the packet reaches its destination, the packet is discarded to prevent it from circulating endlessly in an infinite routing loop.

- Analyze the function of the Time-To-Live (TTL) header field:
- • In complex mesh networks, temporary routing misconfigurations or table updates can occasionally create routing loops (e.g. Router A sends to Router B, which sends back to Router A).
- • To prevent orphaned packets from bouncing back and forth forever and clogging bandwidth, every router that forwards a packet decrements its TTL counter by 1.
- • If a packet's TTL counter drops to 0, the current router discards (drops) the packet and sends an error message (ICMP Time Exceeded) back to the sender.
- • This acts as a self-destruct mechanism that safeguards network capacity.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** TTL decrements towards zero; it does not increment or grant administrative privileges.
✕ **(B):** TTL is a packet-level hop counter, not a cellular billing quota meter.
✕ **(C):** TTL is unrelated to physical hardware thermal sensors.

Question 17 · Hierarchical Structure of IP Addresses

Priority ★★★★★

CORRECT ANSWER: (C) The leading bits (network prefix) identify the specific network or Internet Service Provider (ISP), while the trailing bits identify the specific host device on that network.

- Analyze the hierarchical structure of IP addressing:
- • An IP address is split into two primary parts: the **Network ID (prefix)** and the **Host ID (suffix)**.
- • Global backbone routers only need to inspect the network prefix to route a packet to the destination's ISP or enterprise network (e.g., University of California or Comcast).
- • Once the packet arrives at the local network's gateway router, the local router inspects the host ID to deliver the packet to the exact computer or smartphone.
- • This hierarchy prevents routers from having to memorize billions of individual device locations, keeping global routing tables compact and efficient.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** IP addresses identify network interfaces, not device form factors (mobile vs desktop).
✕ **(B):** IP addresses do not encode raw GPS coordinates.
✕ **(D):** Random addressing would make routing impossible; hierarchy is essential for scalable packet forwarding.

Question 18 · Router Forwarding Table Decisions

Priority ★★★★

CORRECT ANSWER: (A) It compares the packet's destination IP address against its internal routing table to find the longest matching network prefix.

- Analyze router forwarding mechanics:
- • A router maintains a **routing table** containing network destination prefixes and the corresponding next-hop IP address or physical outgoing port.
- • When an incoming packet arrives, the router examines only the packet header's destination IP address.
- • It performs a 'longest prefix match' in its routing table and places the packet on the designated output port.
- • Routers never inspect application-layer payload data (like web text or video frames) to make routing decisions.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Routers only read header metadata (IP addresses); they do not inspect or interpret application payloads.
✕ **(C):** Broadcasting packets everywhere (flooding) causes massive network congestion; routers forward selectively.
✕ **(D):** Client web browsers have no visibility or control over intermediate router hardware.

Question 19 · Public vs. Private IP Addresses & NAT

Priority ★★★★

CORRECT ANSWER: (A) These are designated private IP addresses used only inside local area networks; the home router translates them to a single unique public IP address using Network Address Translation (NAT) when accessing the global Internet.

- Understand Private IP addresses and Network Address Translation (NAT):
- • Certain IP ranges (including `192.168.0.0/16` and `10.0.0.0/8`) are reserved for private, local use.
- • Routers on the public Internet will never route packets addressed to a private IP.
- • Instead, a home or office router uses **Network Address Translation (NAT)**: all local devices share the router's single public IP address when communicating with web servers outside the local network.
- • This conserved millions of IPv4 addresses and allowed millions of private networks to reuse the same local address blocks.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Billions of homes and devices access the internet concurrently.
✕ **(C):** Local networks use standard physical wires or Wi-Fi radio waves, not satellite telepathy.
✕ **(D):** NAT is an addressing translation mechanism, not a dynamic encryption key.

Question 20 · IP Spoofing & Network Security Vulnerabilities

Priority ★★★★★

CORRECT ANSWER: (B) An attacker crafts data packets with a falsified source IP address in the packet header to conceal their identity or impersonate a trusted computing device.

- Define IP spoofing in network security:
- • Because basic IP packets do not have built-in authentication of the source address in the header, a malicious sender can forge (spoof) the source IP address.
- • Uses of IP spoofing:
- - Impersonation: Tricking a server into believing a request comes from an authorized internal device.
- - Distributed Denial of Service (DDoS) Amplification: Sending small requests with the victim's spoofed IP address to third-party servers, causing those servers to send massive responses directly to the victim, overwhelming their bandwidth.
- • Defenses include ingress filtering at routers and cryptographic authentication.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Domain copyright claims are legal/DNS disputes, not packet-level address spoofing.
✕ **(C):** Bandwidth quotas are enforced by ISP throttling, not IP spoofing.
✕ **(D):** Physical cable destruction is physical sabotage, not IP spoofing.

Question 21 · Structure of a Network Packet: Header vs. Payload

Priority ★★★★★

CORRECT ANSWER: (C) The header contains routing metadata (such as source IP, destination IP, packet size, and sequence number), while the payload contains the actual user data being transmitted.

- Analyze packet architecture in the AP CSP curriculum:
- • **Header (Metadata):** Contains critical control instructions needed to deliver and reassemble the packet, including:
- - Sender's (Source) IP address
- - Receiver's (Destination) IP address
- - Packet sequence number (e.g. Packet 4 of 20)
- - Time-To-Live (TTL) and error-checking checksums.
- • **Payload (Data):** The actual chunk of content being transferred (e.g., a portion of an image, text snippet, or audio frame).
- • Intermediate routers read only the header metadata to make routing decisions.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Headers are vital throughout the entire journey; without headers, routers would not know where to send packets.
✕ **(B):** Packet headers do not store credit card credentials; they store network routing information.
✕ **(D):** The header is small control metadata; user data is stored inside the payload.

Question 22 · Out-of-Order Packet Arrival & TCP Reassembly

Priority ★★★★★

CORRECT ANSWER: (D) The Transmission Control Protocol (TCP) on the receiving computer reads the sequence number in each packet's header and reorders them into their original sequence before displaying the image.

- Analyze packet reassembly in TCP:
- • Because packet switching routes each packet independently, different packets experience different router queues and propagation delays.
- • The sender's TCP layer stamps every packet with a sequential identifier (**sequence number**).
- • Upon arrival, the receiver's TCP buffer collects the packets and uses these sequence numbers to reconstruct the data stream into its exact original arrangement.
- • Once completely assembled and verified, the data is delivered cleanly to the application layer.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Packets are not discarded simply for arriving out of order; TCP places them in a buffer and sorts them.
✕ **(B):** Images are never delivered to the screen in raw out-of-order packet sequence.
✕ **(C):** Routers forward packets immediately without buffering or coordinating with other routers to ensure ordered arrival.

Question 23 · Characteristics of the Transmission Control Protocol (TCP)

Priority ★★★★★

CORRECT ANSWER: (A) Connection-oriented and reliable; it establishes a connection, verifies that all packets arrive intact, requests retransmission of lost packets, and reassembles them in order.

- Examine TCP's reliability guarantees:
- • **TCP** is the primary transport-layer protocol for applications requiring 100% data fidelity.
- • Key properties:
- - **Connection-oriented:** Uses a 3-way handshake to establish a synchronized session between sender and receiver.
- - **Reliability:** The receiver sends acknowledgment (ACK) packets for received data.
- - **Retransmission:** If an ACK is not received within a timeout window, TCP assumes the packet was dropped and automatically retransmits it.
- - **In-order Delivery:** Uses sequence numbers to guarantee data is passed to the application in order.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Connectionless and unacknowledged describes UDP (User Datagram Protocol), not TCP.
✕ **(C):** Light modulation is a physical layer function, not a transport-layer protocol.
✕ **(D):** TCP is an open, free standard defined by the IETF, not proprietary software.

Question 24 · Characteristics of the User Datagram Protocol (UDP)

Priority ★★★★★

CORRECT ANSWER: (A) Connectionless and lightweight; it transmits packets rapidly with minimal overhead, without establishing a formal connection, tracking packet order, or retransmitting lost packets.

- Examine UDP's speed and lightweight design:
- • **UDP** prioritizes speed and low latency over perfect reliability.
- • Key properties:
- - **Connectionless:** Senders transmit datagrams immediately without waiting for a handshake.
- - **No Acknowledgments:** Receivers do not send ACKs back to the sender.
- - **No Retransmission:** If a packet is corrupted or dropped due to network congestion, UDP simply ignores it and continues.
- - **Minimal Overhead:** Smaller header size (8 bytes vs. TCP's 20+ bytes) and zero round-trip latency delays.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Domain searches use the DNS protocol (which often runs over UDP, but UDP itself is a transport protocol).
✕ **(C):** Guaranteed in-order delivery and retransmission describes TCP, not UDP.
✕ **(D):** Ethernet pin voltages are defined by IEEE physical layer standards.

Question 25 · Application Selection: TCP vs. UDP

Priority ★★★★★

CORRECT ANSWER: (B) Application 1 should use TCP (guaranteeing 100% accuracy without data corruption); Application 2 should use UDP (minimizing latency and input lag where dropping an occasional packet is acceptable).

- Analyze protocol trade-offs for real-world application domains:
- • **Application 1 (Banking / Financial Transfer):**
- - Absolute data integrity is mandatory. A dropped packet could mean missing digits in a financial account transfer.
- - Waiting a few milliseconds for TCP retransmission is completely acceptable to guarantee 100% accuracy.
- - **Choice: TCP**.
- • **Application 2 (Live Multiplayer Gaming / VoIP):**
- - Immediate responsiveness is paramount. If a player's coordinate packet from 100 milliseconds ago is lost, retransmitting it is useless because the player has already moved.
- - Stalling the game to retransmit stale position data causes severe lag/stutter.
- - **Choice: UDP**.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** TCP routinely transmits all forms of digital data, including numeric financial transactions.
✕ **(C):** Swaps the protocols; banking cannot afford packet loss, and gaming cannot tolerate TCP retransmission latency.
✕ **(D):** UDP is heavily utilized worldwide for video streaming (DNS, VoIP, gaming, YouTube, Netflix).

Question 26 · Network Congestion & Packet Dropping Mechanics

Priority ★★★★

CORRECT ANSWER: (C) When incoming packets arrive faster than the router's outgoing link can transmit, the router's memory buffer overflows and drops incoming packets; TCP detects missing acknowledgments and retransmits them at a reduced sending rate.

- Analyze router queue management and TCP congestion control:
- • Routers have finite internal RAM memory buffers (queues) to hold packets waiting for an outgoing line.
- • When traffic bursts exceed the outgoing bandwidth, the buffer fills up completely. Any further incoming packets must be discarded (**packet drop**).
- • Under TCP, when the sender fails to receive an acknowledgment (ACK) for the dropped packet within a timeout period, it triggers **TCP Congestion Control**:
- - The sender retransmits the lost packet.
- - Crucially, the sender cuts its transmission speed (window size) in half, allowing intermediate routers to clear their backlogs.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Packet drops are routine everyday occurrences on the Internet; they do not reboot the global backbone.
✕ **(B):** Routers have finite memory buffers; they cannot store infinite data.
✕ **(D):** Packet dropping is an automatic physical consequence of buffer overflow, not an intentional behavioral penalty.

Question 27 · The TCP Three-Way Handshake

Priority ★★★★

CORRECT ANSWER: (C) 1. Client sends SYN (Synchronize) 2. Server responds with SYN-ACK (Synchronize-Acknowledge) 3. Client responds with ACK (Acknowledge)

- Trace the 3-Way Handshake protocol steps:
- • Step 1: The client initiates the session by sending a **SYN** packet containing its initial sequence number.
- • Step 2: The server receives the request, agrees to communicate, and responds with a **SYN-ACK** packet (acknowledging the client's sequence number and stating its own initial sequence number).
- • Step 3: The client receives the SYN-ACK and sends back an **ACK** packet, confirming that both sides are synchronized.
- • Once this 3-step negotiation concludes, the connection is established and application data transfer begins.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Data transfer only occurs after the handshake is successfully completed.
✕ **(B):** Ping/Pong are ICMP echo utilities, not the TCP handshake sequence.
✕ **(D):** GET, POST, and DELETE are application-layer HTTP methods, not transport-layer connection establishment signals.

Question 28 · Packet Fragmentation & Retransmission Efficiency

Priority ★★★★★

CORRECT ANSWER: (D) If a single packet is corrupted or dropped due to network noise, only that individual 1,500-byte packet needs to be retransmitted, rather than re-sending the entire 100 MB file.

- Evaluate the error recovery efficiency of packetization:
- • In real-world networks, brief interference (microwave noise, radio fading, brief congestion) can corrupt a few bits.
- • If a file were sent as one continuous 100 MB block, a single flipped bit at minute 9 would invalidate the entire transfer, forcing the sender to restart all 100 MB from scratch.
- • By fragmenting data into small packets (typically bounded by the Maximum Transmission Unit, ~1,500 bytes):
- - Only the specific lost or damaged packet is retransmitted (a fraction of a millisecond of bandwidth).
- - Packets from other users can interleave smoothly, preventing a single large file from monopolizing a physical cable.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Data charges are based on total bytes transmitted, regardless of packet fragmentation size.
✕ **(B):** Nothing travels faster than the speed of light in vacuum; signals in fiber travel at roughly 67% light speed.
✕ **(C):** Modern computers routinely store gigabytes of data in RAM and terabytes on disk.

Question 29 · Packet Checksums & Bit Error Detection

Priority ★★★★

CORRECT ANSWER: (A) To allow the receiving device to perform a mathematical verification to detect whether any bits in the packet were corrupted or altered during transmission.

- Analyze error detection using checksums:
- • When sending a packet, the sender performs a mathematical calculation (adding up the binary numbers of the header/data) and writes the resulting sum into the **checksum** field.
- • When the packet arrives, the receiver executes the exact same mathematical formula on the received bits.
- • If the receiver's computed sum matches the header's checksum, the packet is verified as intact.
- • If a bit flipped during transmission (due to electrical noise or signal degradation), the sums will not match, alerting the receiver that the packet is corrupted and must be dropped.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Checksums are mathematical bit sums; they do not inspect or categorize advertising content.
✕ **(C):** Copyright licenses are legal documents, not network error-checking hashes.
✕ **(D):** Checksums calculate data integrity, not physical geographic distance.

Question 30 · Protocol Data Encapsulation Hierarchy

Priority ★★★★★

CORRECT ANSWER: (B) The HTTP message is placed inside a TCP segment, which is placed inside an IP packet, which is placed inside an Ethernet frame.

- Trace the hierarchical encapsulation process:
- • 1. **Application Layer (HTTP):** Generates the application data (e.g. `GET /index.html HTTP/1.1`).
- • 2. **Transport Layer (TCP):** Wraps the HTTP data with a TCP header (ports, sequence numbers), creating a **TCP Segment**.
- • 3. **Internet Layer (IP):** Wraps the TCP segment with an IP header (source/destination IP addresses), creating an **IP Packet**.
- • 4. **Link / Network Interface Layer (Ethernet/Wi-Fi):** Wraps the IP packet with hardware MAC addresses and frame check sequences, creating an **Ethernet Frame**.
- • The frame is then encoded into physical electrical voltages, radio waves, or light pulses for transmission.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Ethernet frames wrap IP packets, not the other way around.
✕ **(C):** Reverses the hierarchy; HTTP is high-level application data, while IP is lower-level routing.
✕ **(D):** TCP wraps HTTP; HTTP does not wrap TCP.

Question 31 · Domain Name System (DNS) Core Role

Priority ★★★★★

CORRECT ANSWER: (C) It translates human-readable domain names (such as `www.collegeboard.org`) into machine-readable numerical IP addresses (such as `198.51.100.45`).

- Understand the purpose and function of the Domain Name System (DNS):
- • Humans find symbolic, alphabetical names (like `google.com` or `collegeboard.org`) easy to remember and type.
- • However, network routing hardware (routers and switches) operates exclusively using numerical binary/dotted-decimal IP addresses.
- • DNS acts as the Internet's decentralized, distributed phonebook, taking a requested text hostname and resolving it into its corresponding IP address.
- • Without DNS, users would be forced to memorize and enter 32-bit or 128-bit numerical strings for every website.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Data encryption is handled by security protocols such as TLS/HTTPS, not DNS.
✕ **(B):** Physical signal transmission is the role of physical layer media (copper cables, fiber optics, radio waves).
✕ **(D):** File compression is managed by media codecs and application software, not DNS.

Question 32 · Hierarchical Architecture of the DNS System

Priority ★★★★★

CORRECT ANSWER: (C) Root DNS server → Top-Level Domain (TLD) server (`.org`) → Authoritative DNS server (`collegeboard.org`).

- Analyze the DNS resolution hierarchy:
- • 1. **Root DNS Servers:** Direct queries to the appropriate Top-Level Domain (TLD) nameservers based on the extension (`.org`, `.com`, `.edu`).
- • 2. **TLD Servers:** Manage the specific domain registry (e.g., all `.org` domains) and provide the address of the authoritative nameserver for that organization.
- • 3. **Authoritative Nameservers:** Possess the definitive, canonical database records for that specific domain (`collegeboard.org`) and return the exact IP address for the sub-host (`apstudents`).
- • This distributed hierarchy ensures no single server creates a global bottleneck or single point of failure.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** This lists physical networking hardware devices rather than the logical hierarchy of DNS name resolution.
✕ **(B):** TLD servers cannot direct traffic without the Root server identifying the proper registry first.
✕ **(D):** The authoritative server cannot be queried first because its location is not known until resolved by the Root and TLD servers.

Question 33 · DNS Caching & Time-To-Live (TTL)

Priority ★★★★

CORRECT ANSWER: (A) It decreases latency for recurring page visits and significantly reduces network query traffic on upstream DNS root and TLD servers.

- Evaluate the performance advantages of DNS caching:
- • Without caching, every single HTTP request (including every separate image, CSS file, and script on a web page) would require repeated round-trip DNS lookups across the globe.
- • Caching stores the result locally in browser memory or the local recursive resolver.
- • This dramatically cuts down lookup latency (from ~50-100 ms to <1 ms) and prevents global root nameservers from crashing under excessive query volume.
- • The TTL field specifies when cached records expire so that changes in IP addresses eventually propagate.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Devices still require an active physical connection to the network to download web content.
✕ **(C):** DNS caching does not encrypt traffic or protect against eavesdropping.
✕ **(D):** Websites frequently update servers and IP addresses; caching simply pauses lookups for the duration of the TTL.

Question 34 · DNS Spoofing & Cache Poisoning Cyberattack

Priority ★★★★★

CORRECT ANSWER: (D) DNS Spoofing (also known as DNS Cache Poisoning).

- Identify the mechanism of DNS Spoofing / Cache Poisoning:
- • In **DNS Spoofing**, an attacker feeds invalid, forged DNS response data into a resolver's cache.
- • The cache stores the malicious entry for the duration of the TTL.
- • When unsuspecting users query that resolver, they receive the attacker's IP address instead of the genuine server.
- • The URL bar may appear to display the legitimate domain name, but the browser connects to an attacker-controlled server designed to steal credentials (phishing).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Severing cables halts connectivity entirely rather than routing traffic to a clone website.
✕ **(B):** A keylogger is software or hardware that records keystrokes on a local machine, not a server cache corruption.
✕ **(C):** DDoS attacks overwhelm a target server with traffic to render it unavailable; they do not covertly redirect DNS queries.

Question 35 · Hypertext Transfer Protocol: HTTP vs. HTTPS

Priority ★★★★★

CORRECT ANSWER: (B) HTTPS encrypts the data exchanged between the browser and web server using TLS/SSL, preventing unauthorized eavesdroppers on the open Wi-Fi network from reading credit card numbers and passwords.

- Distinguish between HTTP and HTTPS:
- • **HTTP (Hypertext Transfer Protocol):** Transmits web requests and responses in plain text. Anyone monitoring the local wireless airwaves (packet sniffing) can read passwords, session cookies, and personal data.
- • **HTTPS (Hypertext Transfer Protocol Secure):** Wraps HTTP communication inside an encrypted layer (Transport Layer Security / TLS or SSL).
- • Even if eavesdroppers intercept the packets over open public Wi-Fi, the payload appears as undecipherable cryptographic ciphertext.
- • HTTPS also authenticates the web server to ensure users are communicating with the authentic business.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Network protocols cannot protect electronic hardware from electrical power surges.
✕ **(C):** HTTPS involves cryptographic operations that may consume slight CPU processing, but it has no relationship to battery optimization.
✕ **(D):** HTTPS does not double network speed; encryption adds a microscopic computational overhead.

Question 36 · Anatomy of a Uniform Resource Locator (URL)

Priority ★★★★

CORRECT ANSWER: (C) Protocol: `https` | Domain: `apstudents.collegeboard.org` | Port: `443` | Path: `/courses/csp/exam-prep` | Query Parameter: `year=2026`

- Analyze the standardized structure of a URL:
- • **Protocol/Scheme (`https`):** Defines the communication rules and encryption standard used to request the resource.
- • **Domain Name / Host (`apstudents.collegeboard.org`):** Identifies the target computer server to be resolved via DNS.
- • **Port (`443`):** Specifies the standard network endpoint on the host dedicated to secure HTTPS traffic.
- • **Path (`/courses/csp/exam-prep`):** Identifies the specific file path or resource location on the web server.
- • **Query String (`?year=2026`):** Supplies key-value parameters passed to the server-side script or application.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Reverses protocols and domain names; `https` is the protocol and `443` is the port.
✕ **(B):** `collegeboard` is an organizational domain label, not a protocol.
✕ **(D):** Completely scrambles the hierarchical ordering of URL components.

Question 37 · Statelessness of HTTP and Session Management via Cookies

Priority ★★★★

CORRECT ANSWER: (A) The server sends a unique session identifier stored in a client-side **HTTP Cookie or local storage, which the browser automatically attaches to subsequent requests.**

- Understand HTTP statelessness and session tracking:
- • Because HTTP is stateless, the server closes or forgets the connection state after each request-response transaction.
- • To maintain continuity (e.g. login sessions, shopping carts, site preferences), websites issue **Cookies**—small pieces of text data sent by the server and stored on the user's browser.
- • With each subsequent request to that domain, the browser transmits the cookie back to the server, allowing the server to recognize the user and correlate state.
- • This elegant design keeps the core protocol lightweight while allowing stateful applications.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** DNS translates domain names to IP addresses; it does not track personal shopping cart states.
✕ **(C):** Routers store IP routing prefixes, not application-level e-commerce data.
✕ **(D):** Cables are passive transmission media without memory or storage capabilities.

Question 38 · Digital Certificates and Certificate Authorities (CAs)

Priority ★★★★★

CORRECT ANSWER: (A) It verifies the authentic identity of the website owner and issues a cryptographically signed certificate linking the website's public key to its domain name.

- Examine the role of Certificate Authorities (CAs) in public key infrastructure:
- • Without trust verification, an attacker could impersonate a bank website and present their own public encryption key (Man-in-the-Middle attack).
- • A Certificate Authority (CA) is a trusted third-party organization that verifies the identity and domain ownership of an entity.
- • The CA then digitally signs the website's certificate containing the site's public key.
- • Web browsers come pre-installed with the root public keys of trusted CAs, allowing them to verify cryptographic signatures and warn users if a certificate is fake, expired, or invalid.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** CAs have no involvement in utility payments or hardware administration.
✕ **(C):** CAs issue cryptographic trust certificates; they do not host website web pages or data files.
✕ **(D):** CAs do not monitor user web browsing or enforce copyright law.

Question 39 · DNS Scalability, Anycast Routing, and Redundancy

Priority ★★★★

CORRECT ANSWER: (B) Each logical root address is implemented using hundreds of physical servers distributed globally that share the same IP address through Anycast routing and aggressive hierarchical caching.

- Analyze DNS resilience, scalability, and Anycast routing:
- • While there are only 13 logical root server network addresses (due to legacy packet size constraints in UDP DNS responses), each address is backed by a global network of hundreds of redundant physical machines.
- • Using **Anycast routing**, routers automatically direct a query to the geographically and topologically nearest physical instance sharing that IP address.
- • Furthermore, client-side, router, and ISP resolvers cache query results for hours, meaning only a minuscule fraction of queries ever reach root servers.
- • This architecture provides extreme fault tolerance: if dozens of physical root servers fail, traffic instantly routes to surviving machines.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Client devices query DNS resolvers; they are not mandated to host root DNS server infrastructure.
✕ **(C):** DNS operates 24/7 in real time with millisecond responses.
✕ **(D):** There is no daily query restriction on Internet users.

Question 40 · Matching Internet Protocols with Functional Roles

Priority ★★★★★

CORRECT ANSWER: (D) Protocol 1: HTTP | Protocol 2: DNS | Protocol 3: TCP | Protocol 4: IP

- Analyze and correlate the core protocols of the Internet suite:
- • **Protocol 1 is HTTP:** Operates at the application layer to request and deliver web documents (HTML, CSS, images).
- • **Protocol 2 is DNS:** Operates at the application layer to look up and convert text domain hostnames into IP numbers.
- • **Protocol 3 is TCP:** Operates at the transport layer to manage packet numbering, acknowledgments, and reassembly.
- • **Protocol 4 is IP:** Operates at the network/Internet layer to handle addressing and hop-by-hop packet routing.
- • This 4-protocol stack represents the fundamental layered abstraction tested heavily on the AP CSP exam.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Places TCP as web formatting and HTTP as transport delivery.
✕ **(B):** Assigns IP to web formatting and HTTP to routing.
✕ **(C):** Swaps HTTP with DNS and TCP with IP.

Question 41 · Definition and Measurement of Bandwidth

Priority ★★★★★

CORRECT ANSWER: (C) The maximum rate at which data can be transferred across a network connection in a given amount of time, typically measured in bits per second (bps, Mbps, or Gbps).

- Define network bandwidth according to the AP CSP specification:
- • **Bandwidth** represents the transmission capacity of a network pathway.
- • It indicates the volume of data bits that can pass through a channel per second (bitrate).
- • Standard units include: bps (bits per second), Mbps (megabits per second, 10^6 bps), and Gbps (gigabits per second, 10^9 bps).
- • Bandwidth acts like the diameter of a water pipe: a wider pipe allows more gallons of water per second, but does not necessarily speed up the individual molecules.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Uptime or schedule is an operational metric, not network bandwidth.
✕ **(B):** Cable length is physical distance, which influences propagation delay/latency, not bandwidth capacity.
✕ **(D):** Hard drive storage is local non-volatile data capacity, not network transmission speed.

Question 42 · Definition and Measurement of Network Latency

Priority ★★★★★

CORRECT ANSWER: (A) **Latency, which measures the time elapsed for a data packet to travel from the sender to the receiver, typically expressed in milliseconds (ms).**

- Define network latency in AP CSP:
- • **Latency** is the time delay required for a message or packet to travel from the source system to the destination system across physical media, routers, and switches.
- • It is commonly measured as Round-Trip Time (RTT) in **milliseconds (ms)**.
- • Latency is constrained by the speed of light in fiber/copper, physical distance to the server, and queuing delays at intermediate routers.
- • Even on a multi-gigabit bandwidth connection, high latency will cause noticeable lag in interactive, real-time applications.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Pixel density relates to display monitor hardware resolution, not network transit time.
✕ **(C):** CPU clock speed relates to processor computation rate, not network transmission delay.
✕ **(D):** Battery voltage is electrical power, not network latency.

Question 43 · Distinguishing Between Bandwidth and Latency

Priority ★★★★

CORRECT ANSWER: (A) The connection exhibits **high bandwidth (high volume of data transmitted per second) but **high latency** (long propagation delay due to orbital distance).**

- Distinguish between bandwidth and latency characteristics:
- • **Bandwidth (100 Mbps):** High capacity; once the transmission stream is established, 100 million bits of data flow each second.
- • **Latency (650 ms):** High delay; geostationary satellites orbit ~35,786 km above Earth. Signals must travel up to space and back down twice for a round trip (over 140,000 km total transit), introducing inevitable physical propagation delay.
- • Bandwidth is how *wide* the pipe is; latency is how *long* it takes for the first drop of water to travel the length of the pipe.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Bandwidth is high (100 Mbps) and latency is high (650 ms), not low/low.
✕ **(C):** No electromagnetic signal can travel faster than the speed of light in vacuum.
✕ **(D):** Bandwidth is actively present at 100 Mbps; radio waves readily penetrate atmospheric clouds.

Question 44 · Mathematical Calculation of Network Transfer Time

Priority ★★★★★

CORRECT ANSWER: (B) **20 seconds**

- Step-by-step mathematical conversion and transfer time calculation:
- • 1. **Convert File Size from Bytes to Bits:**
- $$\text{Total Bits} = 150\text{ MB} \times 8\text{ bits/Byte} = 1,200\text{ Megabits (Mb)}$$
- • 2. **Apply the Transfer Time Formula:**
- $$\text{Transfer Time} = \frac{\text{Total Data (Megabits)}}{\text{Bandwidth (Mbps)}}$$
- • 3. **Calculate:**
- $$\text{Transfer Time} = \frac{1,200\text{ Mb}}{60\text{ Mbps}} = 20\text{ seconds}$$
- • Common student trap: Forgetting to multiply Bytes by 8 to get bits (which would incorrectly yield $150 / 60 = 2.5\text{ seconds}$).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Mistakenly assumes transfer time equals file size in MB.
✕ **(C):** Incorrectly multiplies 150 by 60 instead of dividing.
✕ **(D):** Incorrectly divides 150 MB by 60 Mbps directly without converting Megabytes to Megabits (missing factor of 8).

Question 45 · Identifying Network Bottlenecks in Sequential Paths

Priority ★★★★★

CORRECT ANSWER: (B) **15 Mbps, because the overall end-to-end bandwidth is constrained by the slowest link (the bottleneck) along the transmission path.**

- Analyze network bottlenecks in sequential routing:
- • When data flows along a single path containing multiple sequential segments, the system's maximum transfer capacity is governed by the link with the smallest bandwidth (the **bottleneck**).
- • Even though Link 1 can emit data at 1,000 Mbps, Router B can only push data across Link 2 at 15 Mbps.
- • Any data arriving faster than 15 Mbps must queue in Router B's buffer or be dropped.
- • Therefore, the maximum sustainable transfer rate is strictly 15 Mbps.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** High initial bandwidth cannot force subsequent narrow links to transmit beyond their physical capacity.
✕ **(C):** Network throughput is not the mathematical average of link speeds; it is capped at the minimum link.
✕ **(D):** Bandwidths in series cannot be summed; data must traverse each link one after another.

Question 46 · Distinguishing Between Bandwidth and Actual Throughput

Priority ★★★★

CORRECT ANSWER: (D) Throughput accounts for real-world factors such as protocol header overhead, network congestion from competing users, and retransmissions caused by packet loss.

- Compare theoretical bandwidth with actual throughput:
- • **Bandwidth:** The maximum theoretical physical capacity of the communication channel under perfect conditions.
- • **Throughput:** The actual rate of successful data delivery achieved in real-world operation.
- • Real throughput is reduced by: (1) Protocol overhead (TCP headers, IP headers, Ethernet frames consuming bandwidth), (2) Network congestion on shared ISP backbone lines, (3) Packet loss triggering TCP retransmissions and window throttling.
- • Throughput is always less than or equal to bandwidth.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Network throughput degradation occurs naturally from traffic congestion and overhead, not arbitrary video throttling.
✕ **(B):** A Byte consists of 8 bits; a Byte is larger than a single bit.
✕ **(C):** Monitor refresh rate (Hz) has no effect on network link throughput.

Question 47 · Network Jitter and Packet Arrival Consistency

Priority ★★★★

CORRECT ANSWER: (A) **Jitter, which represents the variance or fluctuation in packet arrival latency over time.**

- Define network jitter and its impact on real-time streaming:
- • **Jitter** is the variation in packet transit delay across a network.
- • If one packet takes 20 ms to arrive, but the next packet takes 120 ms, and the third takes 35 ms, the high jitter disrupts continuous media playback.
- • Real-time applications (VoIP, video calls, gaming) rely on consistent arrival timing; when jitter is high, media buffers underflow or overflow, causing choppy audio and video stutter.
- • While asynchronous file downloads are immune to jitter, interactive real-time communications are heavily degraded by it.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Attenuation is the gradual loss of signal strength over physical distance, not latency variance.
✕ **(C):** Modulation is signal encoding for transmission media.
✕ **(D):** Duplexing describes bidirectional communication capability.

Question 48 · Network Diagnostic Utilities: Ping and Traceroute

Priority ★★★★

CORRECT ANSWER: (B) It displays the list of every intermediate router (hop) along the packet's path, along with the round-trip latency measured to each router.

- Understand diagnostic network utilities in AP CSP:
- • **Ping:** Uses ICMP Echo requests to measure the basic round-trip time (latency) and packet loss percentage to a target IP address.
- • **Traceroute:** Systematically increments the packet's Time-To-Live (TTL) field starting from 1 to discover every intermediate router (hop) along the route.
- • Each router returns an ICMP 'Time Exceeded' message, allowing traceroute to display the sequence of router IP hostnames and the exact latency to each hop.
- • If a sudden surge in latency or packet drops occurs at Hop 7, the administrator can pinpoint the specific failing router.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Diagnostic tools do not inspect or extract user passwords or private communications.
✕ **(C):** Command-line software cannot physically alter or re-wire physical subsea cables.
✕ **(D):** Traceroute measures path latency; it does not delete emails on mail servers.

Question 49 · Network Congestion and Buffer Queuing Delays

Priority ★★★★

CORRECT ANSWER: (B) Packets accumulate in router memory queues, increasing queuing latency; once memory buffers are completely filled, incoming packets are dropped and must be retransmitted.

- Analyze router behavior during severe network congestion:
- • When incoming packet arrival rate exceeds outgoing link bandwidth, routers store excess packets in temporary hardware memory buffers (queues).
- • As queues grow longer, packets spend more time waiting, drastically increasing network latency (queuing delay / bufferbloat).
- • If congestion persists and buffers fill to 100% capacity, routers are forced to drop arriving packets (packet drop).
- • For TCP connections, packet loss signals congestion, causing senders to reduce transmission rates (TCP congestion control).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Routers do not project packets into the air as ambient room light.
✕ **(C):** Routers process digital data in electronic memory; they do not print paper copies.
✕ **(D):** Dropping packets during congestion is normal standard behavior for IP routers, not a permanent hardware failure.

Question 50 · Comparing File Transfer Scenarios Across Diverse Networks

Priority ★★★★★

CORRECT ANSWER: (C) **Option 4, completing the backup in approximately **25 seconds**.**

- Calculate bulk data transmission across varying bandwidths:
- • 1. **Convert database size to Megabits:**
- $$\text{Total Bits} = 400\text{ MB} \times 8\text{ bits/Byte} = 3,200\text{ Megabits (Mb)}$$
- • 2. **Calculate transfer times for each bandwidth:**
- - Option 1 (16 Mbps): $3,200 / 16 = 200\text{ seconds}$
- - Option 2 (32 Mbps): $3,200 / 32 = 100\text{ seconds}$
- - Option 3 (64 Mbps): $3,200 / 64 = 50\text{ seconds}$
- - Option 4 (128 Mbps): $3,200 / 128 = 25\text{ seconds}$
- • For large bulk transfers, transmission time ($25\text{ s} = 25,000\text{ ms}$) overwhelmingly dominates the initial latency delay ($250\text{ ms} = 0.25\text{ s}$).
- • Therefore, Option 4 finishes vastly faster than all others.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Low latency cannot compensate for an 8x slower bandwidth on a large multi-megabyte file transfer.
✕ **(B):** Option 2 takes 100 seconds, four times longer than Option 4.
✕ **(D):** Option 3 takes 50 seconds, which is twice as long as Option 4 (25 seconds).

Question 51 · Definition of Fault Tolerance in Computer Networks

Priority ★★★★★

CORRECT ANSWER: (C) A system that can continue to function properly and support communication even in the event of the failure of one or more of its components.

- Define Fault Tolerance according to College Board AP CSP guidelines:
- • **Fault Tolerance:** The capability of a system (such as the Internet or a distributed cloud array) to continue operating smoothly without total failure when individual components (routers, links, servers) break down.
- • It is achieved not by building indestructible hardware, but by designing **redundancy** into the architecture.
- • If one path or device fails, backup paths or redundant machines automatically absorb the workload, ensuring seamless operation for end users.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Deleting software upon error is catastrophic failure, the exact opposite of fault tolerance.
✕ **(B):** All electronic computer systems require electrical power to function.
✕ **(D):** Physical hardware components inevitably experience failures; fault tolerance accommodates failures rather than preventing physical degradation.

Question 52 · Role of Redundancy in Achieving Fault Tolerance

Priority ★★★★★

CORRECT ANSWER: (D) It provides alternative transmission routes so that if a primary cable is severed, data traffic can be dynamically rerouted without interrupting network services.

- Understand the purpose of network redundancy:
- • **Redundancy** is the inclusion of extra or backup components (devices, servers, communication lines) that are not strictly necessary when everything is functioning normally.
- • In the event of a fiber cut, equipment malfunction, or power outage on a primary route, traffic is instantly directed through redundant pathways.
- • Redundancy is the primary engineering mechanism that creates fault tolerance in network topologies.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Digital data stays digital; redundancy does not produce physical handwriting.
✕ **(B):** Adding redundant cables does not reduce electrical consumption; it adds infrastructure resilience.
✕ **(C):** Network redundancy has no role in password creation or authentication policy.

Question 53 · Single Point of Failure (SPOF) Identification

Priority ★★★★★

CORRECT ANSWER: (D) Router R is a **Single Point of Failure (SPOF); its failure completely disconnects both Branch Offices from both Server Farms.**

- Analyze Single Point of Failure (SPOF):
- • A **Single Point of Failure (SPOF)** is a critical component whose individual failure halts the operation of the entire system or isolates entire segments.
- • In this topology, every pathway between the Branch Offices (A & B) and the Server Farms (1 & 2) passes through Router R.
- • There are zero alternative routes (no redundancy).
- • When Router R fails, the entire network is partitioned and all inter-branch server communication collapses.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** The network has no redundant bypass paths; it cannot maintain communication if Router R fails.
✕ **(B):** Network hardware failure does not cause client devices to delete their local operating systems.
✕ **(C):** Both Branch A and Branch B depend on Router R; both lose access simultaneously.

Question 54 · Minimum Number of Link Failures to Disconnect a Device

Priority ★★★★★

CORRECT ANSWER: (B) **3 cuts, because Node A has exactly 3 direct physical connections (to B, C, and D), and all 3 must fail to isolate it.**

- Determine node isolation and connectivity degree in a network graph:
- • Node A is connected to 3 neighboring nodes: B, C, and D.
- • If 1 cable cut occurs (e.g., A–B), Node A can still communicate with the rest of the network via Node C and Node D.
- • If 2 cables are cut (e.g., A–B and A–C), Node A can still route packets through Node D.
- • To completely disconnect and isolate Node A from the network, every single one of its direct incident edges (all 3 links) must be severed.
- • Therefore, the minimum number of link cuts required is 3.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Severing links between other nodes (e.g., E–F) does not directly isolate Node A.
✕ **(C):** A single cut leaves two active redundant paths connected to Node A.
✕ **(D):** Severed physical cables cannot transmit signals across physical breaks without active wireless transceivers.

Question 55 · Counting Alternative Paths Between Source and Destination

Priority ★★★★★

CORRECT ANSWER: (C) **3 distinct paths (`S → 1 → D`, `S → 2 → D`, and `S → 3 → D`).**

- Trace independent network pathways:
- • Path 1: S → Node 1 → D
- • Path 2: S → Node 2 → D
- • Path 3: S → Node 3 → D
- • These three paths are node-disjoint (aside from source S and destination D).
- • If Node 1 experiences a hardware crash and the direct link through Node 2 is congested, packets can seamlessly route through Node 3.
- • This multi-path topology provides high fault tolerance and dynamic load balancing.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Intermediate nodes forward packets without erasing them.
✕ **(B):** Networks support multiple simultaneous alternative paths via dynamic routing.
✕ **(D):** Multiplying node counts is mathematically invalid; tracing distinct edge walks yields exactly 3 paths.

Question 56 · Engineering Trade-offs: Economic Cost vs. Redundancy

Priority ★★★★

CORRECT ANSWER: (C) It maximizes fault tolerance and eliminates all single points of failure, but incurs extremely high financial cost and physical complexity to lay and maintain cables between all pairs of buildings.

- Evaluate design trade-offs in network redundancy:
- • For N buildings, a fully connected mesh requires $\frac{N(N - 1)}{2}$ direct connections.
- - For 20 buildings, that requires $\frac{20 \times 19}{2} = 190$ separate dedicated physical lines!
- • **Advantage:** Maximum possible fault tolerance; any building can communicate directly, and many failures can occur without partition.
- • **Trade-off / Disadvantage:** Immense physical infrastructure cost, complex cabling, and high maintenance expenses.
- • In practice, real-world networks use strategic **partial mesh** architectures to achieve high fault tolerance at reasonable cost.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Mesh networks require routers and switches and function continuously 24/7.
✕ **(B):** Signals in cables do not cause friction fires, and latency is never zero (constrained by the speed of light).
✕ **(D):** Network topology does not protect against application malware or prevent file saving.

Question 57 · Dynamic Rerouting and Autonomous Fault Recovery

Priority ★★★★★

CORRECT ANSWER: (A) Routers detect the broken link via missing keepalive signals, immediately update their routing tables, and automatically forward subsequent packets along an alternative redundant path.

- Analyze autonomous dynamic rerouting during live failures:
- • Dynamic routing protocols (such as OSPF and BGP) continuously exchange heartbeat/keepalive telemetry between adjacent routers.
- • When Router 2 stops receiving signals on the severed link to Router 3, it marks that interface as DOWN.
- • Routing tables are instantly updated across the network.
- • Subsequent packets destined for the hospital database are redirected through alternative routes (e.g., through Router 4 or Router 5).
- • Any dropped packets in flight are detected and retransmitted by TCP, making the physical failure virtually invisible to end users.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Severed fiber cables cannot wirelessly transmit optical laser pulses through dirt.
✕ **(C):** Dynamic rerouting is autonomous and handled by network equipment, not manual human rewiring.
✕ **(D):** Networks with redundant paths do not shut down when a single link is severed.

Question 58 · Comparative Fault Tolerance: Star vs. Mesh Topologies

Priority ★★★★

CORRECT ANSWER: (D) If the central switch in Topology X fails, all 30 computers lose network access simultaneously; whereas in Topology Y, failure of a single device can be bypassed through alternative links.

- Compare Star vs Mesh/Redundant Topologies:
- • **Star Topology (X):** Very simple to cable and configure, but the central hub/switch is a **Single Point of Failure (SPOF)**. If the switch loses power or crashes, every connected endpoint is severed.
- • **Redundant/Cross-linked Topology (Y):** Offers alternative data pathways so that a failure in one link or node does not paralyze the remaining network.
- • Understanding this fundamental trade-off between architectural simplicity and fault tolerance is a key concept in AP CSP Unit 4.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Both topologies carry arbitrary digital bitstreams regardless of file format.
✕ **(B):** Star topologies do not cause computer overheating.
✕ **(C):** Star topologies are typically much more cable-efficient than heavily interconnected redundant topologies.

Question 59 · Why the Internet is Designed as a Decentralized Distributed Mesh

Priority ★★★★★

CORRECT ANSWER: (D) A centralized network could be completely incapacitated if enemy forces destroyed the single central hub, whereas a decentralized mesh can survive the destruction of multiple major nodes by routing around the damage.

- Examine the historical origin and architectural philosophy of the Internet:
- • During the Cold War, Paul Baran and ARPA engineers analyzed network survivability against military attack.
- • A **centralized network** (like traditional telephone switches) has a critical Achilles' heel: destroy the central node, and communication collapses.
- • A **decentralized distributed mesh** has no single master control center; routing decisions are distributed across thousands of autonomous routers.
- • If one city or hub is knocked offline, traffic automatically routes through surviving perimeter nodes, providing unmatched fault tolerance.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Decentralized architecture relates to packet delivery survivability, not software malware prevention.
✕ **(B):** Network infrastructure requires electrical power regardless of whether it is centralized or decentralized.
✕ **(C):** Centralized networks are very easy to build and were widely used in early telephone telephony.

Question 60 · Identifying the Critical Bottleneck Node in a Network Graph

Priority ★★★★★

CORRECT ANSWER: (C) Adding direct backup connections between Router 1 and Router 4, and between Router 2 and Router 5, creating alternative routes that bypass Router 3 entirely.

- Analyze network graph redundancy and remediation of single points of failure:
- • In the current configuration, Router 3 is an **articulation point** (bridge node). If Router 3 fails, the graph splits into three disconnected components ({1, 2}, {3}, {4, 5}).
- • To make the network resilient against Router 3's failure, there must exist at least one path between {1, 2} and {4, 5} that does NOT pass through Router 3.
- • Adding direct links (e.g., 1–4 and 2–5) creates parallel bypass routes.
- • If Router 3 subsequently crashes, Router 1 can communicate directly with Router 4, preserving end-to-end connectivity across the network.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Increasing bandwidth does nothing if Router 3 experiences a hardware power failure.
✕ **(B):** Faster cables do not prevent total disconnection if the sole central router crashes.
✕ **(D):** Renaming a router does not add physical or logical routing redundancy.

Question 61 · Sequential vs. Parallel Computing Definitions

Priority ★★★★★

CORRECT ANSWER: (C) In sequential computing, operations are executed one after another in order on a single processor; in parallel computing, a program is split into multiple parts that execute simultaneously on multiple processors or cores.

- Define Sequential and Parallel Computing according to the College Board curriculum:
- • **Sequential Computing:** Executes instructions in strict step-by-step chronological order on a single processing unit. Total execution time equals the sum of all individual step durations.
- • **Parallel Computing:** Divides a computational problem into smaller sub-tasks that are executed simultaneously across multiple processing cores or processors.
- • Parallel computing significantly shortens execution time for computationally intensive tasks, but requires algorithms designed to run concurrently.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Reverses the two definitions: sequential uses one processor, while parallel uses multiple processors.
✕ **(B):** Computing paradigms describe hardware execution architecture, not calendar days of the week.
✕ **(D):** Sequential vs parallel execution relates to processing throughput, not data encryption status.

Question 62 · Multi-Core Processors and Hardware Concurrency

Priority ★★★★★

CORRECT ANSWER: (B) The Central Processing Unit (CPU) contains eight independent physical execution units (cores) that can each read and execute program instructions concurrently.

- Understand multi-core CPU architecture in modern computing systems:
- • A **core** is an individual execution engine capable of performing arithmetic, logic, and control operations.
- • A multi-core processor integrates two or more independent execution units onto a single integrated circuit chip.
- • An 8-core CPU allows up to 8 separate software threads or processes to execute truly simultaneously in hardware without time-slicing.
- • This hardware concurrency forms the foundation for parallel and concurrent software execution.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Cores are microscopic execution engines on the silicon chip, not external input keyboards.
✕ **(C):** Battery life depends on power consumption and battery capacity (watt-hours), not core count.
✕ **(D):** Storage capacity is determined by SSD/HDD size (gigabytes/terabytes), not CPU core count.

Question 63 · Mathematical Calculation of Parallel Speedup

Priority ★★★★★

CORRECT ANSWER: (B) **4.0, meaning the parallel program runs 4 times faster than the sequential version.**

- Apply the College Board AP CSP Speedup formula:
- • 1. **Identify Given Values:**
- - $T_{\text{sequential}} = 120\text{ seconds}$
- - $T_{\text{parallel}} = 30\text{ seconds}$
- • 2. **Apply Formula:**
- $$\text{Speedup} = \frac{T_{\text{sequential}}}{T_{\text{parallel}}} = \frac{120\text{ s}}{30\text{ s}} = 4.0$$
- • 3. **Interpret Result:**
- A speedup of 4.0 on 4 processors represents **linear speedup** (ideal scaling), where the program finishes in exactly one-fourth of the original time.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Calculates the difference in seconds ($120 - 30 = 90$) rather than the speedup ratio.
✕ **(C):** Multiplies execution times together instead of dividing.
✕ **(D):** Inverts the formula ($30 / 120 = 0.25$); speedup represents the factor of acceleration, which is greater than 1.

Question 64 · Amdahl's Law and the Sequential Bottleneck

Priority ★★★★★

CORRECT ANSWER: (D) **Amdahl's Law, which states that the overall speedup of a program is strictly limited by the portion of the program that must be executed sequentially.**

- Understand Amdahl's Law and the sequential limitation in parallel computing:
- • Almost every real-world program contains some instructions that cannot be parallelized (e.g., reading initial input files, coordinating threads, writing final output, or dependent calculations).
- • **Amdahl's Law:** No matter how many processors N you add (even if $N \to \infty$), the parallel execution time cannot drop below the time required to execute the sequential portion.
- • For example, if 20% of a program must be run sequentially, the absolute maximum theoretical speedup possible is $\frac{1}{0.20} = 5.0\text{x}$, even on a million cores!
- • This sequential bottleneck prevents linear scaling across massive core arrays.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Boyle's Law is a chemistry/physics gas law relating pressure and volume.
✕ **(B):** Metcalfe's Law relates to the telecommunications value of a network proportional to N^2.
✕ **(C):** Moore's Law relates to transistor density doubling roughly every two years, not speedup diminishing returns.

Question 65 · Calculating Parallel Runtime with a Sequential Component

Priority ★★★★★

CORRECT ANSWER: (D) Total Time = **40 seconds | Speedup = **2.5****

- Step-by-step calculation of parallel execution time with sequential constraints:
- • 1. **Deconstruct the execution times:**
- - Sequential portion ($T_{\text{seq-part}}$) = 20 seconds (cannot be accelerated).
- - Parallelizable portion ($T_{\text{par-part}}$) = 80 seconds.
- • 2. **Calculate parallelized duration on 4 cores:**
- $$T_{\text{par-work}} = \frac{80\text{ s}}{4\text{ cores}} = 20\text{ seconds}$$
- • 3. **Calculate total parallel runtime:**
- $$T_{\text{total}} = T_{\text{seq-part}} + T_{\text{par-work}} = 20\text{ s} + 20\text{ s} = 40\text{ seconds}$$
- • 4. **Calculate Speedup:**
- $$\text{Speedup} = \frac{T_{\text{original}}}{T_{\text{total}}} = \frac{100\text{ s}}{40\text{ s}} = 2.5$$
- • Notice that despite using 4 cores, the speedup is only 2.5 (not 4.0) due to the 20-second sequential bottleneck.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Assumes the parallelizable portion takes zero time, which is impossible.
✕ **(B):** Incorrectly assumes the entire 100 seconds can be divided by 4 ($100 / 4 = 25$), ignoring the mandatory sequential portion.
✕ **(C):** Incorrectly subtracts or fails to divide the 80-second portion by 4 cores.

Question 66 · Parallel Overhead and Communication Costs

Priority ★★★★

CORRECT ANSWER: (C) The **parallel overhead (time spent creating threads, allocating memory, and synchronizing communication between cores) exceeded the tiny computational time of the original task.**

- Understand parallel overhead in concurrency design:
- • Parallel computing is not free; managing concurrent execution introduces **overhead**:
- - Spawning and scheduling worker threads/processes.
- - Communicating and transferring data between cores/memory caches.
- - Locking and synchronizing shared resources.
- - Collecting and merging partial results.
- • If the problem size is very small (fine-grained task), the overhead cost vastly outweighs the actual computation time.
- • Parallel computing is only beneficial when the computational workload is large enough to dwarf synchronization overhead.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Running threads cannot erase physical CPU silicon cores.
✕ **(B):** Processors regulate power continuously; power dropouts do not triple execution time.
✕ **(D):** Parallel architectures execute arbitrary instructions including comparisons, arithmetic, and memory operations.

Question 67 · Data Dependencies and Concurrency Hazards

Priority ★★★★

CORRECT ANSWER: (B) There is a strict **data dependency; Step 2 cannot calculate `Y` until Step 1 computes `X`, and Step 3 cannot calculate `Z` until Step 2 computes `Y`.**

- Analyze data dependencies in algorithm parallelization:
- • A **data dependency** occurs when the execution of an instruction relies on the result or output of a preceding instruction.
- • Here, each step consumes the output of the prior step:
- - Step 2 needs X (from Step 1).
- - Step 3 needs Y (from Step 2).
- • If all three steps were run concurrently at the exact same instant, Step 2 and Step 3 would read uninitialized or invalid data.
- • Tasks with deep sequential dependencies must be executed sequentially, or restructured into independent sub-problems.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Variable names have no effect on hardware arithmetic capability.
✕ **(C):** Internet protocols have nothing to do with internal CPU core variable reading.
✕ **(D):** Multiplication by 2 is fully supported by all modern processors in any mode.

Question 68 · Embarrassingly Parallel Problems vs. Inherently Sequential Problems

Priority ★★★★

CORRECT ANSWER: (B) Rendering 10,000 individual, independent pixel colors in a 3D animated movie frame.

- Identify embarrassingly parallel computational tasks:
- • In 3D graphics rendering (ray tracing), calculating the color of pixel (x, y) does not depend on the color of pixel $(x+5, y+10)$.
- • Each pixel can be computed independently and concurrently by thousands of GPU cores simultaneously, achieving near-perfect linear scaling.
- • In contrast, calculating iterative sequences (like Fibonacci numbers or sequential bank ledgers) requires the output of previous steps before subsequent steps can begin, making them inherently sequential.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** A running ledger has strict chronological state dependency across transactions.
✕ **(C):** Fibonacci sequence generation is inherently sequential because each term depends directly on prior terms.
✕ **(D):** Reading pages front-to-back is a physical sequential process.

Question 69 · Parallel Efficiency and Processor Utilization

Priority ★★★★

CORRECT ANSWER: (A) Speedup = 4.0 | Efficiency = **0.50 (50%), meaning each processor is effectively utilized at 50% of its theoretical maximum potential.**

- Calculate and interpret Parallel Efficiency:
- • 1. **Calculate Speedup:**
- $$\text{Speedup} = \frac{T_{\text{seq}}}{T_{\text{par}}} = \frac{80\text{ s}}{20\text{ s}} = 4.0$$
- • 2. **Calculate Efficiency across $N = 8$ processors:**
- $$\text{Efficiency} = \frac{\text{Speedup}}{N} = \frac{4.0}{8} = 0.50 = 50\%$$
- • 3. **Interpretation:**
- If all 8 processors achieved 100% ideal linear speedup, the speedup would be 8.0 (runtime = 10 s). Because the speedup is only 4.0, each processor is delivering half of its potential theoretical boost due to sequential portions and synchronization overhead.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Subtracts 20 from 80 instead of calculating the ratio.
✕ **(C):** Speedup is 4.0, not 1.0.
✕ **(D):** Inverts speedup and yields an impossible efficiency above 100% in standard execution.

Question 70 · Evaluating Algorithmic Speedup Potential Across Candidate Tasks

Priority ★★★★★

CORRECT ANSWER: (D) **Alg Z, because it has the smallest sequential bottleneck (only 2 seconds), allowing 98% of its workload to be divided across the 8 cores.**

- Compare parallel execution runtimes using Amdahl's Law:
- • Calculate total parallel runtime for each on 8 cores ($T_{\text{total}} = T_{\text{seq}} + \frac{T_{\text{par}}}{8}$):
- - **Alg W:** $50 + \frac{50}{8} = 50 + 6.25 = 56.25\text{ s}$ (Speedup: $100 / 56.25 \approx 1.78$)
- - **Alg X:** $30 + \frac{70}{8} = 30 + 8.75 = 38.75\text{ s}$ (Speedup: $100 / 38.75 \approx 2.58$)
- - **Alg Y:** $10 + \frac{90}{8} = 10 + 11.25 = 21.25\text{ s}$ (Speedup: $100 / 21.25 \approx 4.71$)
- - **Alg Z:** $2 + \frac{98}{8} = 2 + 12.25 = 14.25\text{ s}$ (Speedup: $100 / 14.25 \approx 7.02$)
- • **Alg Z** finishes fastest (14.25 seconds) and delivers the highest speedup (7.02x) because minimizing the sequential bottleneck allows the parallel cores to maximize their impact.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** 70 is not a multiple of 8, and Alg X finishes in 38.75 seconds.
✕ **(B):** Alg W finishes in 56.25 seconds, nearly 4 times slower than Alg Z.
✕ **(C):** Different algorithms with different sequential fractions have vastly different runtimes on multi-core systems.

Question 71 · Definition of Distributed Computing

Priority ★★★★★

CORRECT ANSWER: (D) A model in which multiple autonomous computers located in different physical areas communicate across a network to share the workload and collaborate on solving a massive computational problem.

- Define Distributed Computing according to AP CSP guidelines:
- • **Distributed Computing:** A computational architecture where independent computers (nodes) connected via a network coordinate their actions and share data via message passing.
- • Instead of buying an impossibly expensive centralized supercomputer, tasks are divided into thousands of discrete chunks and dispatched across a cluster of networked machines.
- • Each node computes its assigned portion and transmits results back to a coordinator.
- • This architecture enables immense computational power and high geographic scalability.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Multiple peripheral input devices on one machine is not distributed computing.
✕ **(B):** Computers execute logical instructions forward; time reversal is physically impossible.
✕ **(C):** Microfiche is an analog microfilm archiving format, not digital computing.

Question 72 · Key Architectural Differences: Parallel vs. Distributed Computing

Priority ★★★★★

CORRECT ANSWER: (B) Parallel computing typically uses multiple cores sharing common high-speed local memory (RAM) with microscopic nanosecond communication latency, whereas distributed computing connects distinct computers over a network where communication latency is substantially higher (milliseconds).

- Contrast Parallel and Distributed Computing:
- • **Parallel Computing (Single Machine):** Cores reside on the same silicon die or motherboard, communicating via shared memory buses at nanosecond speeds. Communication overhead is very low.
- • **Distributed Computing (Networked Machines):** Nodes are autonomous computers, each with their own separate memory and CPU, communicating via network switches and cables (TCP/IP).
- • Network latency (milliseconds) is orders of magnitude slower than local memory buses (nanoseconds).
- • Therefore, distributed computing is best suited for coarse-grained tasks where the computation time on each node far exceeds the time required to send data over the network.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Both paradigms perform Turing-complete arbitrary mathematical and logical computations.
✕ **(C):** Distributed nodes are physically separated and communicate across local or global networks.
✕ **(D):** Reverses network requirements; distributed computing intrinsically requires network connectivity.

Question 73 · Volunteer Computing and Citizen Science Networks

Priority ★★★★★

CORRECT ANSWER: (B) **Volunteer Computing (or Citizen Science); it enables scientific researchers to harness the aggregate processing power of millions of consumer devices at a fraction of the cost of building a dedicated supercomputer.**

- Examine Volunteer Computing and Citizen Science in AP CSP:
- • **Volunteer Computing:** A form of distributed computing where members of the public donate unused CPU/GPU cycles on their personal computers or smartphones to assist scientific research.
- • Projects like *Folding@home* simulate biomedical disease mechanisms (like Alzheimer's, cancer, and COVID-19).
- • Millions of volunteers form an emergent distributed computing system that rivals or exceeds the world's fastest supercomputers in aggregate floating-point operations.
- • It represents a major positive societal impact of networked computing systems.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Volunteer computing runs digital scientific simulations, not analog fax operations.
✕ **(C):** Phishing is malicious social engineering to steal private credentials, not legitimate scientific research.
✕ **(D):** Bandwidth throttling is intentional ISP speed reduction, not scientific computation.

Question 74 · Cloud Computing and Elastic Infrastructure

Priority ★★★★

CORRECT ANSWER: (A) **Elasticity (On-Demand Scalability), which allows computing resources (virtual servers, storage, bandwidth) to automatically scale up dynamically during peak traffic and scale down when traffic subsides, paying only for resources used.**

- Understand Cloud Computing and Elastic Scalability:
- • **Cloud Computing:** The delivery of on-demand computing services (servers, storage, databases, networking, software) over the Internet.
- • **Elasticity:** The ability of cloud infrastructure to automatically provision and de-provision computing capacity in response to real-time workload fluctuations.
- • Companies avoid massive upfront capital expenditures (buying physical server hardware) and pay only for operational compute-seconds used.
- • This scalability is powered by massive, distributed cloud data centers worldwide.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Cloud computing provides computing infrastructure; it cannot dictate consumer shopping return behaviors.
✕ **(C):** Cloud data centers consume enormous amounts of electricity; cloud services are commercial and paid.
✕ **(D):** Cloud-hosted applications continue to use standard web protocols (HTTP, HTML, CSS, JavaScript).

Question 75 · Malicious Distributed Computing: Botnets and DDoS Attacks

Priority ★★★★★

CORRECT ANSWER: (C) A **Distributed Denial of Service (DDoS) attack.**

- Analyze the mechanics of a Distributed Denial of Service (DDoS) attack:
- • In a **Denial of Service (DoS)** attack, an attacker attempts to render a machine or network resource unavailable to its intended users.
- • In a **Distributed DoS (DDoS)** attack, the traffic flooding the victim originates from thousands of compromised endpoints across the Internet (a botnet).
- • Because the attacking traffic comes from hundreds of thousands of legitimate residential IP addresses globally, it is virtually impossible for the victim's firewall to simply block a single IP.
- • The victim's incoming bandwidth pipe and CPU connection tables are saturated, denying service to real patients and doctors.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** A wiretap passively monitors signals on a physical cable, not distributed flooding.
✕ **(B):** Cryptographic handshakes establish secure encrypted connections, not attack vectors.
✕ **(D):** Brute-force password guessing targets user authentication, not bandwidth flooding.

Question 76 · Scalability Strategies: Horizontal vs. Vertical Scaling

Priority ★★★★

CORRECT ANSWER: (D) Strategy 1 is **Vertical Scaling (Scaling Up), while Strategy 2 is **Horizontal Scaling (Scaling Out)**.**

- Distinguish between Horizontal and Vertical Scaling in distributed systems:
- • **Vertical Scaling (Scale Up):** Adding more power (CPU cores, faster clock speed, larger RAM, faster SSDs) to an existing single machine.
- - Limitation: Eventually hits physical hardware limits and becomes exponentially expensive.
- • **Horizontal Scaling (Scale Out):** Adding more machine nodes to the pool of resources (e.g. 50 commodity servers sharing the database).
- - Advantage: Virtually unlimited theoretical expansion, higher fault tolerance (if 1 server dies, 49 continue running), and cost-effectiveness.
- • Modern large-scale internet services (Google, Amazon, Meta) rely predominantly on horizontal scaling.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Inverts the definitions of vertical and horizontal scaling.
✕ **(B):** Strategy 1 uses 128 cores (multi-core parallel) and Strategy 2 uses 50 networked machines (distributed).
✕ **(C):** Network infrastructure expands continuously through routers and switches.

Question 77 · Fault Recovery and Worker Redundancy in Distributed Systems

Priority ★★★★

CORRECT ANSWER: (D) The coordinator detects that worker #482 stopped sending heartbeat signals, marks its sub-task as incomplete, and reassigns that exact sub-task to an available healthy worker node without restarting the entire 1,000-task job.

- Analyze fault recovery in distributed computing frameworks:
- • In large clusters with thousands of commodity machines, hardware failures (disk crashes, power supply failures, memory errors) are a statistical certainty every single day.
- • Robust distributed systems are designed assuming workers will fail.
- • The coordinator periodically pings workers with heartbeat messages.
- • If worker #482 drops off, only its specific chunk of work is reassigned to another active worker.
- • The other 999 workers continue processing undisturbed, preserving overall computational progress.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Restarting or deleting the whole job would destroy system efficiency in large-scale computing.
✕ **(B):** Distributed frameworks ensure data integrity and refuse to accept corrupted partial data.
✕ **(C):** Network cables cannot transmit high-voltage power to physically reboot burnt-out power supplies.

Question 78 · Data Locality and Network Bandwidth Constraints

Priority ★★★★

CORRECT ANSWER: (A) **Data Locality (Moving Computation to Data): Sending the lightweight analysis code to execute directly on the nodes where the data is already stored, transmitting only the tiny summary results back across the network.**

- Understand Data Locality in distributed architectures:
- • Moving massive volumes of data (terabytes/petabytes) across network switches consumes immense bandwidth and introduces severe transfer bottlenecks.
- • In contrast, a program's executable script or query code is tiny (often just a few kilobytes).
- • **Data Locality:** By distributing the processing code to run locally on the exact machines holding the data chunks, disk reads happen over ultra-fast internal system buses.
- • Only the final aggregated results (a few megabytes) travel across the network, optimizing overall system throughput.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** FM radio bandwidth is far too narrow and unsuited for digital data center computation.
✕ **(C):** Manual paper transcription is physically impossible for 100 Terabytes of data.
✕ **(D):** Information theory dictates that 100 Terabytes cannot be compressed into a single byte while retaining meaningful data.

Question 79 · Societal and Scientific Applications of Distributed Computing

Priority ★★★★

CORRECT ANSWER: (A) It allows molecular docking simulations for millions of candidate chemicals to be calculated simultaneously across thousands of networked computers, shrinking decades of physical lab screening into weeks.

- Explore the scientific impact of distributed computing in biotechnology:
- • Molecular docking is a classic distributed computing application (*in silico* screening).
- • Each candidate drug molecule calculation is computationally independent from all other candidates (embarrassingly parallel).
- • A cluster of 10,000 networked computers can simulate 10,000 distinct chemical compounds simultaneously.
- • Researchers can quickly identify the top 50 most promising molecules, focusing scarce physical laboratory resources exclusively on viable candidates.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Computational drug screening aids medical science; it does not replace medical practitioners.
✕ **(C):** Computers run digital simulations; they do not physically synthesize biochemical compounds.
✕ **(D):** Wi-Fi signals carry digital data, not medicinal treatments.

Question 80 · Matching Computational Tasks with the Optimal Computing Paradigm

Priority ★★★★★

CORRECT ANSWER: (C) Task 1: Sequential Computing | Task 2: Parallel Computing | Task 3: Distributed Computing | Task 4: Malicious Distributed Computing (DDoS)

- Analyze and classify computational paradigms:
- • **Task 1 is Sequential Computing:** Strict chronological dependencies between ledger transactions prevent concurrent execution.
- • **Task 2 is Parallel Computing:** Thousands of independent pixel calculations run concurrently across GPU cores in a single shared-memory machine.
- • **Task 3 is Distributed Computing:** 50 Petabytes exceeds single-machine capacity; the workload is partitioned across 5,000 networked cluster servers.
- • **Task 4 is Malicious Distributed Computing (DDoS):** Thousands of distributed compromised nodes coordinate over the internet to overwhelm a victim.
- • This comprehensive classification reflects core AP CSP Big Idea 4 learning objectives.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Uses extraneous buzzwords unrelated to the fundamental AP CSP curriculum.
✕ **(B):** Misclassifies dependent ledger tracking as distributed and single-machine pixel rendering as sequential.
✕ **(D):** Scrambles the operational constraints of the four tasks.

Question 81 · Symmetric Encryption and the Key Distribution Problem

Priority ★★★★★

CORRECT ANSWER: (B) The sender and receiver use the exact same secret key to both encrypt and decrypt data; the primary challenge is securely transmitting that secret key to the receiver without an eavesdropper intercepting it.

- Analyze Symmetric Key Encryption and its historical limitation:
- • **Symmetric Encryption (e.g. AES):** A single cryptographic key is shared between both parties. Alice encrypts the plaintext with Key K, and Bob decrypts the ciphertext with the identical Key K.
- • **Advantage:** Extremely fast computationally, ideal for encrypting gigabytes of data.
- • **The Key Distribution Problem:** If Alice and Bob have never met in person and communicate across an untrusted, public Internet, how can Alice send Key K to Bob without an eavesdropper intercepting the key and decrypting all future messages?
- • This fundamental vulnerability led to the invention of public-key (asymmetric) cryptography.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Encryption algorithms are purely mathematical and execute on standard hardware over any media.
✕ **(C):** Cryptographic algorithms operate on raw binary bits, indifferent to letter casing.
✕ **(D):** Using separate public and private keys describes asymmetric cryptography, not symmetric encryption.

Question 82 · Architecture of Asymmetric (Public Key) Cryptography

Priority ★★★★★

CORRECT ANSWER: (B) Each user generates a mathematically linked key pair: a **Public Key that can be freely shared with the world, and a **Private Key** that is kept strictly secret and never shared.**

- Define Asymmetric (Public Key) Cryptography in AP CSP:
- • Asymmetric systems generate two mathematically related keys:
- - **Public Key:** Made publicly accessible to anyone. Used by senders to encrypt messages intended for that recipient.
- - **Private Key:** Kept confidential and secure by the owner on their local device. Used to decrypt messages that were encrypted with the corresponding public key.
- • **One-way Trapdoor Property:** Data encrypted with the public key can *only* be decrypted using the private key. Even the person who encrypted the message cannot decrypt it without the private key!
- • This completely eliminates the need to transmit private secret keys across untrusted networks.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Public key cryptography operates entirely digitally over electronic networks.
✕ **(C):** Using a single default password creates massive vulnerability and is not asymmetric cryptography.
✕ **(D):** If a single key were published openly, anyone could decrypt all messages, destroying confidentiality.

Question 83 · Asymmetric Encryption Workflow: Alice Sending to Bob

Priority ★★★★★

CORRECT ANSWER: (D) Alice encrypts the document using **Bob's Public Key; Bob decrypts the document using **Bob's Private Key**.**

- Trace the fundamental confidentiality workflow in asymmetric encryption:
- • 1. **Sender's Action:** Alice wants only Bob to read the document. She retrieves **Bob's Public Key** (which Bob made available to everyone) and encrypts the document.
- • 2. **Transit:** The encrypted ciphertext travels across routers on the Internet. Even if eavesdroppers intercept the packets, they cannot decrypt it.
- • 3. **Recipient's Action:** Upon arrival, Bob uses **Bob's Private Key** (which only Bob possesses) to decrypt the ciphertext back into plaintext.
- • Key rule to memorize for AP CSP: *Encrypt with the recipient's public key; decrypt with the recipient's private key.*

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Data encrypted with Alice's public key can only be decrypted by Alice's private key, not Bob.
✕ **(B):** Alice does not have access to Bob's private key; private keys are never shared.
✕ **(C):** Encrypting with Alice's private key allows anyone with Alice's public key to decrypt it (this is a digital signature, which provides authenticity, not confidentiality).

Question 84 · Digital Signatures and Message Integrity

Priority ★★★★★

CORRECT ANSWER: (A) The vendor encrypts a hash of the software using the **vendor's Private Key; client computers decrypt and verify the signature using the **vendor's Public Key**.**

- Analyze the mechanics of Digital Signatures:
- • **Generation:** The author hashes the file and encrypts the hash using their own **Private Key** to create a Digital Signature.
- • **Verification:** Anyone can decrypt the signature using the author's **Public Key** and compare the resulting hash to a fresh hash of the downloaded file.
- • **Guarantees:**
- - **Authentication / Non-repudiation:** Proves beyond doubt that the file came from the owner of the private key.
- - **Integrity:** Proves that not a single bit of the file was modified or corrupted in transit.
- • If an attacker alters even one bit, the decrypted hash will not match, and the operating system rejects the update.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Digital signatures are mathematical cryptographic algorithms, not physical ink markings.
✕ **(C):** Private keys must never be shared; publishing a private key compromises the vendor's security permanently.
✕ **(D):** Software executables are binary programs, not audio files.

Question 85 · Man-in-the-Middle (MitM) Attacks and PKI Trust Verification

Priority ★★★★★

CORRECT ANSWER: (C) A **Man-in-the-Middle (MitM) attack; prevented because browsers verify the bank's digital certificate signed by a trusted **Certificate Authority (CA)** to ensure the public key belongs to the real bank.**

- Examine Man-in-the-Middle (MitM) attacks and the necessity of Certificate Authorities:
- • In a **Man-in-the-Middle attack**, an adversary transparently intercepts and relays communications between two parties who believe they are talking directly to each other.
- • While asymmetric encryption secures data in transit, it does NOT inherently verify *who owns* the public key.
- • **The Solution (PKI / CAs):** A trusted Certificate Authority signs a digital certificate that binds the bank's real identity (`bank.com`) to its public key.
- • Browsers check this cryptographic signature against pre-installed root certificates. If an attacker presents a fraudulent key, the browser triggers a prominent security warning and blocks the connection.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** The attack covertly eavesdrops and alters credentials; it is not a volumetric DoS outage.
✕ **(B):** The attack occurs across network data packets, not via physical hardware burglary.
✕ **(D):** Display brightness has zero effect on cryptographic packet verification.

Question 86 · Rogue Wi-Fi Access Points and Evil Twin Attacks

Priority ★★★★

CORRECT ANSWER: (B) The attacker can monitor and capture all unencrypted network traffic (HTTP requests, DNS lookups, and cleartext credentials) transmitted by the laptop.

- Analyze Rogue Access Points (Evil Twin hotspots) in cybersecurity:
- • A **Rogue Access Point** is an unauthorized wireless access point configured to mimic a legitimate network to lure unsuspecting users into connecting.
- • All user packets pass through the attacker's hardware before reaching the Internet.
- • **Threats:** The attacker can run packet sniffers (Wireshark) to read unencrypted traffic, log DNS queries to see which websites the user visits, and inject spoofed web pages.
- • **Defense:** Avoid open untrusted Wi-Fi, enforce HTTPS everywhere, and utilize a Virtual Private Network (VPN).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Wi-Fi signals operate under strict low-power FCC limits and cannot cause battery explosions.
✕ **(C):** Connecting to a rogue hotspot does not disable future wireless networking capability.
✕ **(D):** Network access points do not have access to trigger unauthorized physical print jobs of private SSD files.

Question 87 · Role and Operation of Virtual Private Networks (VPNs)

Priority ★★★★★

CORRECT ANSWER: (B) It establishes an encrypted tunnel between the employee's device and the VPN server, preventing anyone on the local Wi-Fi network or the local ISP from inspecting the contents or destinations of their web traffic.

- Understand the function and security benefits of a Virtual Private Network (VPN):
- • A **VPN** encapsulates and encrypts all network packets leaving the client device.
- • When connected to an untrusted public Wi-Fi network, anyone snooping on the local wireless airwaves (or the coffee shop's ISP) only sees an unintelligible stream of encrypted packets travelling to a single VPN server IP address.
- • The VPN server decrypts the traffic and forwards it to the actual Internet destination, and vice versa.
- • A VPN conceals both the payload data and the specific websites being visited from local eavesdroppers and ISPs.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** A VPN protects digital network packets; it does not render physical hardware invisible.
✕ **(C):** Software VPNs do not alter or replace physical coffee shop routers.
✕ **(D):** A VPN is a networking protocol suite; it does not disable user keyboard input.

Question 88 · Multi-Factor Authentication (MFA) Principles

Priority ★★★★

CORRECT ANSWER: (A) **Something you know (the password) and **Something you have** (the registered smartphone).**

- Classify authentication factors according to cybersecurity standards:
- • Standard Authentication Factors:
- - 1. **Knowledge (Something you know):** Passwords, PINs, security passphrases.
- - 2. **Possession (Something you have):** Physical smartphones, hardware security keys (YubiKey), smart cards.
- - 3. **Inherence (Something you are):** Biometrics (fingerprints, facial recognition, iris scans).
- • In this university scenario, entering a password is *Something you know*, and confirming on a physical smartphone is *Something you have*.
- • Using two different *categories* ensures that if a password is stolen in a remote data breach, the attacker cannot log in without physically possessing the victim's phone.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** A password and a security question are both 'something you know', which represents two-step single-factor authentication, not true multi-factor.
✕ **(C):** Food and clothing are not digital authentication factors.
✕ **(D):** No biometrics (fingerprint/face) or physical geographic address verification was used in the prompt.

Question 89 · Hybrid Encryption Architecture in TLS/HTTPS

Priority ★★★★

CORRECT ANSWER: (D) It uses **asymmetric encryption during the initial handshake to securely exchange a temporary secret session key, and then switches to fast **symmetric encryption** using that session key to encrypt all subsequent bulk web traffic.**

- Understand Hybrid Encryption in TLS/HTTPS:
- • **The Dilemma:** Symmetric encryption is ultra-fast but has a key exchange problem. Asymmetric encryption solves the key exchange problem but is computationally expensive and slow for large files.
- • **The Hybrid Solution (TLS Handshake):**
- - Step 1: Client and server use **asymmetric encryption** (public/private keys) to authenticate each other and securely agree upon a shared, temporary **session key**.
- - Step 2: Once the session key is established, all subsequent web pages, images, and data streams are encrypted using ultra-fast **symmetric encryption** (AES) using that session key.
- • This gives the Internet the best of both worlds: rock-solid key security with near-instant transfer speeds.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Cryptographic computations are executed autonomously by computer hardware, not manual human calculation.
✕ **(B):** HTTPS encrypts the entire HTTP payload, including all text, images, scripts, and styling.
✕ **(C):** Encryption protocols switch dynamically within milliseconds during connection setup, not by calendar days.

Question 90 · Comprehensive Cybersecurity Techniques and Guarantees Synthesis

Priority ★★★★★

CORRECT ANSWER: (C) All four technologies work synergistically across different layers to provide comprehensive confidentiality, integrity, authentication, and access control across modern computer systems.

- Synthesize core cybersecurity principles in AP CSP Big Idea 4:
- • Modern computer network security requires **defense in depth**:
- - **Asymmetric Encryption:** Solves secure key exchange and confidentiality.
- - **Digital Certificates & CAs:** Provide identity verification and trust authentication.
- - **VPNs:** Shield network packets from snooping on local Wi-Fi and ISP backbones.
- - **MFA:** Enforces robust user access control.
- • Together, these layered security mechanisms protect billions of users daily across the global Internet.
- • This synthesis question solidifies the 90 Multiple Choice Questions covering all topics in Unit 4.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** All four technologies are open, international industry standards recognized and recommended worldwide.
✕ **(B):** A VPN encrypts network transit; it cannot protect a user who willingly gives their password away to scammers.
✕ **(D):** Digital certificates are cryptographic credentials; they do not prevent physical hard drive hardware wear.

Question 91 · Model Solution & Scoring Rubric

Network Packet Inspection & Firewall Filter ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

• **Program Purpose:** The purpose of `NetGuard` is to protect local school networks from cyberattacks by autonomously inspecting packet headers and blocking unauthorized or hostile incoming network traffic.

• **Program Function:** A network administrator inputs raw packet logs or observes live network capture streams. The program displays an interactive dashboard showing the total packets processed, the number of dropped malicious packets, and real-time alert logs.

Part B: Data Abstraction & Complexity Management

• **List Identified:** The list is `incomingPackets`, which stores structured packet objects containing fields: `sourceAddress`, `destPort`, `packetSize`, and `payload`.

• **Managing Complexity:** Using `incomingPackets` allows the system to dynamically process an arbitrary number of network packets (from 10 to 100,000) using a single standardized data structure. Without this list, the programmer would need to define individual variables for every single packet (`packet1_ip`, `packet2_ip`, etc.), which would make dynamic iteration, sorting, and live continuous streaming impossible.

Part C: Managing Complexity & Algorithmic Procedure

• **Procedure Header & Parameters:** `filterPacketStream(packetList, blacklistedPrefixes)` takes the list of incoming packet objects and a list of blacklisted network prefixes.

• **Algorithmic Logic:**

1. Initialize an empty list `approvedPackets = []`.
2. `FOR EACH packet IN packetList:` (Iteration)

a. Set `isMalicious = false`.

b. `FOR EACH prefix IN blacklistedPrefixes:`

- `IF packet.sourceAddress.startsWith(prefix) OR packet.destPort == 23 THEN` (Selection)
- Set `isMalicious = true`

c. `IF isMalicious == false THEN`

- Append `packet` to `approvedPackets`

d. `ELSE:`

- Log security alert with packet metadata.

3. Return `approvedPackets` (Sequencing and return value).

Part D: Testing, Edge Cases & Verification

• **Test Call 1 (Legitimate Traffic):**

- *Arguments:* `filterPacketStream([Packet('192.168.1.50', port=443)], ['10.0.0.', '172.16.'])`
- *Condition Tested:* Evaluates a secure HTTPS packet on port 443 with a clean IP address not in the blacklist.
- *Outcome:* `isMalicious` remains `false`; packet is appended to `approvedPackets` and returned successfully.

• **Test Call 2 (Blocked Attack Traffic):**

- *Arguments:* `filterPacketStream([Packet('10.0.0.99', port=23)], ['10.0.0.', '172.16.'])`
- *Condition Tested:* Evaluates an insecure Telnet packet on port 23 matching a blacklisted subnet prefix.
- *Outcome:* `isMalicious` evaluates to `true`; packet is dropped and a security alert is recorded; returns empty list.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Clearly describes the overall real-world purpose of the program (securing a network by filtering malicious packets).	**1 pt**
Row 2: Program Function: Accurately describes how the program operates from an input/output perspective during execution.	**1 pt**
Row 3: Data Abstraction Identified: Explicitly names a list (e.g., `incomingPackets` or `blacklistedSubnets`) and states the type of data stored inside it.	**1 pt**
Row 4: Complexity Management Explained: Explains in detail how the list manages complexity, contrasting it with maintaining hundreds of individual scalar variables.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents a student-developed procedure with at least two explicit parameters that impact the procedure's logic.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Demonstrates sequencing, conditional `IF/ELSE` branching, and a `FOR EACH` or `WHILE` loop iterating over the collection.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Explains step-by-step how the procedure evaluates header IP prefixes and port rules to accept or drop packets.	**1 pt**
Row 8: Distinct Test Calls Provided: Specifies two distinct test calls with concrete, differing argument values.	**1 pt**
Row 9: Conditions & Outcomes Detailed: Explains what specific conditions were triggered (e.g., blacklisted IP drop vs legitimate port pass) and the exact expected output.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 92 · Model Solution & Scoring Rubric

Dynamic Routing & Fault-Tolerant Path Optimizer ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

• **Program Purpose:** `RouteMesh` provides an automated routing control program that maintains reliable network communication by calculating lowest-latency paths and routing around broken links.

• **Program Function:** The user selects a source and destination router. The program displays the current active path. When a simulated cable is cut, the program automatically reroutes traffic along redundant nodes within milliseconds.

Part B: Data Abstraction & Complexity Management

• **List Identified:** `networkGraph`, an adjacency list storing router records where each router holds a list of connected neighbors and link latencies in milliseconds.

• **Managing Complexity:** By using a nested list structure, any arbitrary network topology (with 5 or 500 routers) can be traversed with a single generalized graph search algorithm. Without it, hardcoded branching for every possible route combination would require thousands of messy `IF/ELSE` statements.

Part C: Managing Complexity & Algorithmic Procedure

• **Procedure Header & Parameters:** `findOptimalPath(networkGraph, startNode, endNode, failedLinks)` takes the network map, endpoints, and a list of dead links.

• **Algorithmic Logic:**

1. Initialize `visitedNodes = []`, `priorityQueue = [[startNode, 0, [startNode]]]`.

2. `WHILE priorityQueue IS NOT EMPTY:` (Iteration)

a. Extract path with lowest cumulative latency.

b. `IF currentNode == endNode THEN` return path (Selection).

c. `FOR EACH neighbor IN networkGraph[currentNode]:`

- `IF [currentNode, neighbor] NOT IN failedLinks AND neighbor NOT IN visitedNodes THEN`

- Calculate new latency and append neighbor to candidate path queue.

3. Return `"NO_PATH_AVAILABLE"` if destination cannot be reached.

Part D: Testing, Edge Cases & Verification

• **Test Call 1 (Bypassing Failed Link):**

- *Arguments:* `findOptimalPath(meshNet, 'NodeA', 'NodeD', [['NodeA', 'NodeB']])`

- *Condition Tested:* Primary direct link `A-B` is severed; checks if redundant route `A → C → D` is chosen.

- *Outcome:* Procedure skips severed edge and returns `['NodeA', 'NodeC', 'NodeD']` with total latency 35 ms.

• **Test Call 2 (Total Network Partition):**

- *Arguments:* `findOptimalPath(meshNet, 'NodeA', 'NodeD', [['NodeA', 'NodeB'], ['NodeA', 'NodeC']])`

- *Condition Tested:* All outgoing links from `NodeA` are in `failedLinks`.

- *Outcome:* Queue empties without finding a path; procedure returns `"NO_PATH_AVAILABLE"`.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: simulating fault-tolerant dynamic routing and path optimization.	**1 pt**
Row 2: Program Function: Explains runtime behavior: recalculating paths dynamically upon link drops.	**1 pt**
Row 3: Data Abstraction Identified: Identifies list/dictionary `networkGraph` storing adjacency edges and link weights.	**1 pt**
Row 4: Complexity Management Explained: Explains how the graph list abstracts node connectivity without hardcoded variables.	**1 pt**
Row 5: Procedure Identified & Parameterized: Provides `findOptimalPath` with start/end nodes and failed link parameters.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Includes queue/loop iteration, edge validation conditionals, and sequential updates.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Step-by-step path exploration logic bypassing severed links.	**1 pt**
Row 8: Distinct Test Calls Provided: Two test cases with differing link failure arguments.	**1 pt**
Row 9: Conditions & Outcomes Detailed: Contrasts successful alternative path selection with disconnected error alert.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 93 · Model Solution & Scoring Rubric

DNS Cache Resolver & TTL Invalidation Manager ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

• **Program Purpose:** `SmartResolver` speeds up web browsing and conserves network bandwidth by serving domain-to-IP resolutions from a local memory cache before querying global DNS servers.

• **Program Function:** A user enters a website URL. The program searches its cache table. If a valid record exists, it immediately returns the numerical IP; otherwise, it queries the authoritative server and caches the new record.

Part B: Data Abstraction & Complexity Management

• **List Identified:** `dnsCacheTable`, a list of records where each entry is a list containing `[domainName, resolvedIP, expirationTimestamp]`.

• **Managing Complexity:** Storing DNS entries in a list allows the program to add, update, and evict an arbitrary number of domain mappings dynamically. Without this list, every single website on the Internet would require three dedicated scalar variables, making program maintenance impossible.

Part C: Managing Complexity & Algorithmic Procedure

• **Procedure Header & Parameters:** `resolveDomain(queryDomain, currentTime, dnsCacheTable)` takes the target domain, current system time, and the cache list.

• **Algorithmic Logic:**

1. `FOR EACH entry IN dnsCacheTable:` (Iteration)

a. `IF entry[0] == queryDomain THEN` (Selection - Match)

- `IF currentTime < entry[2] THEN` (Selection - TTL Valid)

- Return entry[1] (Cache Hit - immediate return)

- `ELSE:`

- Remove `entry` from `dnsCacheTable` (Evict stale record).

2. Set `newIP = queryAuthoritativeServer(queryDomain)` (Sequencing - Cache Miss).

3. Append `[queryDomain, newIP, currentTime + 3600]` to `dnsCacheTable`.

4. Return `newIP`.

Part D: Testing, Edge Cases & Verification

• **Test Call 1 (Fresh Cache Hit):**

- *Arguments:* `resolveDomain('collegeboard.org', 1000, [['collegeboard.org', '198.51.100.10', 2500]])`

- *Condition Tested:* `currentTime (1000)` is less than `expiration (2500)`.

- *Outcome:* Returns `'198.51.100.10'` immediately in <1 ms without contacting external root servers.

• **Test Call 2 (Expired TTL - Cache Miss):**

- *Arguments:* `resolveDomain('collegeboard.org', 3000, [['collegeboard.org', '198.51.100.10', 2500]])`

- *Condition Tested:* `currentTime (3000)` exceeds `expiration (2500)`; record is stale.

- *Outcome:* Stale record is evicted; authoritative server is queried; returns newly fetched IP and updates cache.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: accelerating DNS resolution and reducing upstream server load.	**1 pt**
Row 2: Program Function: Describes lookup behavior from client request to IP return.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `dnsCacheTable` list storing domain, IP, and expiration timestamp.	**1 pt**
Row 4: Complexity Management Explained: Explains complexity management: maintaining dynamic lookup tables without hardcoded variables.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `resolveDomain` with domain, time, and cache list parameters.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses iteration across cache, selection checking domain match and TTL validity.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Explains cache hit vs cache miss logic and eviction.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (fresh cache hit vs expired record).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Explains conditional branching outcomes and resulting IP addresses.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 94 · Model Solution & Scoring Rubric

Parallel Batch Processing & Speedup Simulator ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

• **Program Purpose:** `ParallelSim` helps software developers optimize multithreaded applications by predicting runtime reductions and demonstrating how sequential bottlenecks limit multi-core speedup.

• **Program Function:** The user inputs a list of task durations and selects a core count. The program calculates and displays sequential time, parallel time, speedup ratio, and core utilization efficiency.

Part B: Data Abstraction & Complexity Management

• **List Identified:** `taskDurationList`, which contains numerical values representing the duration in seconds for each independent data processing sub-task.

• **Managing Complexity:** Using `taskDurationList` enables the program to aggregate any batch size (whether 5 tasks or 5,000 tasks) with a single iterative loop. Without this list, separate code would be required for every individual task, preventing dynamic batch benchmarking.

Part C: Managing Complexity & Algorithmic Procedure

• **Procedure Header & Parameters:** `calculateParallelMetrics(taskDurationList, seqFraction, coreCount)` takes the list of task times, fraction that cannot be parallelized, and number of cores.

• **Algorithmic Logic:**

1. Set `totalWork = 0`.
2. `FOR EACH task IN taskDurationList:` (Iteration)
- `totalWork = totalWork + task`
3. Set `tSeq = totalWork * seqFraction` (Sequential component).
4. Set `tPar = (totalWork * (1 - seqFraction)) / coreCount` (Parallel component).
5. Set `totalParTime = tSeq + tPar`.
6. `IF totalParTime > 0 THEN` (Selection - Guard against zero division)
- Set `speedup = totalWork / totalParTime`
- Set `efficiency = speedup / coreCount`
7. Return `[totalParTime, speedup, efficiency]`.

Part D: Testing, Edge Cases & Verification

• **Test Call 1 (Single Core Baseline):**
- *Arguments:* `calculateParallelMetrics([10, 20, 30, 40], 0.20, 1)`
- *Condition Tested:* `totalWork = 100`, `coreCount = 1`.
- *Outcome:* `totalParTime = 20 + 80 = 100` seconds; `speedup = 1.0`; `efficiency = 100%`.

• **Test Call 2 (8 Cores with 20% Sequential Bottleneck):**
- *Arguments:* `calculateParallelMetrics([10, 20, 30, 40], 0.20, 8)`
- *Condition Tested:* `coreCount = 8`; parallel portion divided by 8 ($80 / 8 = 10$).
- *Outcome:* `totalParTime = 20 + 10 = 30` seconds; `speedup = 100 / 30 = 3.33x`; `efficiency = 3.33 / 8 = 41.6%`.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: analyzing parallel computing speedup and Amdahl's Law limitations.	**1 pt**
Row 2: Program Function: Describes simulation input of tasks and output of runtime and speedup metrics.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `taskDurationList` storing numerical durations in seconds.	**1 pt**
Row 4: Complexity Management Explained: Explains how the list permits summing and partitioning arbitrary task counts.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `calculateParallelMetrics` with tasks, sequential fraction, and core count.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses loop to accumulate total work, selection for zero-division guard, and metric sequencing.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Accurately applies speedup and runtime formulas with sequential components.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (single-core baseline vs multi-core).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Details mathematical calculations and resulting speedup numbers.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 95 · Model Solution & Scoring Rubric

Distributed Sensor Data Aggregation & Volunteer Network ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

- **Program Purpose:** `EcoGrid` crowdsources environmental telemetry from thousands of volunteer citizen sensors to generate high-resolution real-time air pollution maps without expensive government hardware.
- **Program Function:** Volunteer devices transmit periodic sensor packets over the Internet. The coordinator node filters out inactive sensors, computes regional particulate averages, and publishes public health safety alerts.

Part B: Data Abstraction & Complexity Management

- **List Identified:** `sensorNodesList`, a list of sensor objects where each entry contains `[sensorID, latitude, longitude, pm25Value, lastHeartbeat]`.
- **Managing Complexity:** Using `sensorNodesList` allows the system to process incoming data from any number of citizen devices as they connect or disconnect dynamically. Without this list, thousands of separate variables would have to be manually declared and updated, making real-world distributed scaling impossible.

Part C: Managing Complexity & Algorithmic Procedure

- **Procedure Header & Parameters:** `aggregateAirQuality(sensorNodesList, maxAgeSeconds, currentTimestamp)` takes the sensor list, timeout limit, and current time.
- **Algorithmic Logic:**

1. Initialize `validSum = 0`, `validCount = 0`.
2. `FOR EACH sensor IN sensorNodesList:` (Iteration)

a. `IF (currentTimestamp - sensor.lastHeartbeat) <= maxAgeSeconds THEN` (Selection - Node is active)

- `validSum = validSum + sensor.pm25Value`
- `validCount = validCount + 1`

b. `ELSE:`

- Log warning that sensor node is offline/dead.

3. `IF validCount > 0 THEN` (Selection - Guard against empty data)

- Return `validSum / validCount`

4. `ELSE:` return `"NO_ACTIVE_SENSORS"`.

Part D: Testing, Edge Cases & Verification

- **Test Call 1 (All Active Sensors):**

- *Arguments:* `aggregateAirQuality([Sensor('S1', pm=12, time=500), Sensor('S2', pm=18, time=490)], 60, 510)`
- *Condition Tested:* Both sensors checked in within 60 seconds of time 510 ($510 - 500 = 10 \le 60$, $510 - 490 = 20 \le 60$).
- *Outcome:* Both readings included; returns average PM2.5 index: $(12 + 18) / 2 = 15.0$ $\mu\text{g/m}^3$.

- **Test Call 2 (Filtering Out Dead Sensor):**

- *Arguments:* `aggregateAirQuality([Sensor('S1', pm=12, time=500), Sensor('S2', pm=90, time=300)], 60, 510)`
- *Condition Tested:* Sensor S2 last pinged at time 300 ($510 - 300 = 210 > 60$); exceeds `maxAgeSeconds`.
- *Outcome:* S2 is flagged dead and ignored; returns only S1 reading: 12.0 $\mu\text{g/m}^3$.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: distributed volunteer environmental data collection and air quality calculation.	**1 pt**
Row 2: Program Function: Describes volunteer telemetry intake and public air index reporting.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `sensorNodesList` holding node ID, PM2.5 reading, and last ping timestamp.	**1 pt**
Row 4: Complexity Management Explained: Explains how the collection abstracts thousands of remote sensor endpoints.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `aggregateAirQuality` with sensors, time threshold, and current time.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Demonstrates looping through sensors, conditional timestamp checks, and statistical math.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Detailed explanation of valid sensor selection and mean calculation.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (all healthy sensors vs stale sensor presence).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Details expected mathematical index results under varying node health.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 96 · Model Solution & Scoring Rubric

Symmetric vs Asymmetric Hybrid Key Exchange Module ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

- **Program Purpose:** `CryptoVault` provides an end-to-end encrypted messaging engine that combines asymmetric security (for key exchange) with symmetric speed (for bulk data transfer) while preventing Man-in-the-Middle attacks.
- **Program Function:** When a client initiates contact, the program validates the server's digital certificate against trusted authorities, encrypts a newly generated AES session key using the server's public key, and returns the encrypted session token.

Part B: Data Abstraction & Complexity Management

- **List Identified:** `trustedCertificatesList`, which stores authorized Certificate Authority (CA) root signatures and accredited domain hashes.
- **Managing Complexity:** Using `trustedCertificatesList` allows the client application to dynamically update, add, or revoke trust in hundreds of global certification authorities using a standardized loop. Without this list, every single trust validation would require separate hardcoded conditional checks.

Part C: Managing Complexity & Algorithmic Procedure

- **Procedure Header & Parameters:** `negotiateSecureSession(clientHello, serverCert, trustedCertificatesList)` takes client parameters, server certificate, and the trusted CA list.
- **Algorithmic Logic:**

1. Set `isTrustedCA = false`.
2. `FOR EACH ca IN trustedCertificatesList:` (Iteration)
- `IF serverCert.issuer == ca.name AND verifySignature(serverCert, ca.publicKey) THEN` (Selection)
- `isTrustedCA = true`
3. `IF isTrustedCA == true THEN`
- Generate random 256-bit symmetric key `sessionKey`.
- Set `encryptedKey = rsaEncrypt(sessionKey, serverCert.publicKey)`.
- Return `['SUCCESS', encryptedKey, sessionKey]`.
4. `ELSE:`
- Return `['ERROR_UNTRUSTED_CERTIFICATE', null, null]`.

Part D: Testing, Edge Cases & Verification

- **Test Call 1 (Legitimate Server Certificate):**
- *Arguments:* `negotiateSecureSession(hello, Cert('bank.com', issuer='DigiCert'), [CA('DigiCert')])`
- *Condition Tested:* Certificate issuer matches trusted CA list and cryptographic signature is valid.
- *Outcome:* Returns `['SUCCESS', encryptedKey, sessionKey]`; secure encrypted communication proceeds.
- **Test Call 2 (Untrusted / Forged Certificate):**
- *Arguments:* `negotiateSecureSession(hello, Cert('bank.com', issuer='EvilAttackerCA'), [CA('DigiCert')])`
- *Condition Tested:* Certificate issuer is NOT present in `trustedCertificatesList`.
- *Outcome:* `isTrustedCA` remains `false`; returns `['ERROR_UNTRUSTED_CERTIFICATE']` and closes connection.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: establishing confidential encrypted channels using hybrid cryptography.	**1 pt**
Row 2: Program Function: Describes handshake verification and transition to symmetric session encryption.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `trustedCertificatesList` holding CA root certificates and public keys.	**1 pt**
Row 4: Complexity Management Explained: Explains how the list manages root authority validation without manual code edits.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `negotiateSecureSession` with client hello, server certificate, and trust list.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses iteration across trust list, selection for signature validity, and key generation sequencing.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Detailed explanation of asymmetric encryption of symmetric session key.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (trusted CA certificate vs untrusted certificate).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Contrasts successful session key issuance with security exception drop.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 97 · Model Solution & Scoring Rubric

Network Latency, Bandwidth & Bottleneck Analyzer ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

- **Program Purpose:** `NetPathSim` diagnoses network performance degradation by identifying link bottlenecks along multi-hop pathways and estimating total file transfer durations.
- **Program Function:** A network engineer inputs the sequential router hops and the file size to be transferred. The program outputs the bottleneck link, total path latency, and expected transfer time in seconds.

Part B: Data Abstraction & Complexity Management

- **List Identified:** `hopList`, where each element is a list representing a network link: `[hopName, bandwidthMbps, latencyMs]`.
- **Managing Complexity:** Using `hopList` allows the simulation to evaluate paths with any number of hops (from 2 local switches to 30 global routers) using a single loop. Without this list, a developer would have to write custom sequential code for every distinct hop length.

Part C: Managing Complexity & Algorithmic Procedure

- **Procedure Header & Parameters:** `calculateTransferTime(hopList, fileSizeBytes)` takes the list of hop objects and the file size in Bytes.
- **Algorithmic Logic:**

1. Set `totalLatencyMs = 0`, `bottleneckBandwidth = hopList[0].bandwidthMbps`.
2. `FOR EACH hop IN hopList:` (Iteration)
- `totalLatencyMs = totalLatencyMs + hop.latencyMs`
- `IF hop.bandwidthMbps < bottleneckBandwidth THEN` (Selection - Find bottleneck)
- `bottleneckBandwidth = hop.bandwidthMbps`
3. Convert file size to Megabits: `fileSizeMegabits = (fileSizeBytes * 8) / 1000000`.
4. Calculate transmission duration: `transmissionSeconds = fileSizeMegabits / bottleneckBandwidth`.
5. Set `totalTimeSeconds = (totalLatencyMs / 1000) + transmissionSeconds`.
6. Return `[bottleneckBandwidth, totalTimeSeconds]`.

Part D: Testing, Edge Cases & Verification

- **Test Call 1 (Small 100-Byte Ping Packet):**
- *Arguments:* `calculateTransferTime([[Hop('H1', 1000, 20), Hop('H2', 10, 30)], 100)`
- *Condition Tested:* File size is negligible; latency ($20 + 30 = 50\text{ ms} = 0.05\text{ s}$) dominates.
- *Outcome:* Bottleneck = 10 Mbps; total time $\approx 0.05008\text{ s}$; confirms latency governs small packet speed.
- **Test Call 2 (Large 100-Megabyte File):**
- *Arguments:* `calculateTransferTime([[Hop('H1', 1000, 20), Hop('H2', 10, 30)], 100000000)`
- *Condition Tested:* Large file (800 Mb); bottleneck bandwidth (10 Mbps) dominates ($800 / 10 = 80\text{ s}$).
- *Outcome:* Total time $= 0.05 + 80 = 80.05\text{ s}$; confirms narrowest link determines bulk transfer rate.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: analyzing network path bottlenecks and estimating end-to-end transfer times.	**1 pt**
Row 2: Program Function: Describes calculation of transit times from hop latency and bandwidth inputs.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `hopList` holding bandwidth (Mbps) and latency (ms) for each link.	**1 pt**
Row 4: Complexity Management Explained: Explains how the list allows traversing paths of any hop count cleanly.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `calculateTransferTime` with hop list and file size in Bytes.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses iteration across hops, selection finding minimum bandwidth, and unit conversion math.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Step-by-step logic explaining $T = \text{latency} + (\text{bits} / \text{bottleneck})$.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (small ping payload vs large multi-megabyte file).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Shows exact mathematical calculations and time differences.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 98 · Model Solution & Scoring Rubric

DDoS Traffic Detection & Rate Limiter ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

- **Program Purpose:** `DefendDDoS` shields web applications from denial-of-service outages by dynamically detecting and blocking automated botnet traffic without denying service to legitimate human visitors.
- **Program Function:** As web requests arrive, the program checks the requester's IP history over the past 60 seconds. If the request volume exceeds the safety threshold, the program temporarily drops the traffic and returns an HTTP 429 error.

Part B: Data Abstraction & Complexity Management

- **List Identified:** `ipRequestLog`, a list of records where each entry contains `[clientIP, timestampList]` tracking the history of arrival times for that specific IP.
- **Managing Complexity:** Using `ipRequestLog` allows the rate limiter to scale to track any number of visiting IP addresses concurrently. Without this list, a developer would have to hardcode individual tracking variables for specific IP addresses, which is impossible on the open Internet.

Part C: Managing Complexity & Algorithmic Procedure

- **Procedure Header & Parameters:** `evaluateRateLimit(incomingIP, ipRequestLog, maxRequestsPerMinute, currentTime)` takes the visitor IP, log table, threshold, and current system time.
- **Algorithmic Logic:**

1. Set `windowStart = currentTime - 60`.
2. `FOR EACH record IN ipRequestLog:` (Iteration)
- `IF record[0] == incomingIP THEN` (Selection - IP found)
a. Filter `record[1]` to keep only timestamps $> windowStart$.
b. `IF length(record[1]) >= maxRequestsPerMinute THEN` (Selection - Threshold exceeded)
- Return `"BLOCK_RATE_LIMIT_EXCEEDED"`.
c. `ELSE:`
- Append `currentTime` to `record[1]`.
- Return `"ALLOW_REQUEST"`.
3. If IP not found, append `[incomingIP, [currentTime]]` to `ipRequestLog` and return `"ALLOW_REQUEST"`.

Part D: Testing, Edge Cases & Verification

- **Test Call 1 (Legitimate Human Traffic):**
- *Arguments:* `evaluateRateLimit('192.0.2.1', [['192.0.2.1', [10, 25]]], 10, 40)`
- *Condition Tested:* IP has made 2 requests in the last 60 seconds; threshold is 10 ($2 < 10$).
- *Outcome:* Returns `"ALLOW_REQUEST"`; request is processed normally.
- **Test Call 2 (Botnet Request Flood):**
- *Arguments:* `evaluateRateLimit('198.51.100.7', [['198.51.100.7', [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]]], 10, 11)`
- *Condition Tested:* IP has already logged 10 requests within 10 seconds; reaches threshold ($10 \ge 10$).
- *Outcome:* Returns `"BLOCK_RATE_LIMIT_EXCEEDED"`; packet dropped, protecting server CPU resources.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: mitigating Distributed Denial of Service (DDoS) floods.	**1 pt**
Row 2: Program Function: Describes request frequency tracking and dynamic rate-limiting behavior.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `ipRequestLog` list holding IP addresses and request timestamps.	**1 pt**
Row 4: Complexity Management Explained: Explains how the log list enables dynamic tracking of thousands of client IPs.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `evaluateRateLimit` with incoming IP, log list, threshold, and current time.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses iteration filtering timestamps, selection comparing against threshold, and list append.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Detailed explanation of rolling window rate evaluation.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (normal browsing rate vs botnet flood exceeding limit).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Contrasts HTTP 200 pass with HTTP 429 Too Many Requests block.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 99 · Model Solution & Scoring Rubric

TCP Packet Sequencing & Out-of-Order Reassembly Buffer ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

• **Program Purpose:** `TCPReassemble` demonstrates the reliability mechanisms of the Transmission Control Protocol (TCP) by buffering packets that arrive out of order, verifying sequence integrity, and assembling complete data payloads.

• **Program Function:** Arriving packets with sequence numbers are pushed into a memory buffer. When triggered, the program checks if all packets from 1 to N are present. If complete, it stitches the text payloads together in proper order; if a sequence number is missing, it signals a retransmission request.

Part B: Data Abstraction & Complexity Management

• **List Identified:** `packetBuffer`, a list of packet objects where each item contains `[sequenceNumber, payloadData]`.

• **Managing Complexity:** Using `packetBuffer` enables the receiver to store packets arriving in any arbitrary permutation and sort them systematically. Without this list, the programmer would need to create predetermined variables for each expected packet position, failing completely if packets arrive scrambled or if file sizes vary.

Part C: Managing Complexity & Algorithmic Procedure

• **Procedure Header & Parameters:** `assembleDataStream(packetBuffer, expectedTotalPackets)` takes the buffer list and total packet count.

• **Algorithmic Logic:**

1. Set `reconstructedMessage = ""`, `missingPackets = []`.
2. `FOR seqNum FROM 1 TO expectedTotalPackets:` (Iteration)
a. Set `foundPacket = null`.
b. `FOR EACH packet IN packetBuffer:`
- `IF packet[0] == seqNum THEN` (Selection - Sequence match)
- `foundPacket = packet`
c. `IF foundPacket != null THEN`
- `reconstructedMessage = reconstructedMessage + foundPacket[1]`
d. `ELSE:`
- Append `seqNum` to `missingPackets`.
3. `IF length(missingPackets) == 0 THEN` return `['SUCCESS', reconstructedMessage]`.
4. `ELSE:` return `['RETRANSMIT_REQUEST', missingPackets]`.

Part D: Testing, Edge Cases & Verification

• **Test Call 1 (Complete Out-of-Order Arrival):**
- *Arguments:* `assembleDataStream([[3, 'WORLD!'], [1, 'HELLO '], [2, 'BEAUTIFUL ']], 3)`
- *Condition Tested:* All sequence numbers 1, 2, 3 are present in buffer, but arrived scrambled.
- *Outcome:* Algorithm sorts by sequence number and returns `['SUCCESS', 'HELLO BEAUTIFUL WORLD!']`.

• **Test Call 2 (Dropped Packet Missing from Stream):**
- *Arguments:* `assembleDataStream([[1, 'HELLO '], [3, 'WORLD!']], 3)`
- *Condition Tested:* Packet 2 was lost in network transit; `foundPacket` is null for `seqNum = 2`.
- *Outcome:* Returns `['RETRANSMIT_REQUEST', [2]]`; TCP signals sender to retransmit packet #2.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: ensuring reliable, ordered data delivery across packet-switched networks.	**1 pt**
Row 2: Program Function: Describes buffer reception, sequence sorting, and reconstructed text output.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `packetBuffer` list storing sequence numbers and payload strings.	**1 pt**
Row 4: Complexity Management Explained: Explains how the list manages out-of-order buffering dynamically.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `assembleDataStream` with packet buffer and expected packet count.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses iteration across sequence indices, selection checking missing numbers, and string concatenation.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Detailed explanation of sequence verification and missing packet detection.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (complete out-of-order set vs missing packet set).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Contrasts successful reassembled string with missing packet request alert.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 100 · Model Solution & Scoring Rubric

Multi-Factor Authentication (MFA) & Secure Token Validator ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Function

• **Program Purpose:** `AuthShield` provides an enterprise identity defense system that prevents unauthorized account takeovers by requiring two distinct authentication factors (something you know and something you have).

• **Program Function:** A user enters their username, password, and the 6-digit code from their smartphone authenticator app. The system verifies both factors. If valid, the user gains access; if either fails or if too many attempts occur, access is denied.

Part B: Data Abstraction & Complexity Management

• **List Identified:** `userAuthDatabase`, a list of user account records where each record contains `[username, storedPasswordHash, secretSeed, failedAttempts, isLocked]`.

• **Managing Complexity:** Using `userAuthDatabase` allows the application to manage thousands of user accounts dynamically. Without this list, every user's credential state and attempt counter would require separate variables, making scalable authentication impossible.

Part C: Managing Complexity & Algorithmic Procedure

• **Procedure Header & Parameters:** `validateMFA(username, enteredPasswordHash, enteredToken, currentTime, userAuthDatabase)` takes login inputs and the user database.

• **Algorithmic Logic:**

1. `FOR EACH user IN userAuthDatabase:` (Iteration)

- `IF user[0] == username THEN` (Selection - User match)

a. `IF user[4] == true THEN` return `"ACCOUNT_LOCKED"`.

b. Set `expectedToken = computeTOTP(user[2], currentTime)`.

c. `IF enteredPasswordHash == user[1] AND enteredToken == expectedToken THEN` (Both factors valid)

- `user[3] = 0` (Reset failed attempts).

- Return `"AUTHENTICATION_SUCCESSFUL"`.

d. `ELSE:`

- `user[3] = user[3] + 1` (Increment failed attempts).

- `IF user[3] >= 5 THEN user[4] = true` (Lock account).

- Return `"AUTHENTICATION_FAILED"`.

2. Return `"USER_NOT_FOUND"`.

Part D: Testing, Edge Cases & Verification

• **Test Call 1 (Valid Password and Valid Token):**

- *Arguments:* `validateMFA('alex26', 'hash_pass123', '582910', 1000, db)` where `expectedToken` at time 1000 is `'582910'` and password hash matches.

- *Condition Tested:* Both Factor 1 (password) and Factor 2 (token) are correct; `user[4]` is false.

- *Outcome:* Returns `"AUTHENTICATION_SUCCESSFUL"`; failed attempt counter resets to 0.

• **Test Call 2 (Valid Password with Stale/Expired Token):**

- *Arguments:* `validateMFA('alex26', 'hash_pass123', '111222', 1000, db)` where entered token `'111222'` does not match expected `'582910'`.

- *Condition Tested:* Factor 1 passes, but Factor 2 fails due to token expiration or typo.

- *Outcome:* Returns `"AUTHENTICATION_FAILED"`; `user[3]` increments to record the failed login.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Program Purpose: Explains purpose: protecting user accounts through Multi-Factor Authentication (MFA).	**1 pt**
Row 2: Program Function: Describes two-step verification process and account access decision.	**1 pt**
Row 3: Data Abstraction Identified: Identifies `userAuthDatabase` list holding usernames, password hashes, shared secrets, and failed attempts.	**1 pt**
Row 4: Complexity Management Explained: Explains how the list manages authentication states across all registered users.	**1 pt**
Row 5: Procedure Identified & Parameterized: Presents `validateMFA` with username, password hash, token, time, and database list.	**1 pt**
Row 6: Algorithmic Sequencing, Selection & Iteration: Uses iteration across user database, selection checking factors and lockout thresholds, and attempt counter updates.	**1 pt**
Row 7: Algorithmic Correctness & Logic Explanation: Detailed explanation of verifying both distinct authentication factors.	**1 pt**
Row 8: Distinct Test Calls Provided: Provides two distinct test calls (valid password + valid token vs valid password + expired/invalid token).	**1 pt**
Row 9: Conditions & Outcomes Detailed: Contrasts successful authenticated session creation with security denial and failure increment.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

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Thank You, Future Systems Architects & Network Engineers!

YOU HAVE COMPLETED THE 100-QUESTION UNIT 4 MASTER EXAMINATION

Congratulations on rigorously conquering all 100 questions of **Unit 4: Computer Systems and Networks**. From mastering packet switching, IP addressing hierarchies, and fault-tolerant redundant routing, to quantifying parallel speedup via Amdahl's Law and navigating public-key cryptography, you have developed a profound understanding of the interconnected systems powering the modern digital universe.

The Internet is one of humanity's greatest engineering achievements—a global, decentralized network built on trust, open standards, and fault tolerance. With these computational insights and analytical skills, you are exceptionally prepared to excel on the AP Computer Science Principles examination and shape the next generation of computing infrastructure.

"We reject: kings, presidents, and voting. We believe in: rough consensus and running code." — David Clark, IETF